Economic Studies 235

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# Algorithms, Incentives, and Information

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ECONOMICS AT UPPSALA UNIVERSITY

The Department of Economics at Uppsala University has a long history. The first chair in Economics in the Nordic countries was instituted at Uppsala University in 1741.

The main focus of research at the department has varied over the years but has typically been oriented towards policy-relevant applied economics, including both theoretical and empirical studies. The currently most active areas of research can be grouped into six categories:

* Labour economics
* Public economics
* Macroeconomics
* Microeconometrics
* Environmental economics
* Housing and urban economics

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Lovisa Rambjer

# Algorithms, Incentives, and Information

## Essays on Labor Markets and Public Service Provision



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Dissertation presented at Uppsala University to be publicly examined in Hörsal 2, Ekonomikum, Kyrkogårdsgatan 10, Uppsala, Monday, 1 June 2026 at 13:15 for the degree of Doctor of Philosophy. The examination will be conducted in English. Faculty examiner: Associate Professor Nikolaj Harmon (Department of Economics at University of Copenhagen).

**Abstract**
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**Essay 1:** Can caseworkers improve on algorithmic assignment of unemployed jobseekers to active labor market programs (ALMPs) at the Public Employment Service (PES)? Because ALMP assignment involves a trade-off between targeting jobseekers at highest unemployment risk and those with the largest expected treatment effects, caseworkers may add value beyond prediction alone. Using as-if random assignment of jobseekers to caseworkers, I reconstruct the algorithmic counterfactual and estimate the effects of deviations from the algorithm's recommendations. I find that caseworker discretion weakens alignment with risk-based targeting and lowers overall employment.**Essay 2 (with Arne Uhlendorff and Johan Vikström):** Unemployed job seekers must comply with job-search requirements to be eligible for unemployment insurance benefits. We exploit variation in strictness across caseworkers who conduct monthly reviews of these requirements to examine the effects of stricter enforcement on job finding, search intensity, and job quality. This variation arises from central office caseworkers who are randomly assigned to review job seekers' activity reports and whose only interaction with job seekers occurs through this process. We find that stricter enforcement increases the likelihood of reported violations and benefit sanctions, leading to persistently higher job-search intensity and shorter unemployment durations. We find no evidence of negative effects on job quality. These positive employment effects are more persistent among job seekers from low-growth industries and those with weak labor market attachment.**Essay 3 (with Umrbek Allakulov, Serena Cocciolo, Binayak Das, Md. Ahasan Habib, and Anna Tompsett):** Can transparency interventions improve WASH service provision? We use a randomized experiment to evaluate the impacts of a transparency intervention, a deliberative multi-stakeholder workshop initiated with a community scorecard exercise, in schools in rural Bangladesh. To measure impacts, we combine survey data, direct observations, and administrative data. The intervention leads to moderate but consistent improvements in knowledge of WASH standards and practices, and institutions for WASH service management, but does not improve school WASH service provision or change WASH facility use patterns. Drawing on rich descriptive data, we suggest several reasons why the intervention we evaluate did not improve WASH service outcomes and propose ways to improve the design of future interventions.

*Keywords:* Statistical Profiling, Prediction, Unemployment Insurance, Job Search, Monitoring, Water, Sanitation, Governance, Transparency

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# Introduction

Public provision of goods and services plays a fundamental role in modern economies. Where markets fail—whether due to externalities, public goods problems, or missing insurance markets—governments intervene to improve welfare and support growth (Pigou, 2017; Samuelson, 1954). But the case for intervention does not guarantee its success. Even well-intended programs may be undermined by incentive distortions, governance failures, and information frictions. A central question is thus how policymakers can improve the efficiency of public services—and which strategies actually work. This dissertation examines three instruments policymakers can use to improve public service provision: algorithms, incentives, and information. Spanning two distinct policy settings and countries, it asks when these instruments work, when they fall short, and why they do so.

The first essay, *Prediction versus Discretion: Human-Algorithm Collaboration in Assignment of Unemployed Jobseekers*, studies the use of AI to target public interventions in an important labor market domain: the assignment of active labor market programs (ALMPs) to unemployed jobseekers. Sweden, like many OECD countries, relies heavily on ALMPs to support workers through labor market transitions. Because ALMPs involve substantial public spending and their effects are highly heterogeneous, effective targeting is a first-order policy concern. The Swedish Public Employment Service uses an algorithm to recommend treatment, while caseworkers retain final discretion. This raises a central question for the design of human-algorithm systems: when caseworkers override the algorithm, do they improve outcomes or worsen them? More broadly, the essay highlights a fundamental trade-off in policy targeting between prioritizing those most in need and those most likely to benefit.

The second essay, *Strictness in the Evaluation of Job Search Effort and Employment Outcomes*, examines the importance of monitoring in the unemployment insurance system in Sweden. The unemployment insurance system is a central institution in modern welfare states. By providing income support after involuntary job loss, it protects workers against large earnings shocks. But it also weakens incentives to search for work, creating a classic trade-off between insurance and moral hazard. The essay focuses on one central policy tool used to manage this trade-off: the monitoring of job-search effort. It studies whether stricter monitoring improves job-finding and how it affects post-unemployment job quality. It also asks whether the benefits and costs of monitoring vary across jobseekers facing different individual unemployment risks and local labor market conditions.

The third essay, *Transparency, Governance, and Water and Sanitation: Experimental Evidence from Schools in Rural Bangladesh*, studies public service delivery in the context of water, sanitation, and hygiene (WASH) in rural Bangladeshi schools. In developing economies, poor sanitation and hygiene remain a major obstacle to human capital formation. Schools are a particularly important setting: insufficient WASH facilities can reduce attendance and attainment, both through illness and through female absenteeism linked to poor menstrual hygiene management. Transparency and accountability interventions have attracted substantial policy interest because they are seen as a potentially low-cost method of improving service quality by informing and empowering users. Yet the empirical evidence on their effectiveness remains mixed. The essay evaluates whether a community scorecard workshop can improve WASH service provision in schools and, more broadly, sheds light on when information alone is sufficient to improve governance and service delivery.

# 1. Literature

The three essays in this thesis contribute to three distinct literatures—on algorithmic decision-making, unemployment insurance design, and transparency and accountability in public service delivery.

Essay 1 contributes to the growing literature on algorithmic decision-making, which studies the use of machine learning to support or replace human experts. In narrowly defined prediction tasks—such as medical diagnosis, pretrial risk assessment, and hiring—algorithms are often found to outperform human decision-makers in terms of accuracy and bias (Kleinberg et al., 2018; Angelova et al., 2025; Mullainathan and Obermeyer, 2022; Hoffman et al., 2018). Yet in many public policy settings, prediction is only one part of a broader allocation problem. Policymakers often face competing objectives, such as whether to target individuals at highest risk or those most likely to benefit from an intervention. Essay 1 studies exactly such a setting: one in which the algorithm targets unemployment risk, but policymakers also care about increasing overall employment. It provides causal evidence on how caseworker discretion over algorithmic recommendations affects alignment with both objectives, and highlights that improved prediction does not automatically imply better policy targeting.

Essay 1 also relates to a strand of literature on the specific use of predictive algorithms in public employment services. Around 40 percent of OECD countries use so called statistical profiling tools in their PES (OECD, 2019, 2024). Most work in this area has focused on model development, predictive performance, and comparisons between statistical models and historical caseworker decisions (Frölich et al., 2003; Lechner and Smith, 2007; van den Berg et al., 2023). More recent work examines the actual deployment of profiling tools.

(Ernst et al., 2024). Essay 1 is the first paper to provide causal evidence on the effects of caseworker discretion over algorithmic recommendations in a PES setting.

Essay 2 contributes to the literature on optimal unemployment insurance design. This literature highlights a central trade-off: unemployment insurance provides valuable insurance against income loss, but may also weaken job-search incentives and prolong unemployment (Baily, 1978; Chetty, 2006). Policymakers can respond along two main margins: benefit generosity and duration, or the use of job-search requirements and monitoring. Existing evidence suggests that stricter requirements, monitoring, and sanctions shorten unemployment duration (Arni and Schiprowski, 2019), although effects may be offset by substitution from informal to formal search methods (van den Berg and van der Klaauw, 2006) and may come at the cost of lower job quality (Arni et al., 2013).

The essay makes two contributions. First, it combines credible identification—based on as-if random assignment of jobseekers to caseworkers with different levels of strictness—with a real-world setting that strengthens external validity. Importantly, because caseworkers have no direct contact with jobseekers, we can isolate the effect of monitoring strictness from other caseworker behaviors. Second, rich administrative data allow us to study heterogeneity in the effects of stricter monitoring across jobseekers facing different individual unemployment risk and local labor market conditions—highlighting the scope for more targeted monitoring policies.

Essay 3 contributes to the literature on transparency and accountability interventions aimed at improving public service provision by informing and empowering users. These interventions seek to make service quality more visible and to stimulate participation and accountability by engaging local stakeholders. They have been widely embraced by policymakers, including in the WASH sector, where community involvement is often viewed as especially important. Yet the empirical evidence on their effectiveness remains mixed.

Kosack and Fung (2014) review transparency interventions aimed at improving government services and conclude that they are most promising in settings where users can act on the information they receive, for example by switching providers. In settings where users face only a single provider, successful interventions tend to combine salient, actionable information with incentives or support that enable users to act. In settings with reluctant providers, top-down pressure from policymakers or allied actors may also be important. Successful examples include Björkman and Svensson (2009); Barr et al. (2012); Mohanan et al. (2020).

The essay contributes to this literature by providing the first evidence on the effectiveness of a transparency intervention in the WASH setting. Low-cost transparency interventions, such as one-off stakeholder workshops, may be attractive because they are inexpensive and scalable. Yet in settings where service improvements require costly investments or collective action, local actors

may face substantial barriers to responding—raising the question of whether transparency alone is sufficient to improve service delivery.

# 2. Methodology

Modern empirical microeconomics is centered on identifying causal effects, emphasizing credible research design, the use of microdata, and transparent empirical analysis. This credibility revolution has highlighted randomization as the gold standard for causal inference, while also stressing the importance of testable identifying assumptions, transparency, and replicability (Card, 2022). All three essays in this dissertation build on this tradition.

## Identification

In both Essays 1 and 2, the key empirical challenge is that treatment is not randomly assigned. In Essay 1, all caseworkers exercise discretion over algorithmic recommendations, meaning all jobseekers are potentially affected—leaving no untreated comparison group. In Essay 2, monitoring strictness in the individual case is likely influenced by jobseeker characteristics—rather than being externally imposed. In both Essay 1 and 2 I use an *examiner tendency design* (Chyn et al., 2025) to solve these problems: because caseworkers differ systematically in their decisions when faced with similar cases, caseworker identity generates plausibly exogenous variation in treatment. The approach dates back to the 1930s, when criminologists documented substantial variation across judges in sentencing behavior (Gaudet et al., 1932). Pioneered in economics by Kling (2006), it is now widely used across fields in empirical economics.

The Swedish Public Employment Service provides an attractive setting for the examiner tendency design. The PES's high degree of task specialization makes it possible to isolate the effects of specific decisions from other caseworker behaviors. Moreover, cases are assigned to caseworkers in a manner that is effectively random, so that caseworkers face comparable pools of jobseekers on average. In Essay 1, I exploit variation across caseworkers in their propensity to assign jobseekers to different programs and in their propensity to deviate from algorithmic recommendations. Essay 2 exploits variation in how strictly caseworkers evaluate job-search requirements.

Essay 3 uses a randomized controlled trial, in which schools are randomly assigned to a treatment and control group. Because assignment is controlled by the researcher, this design provides a particularly credible basis for causal inference. Nonetheless, all three essays require careful empirical validation, and statistical tests are used throughout to assess whether the relevant randomization or as-if random assignment assumptions are supported in the data.

### Reproducibility

Credible causal research requires not only sound identification, but transparency in how the empirical design is implemented and reported. The three essays reflect this in different ways, depending on their research design. Essay 3 follows best practice for experimental work by pre-registering the study design and a pre-analysis plan at the AEA RCT registry,[1] reducing scope for data mining and selective reporting. The main analysis was specified in advance, exploratory analyses are clearly marked in the paper, and a full replication package including anonymized data and code is available through the publisher's website.

In non-experimental work, credibility is built through transparent research designs, careful robustness analyses, and reproducible code. Essays 1 and 2 rely on Swedish administrative data and describe sample construction in detail. Crucially, the empirical design is linked closely to the institutional setting, making the identifying assumptions as transparent and plausible as possible. Supplementary appendices provide extensive robustness checks.

## Main Findings

Each essay examines how a specific instrument—algorithmic targeting, monitoring, or transparency—shapes the efficiency of public service provision.

Essay 1 evaluates the algorithm used by the Swedish Public Employment Service (PES), measuring performance by how well it ranks jobseekers by their baseline unemployment risk and how accurately it identifies those who would have found work without assistance. The performance of the algorithm is only slightly better than a random assignment, and notably worse than the PES's own evaluations suggest, possibly because the PES relies on lower-quality internal data and does not fully account for the fact that some observed outcomes are affected by subsequent treatment assignment.

The essay then compares actual caseworker assignments with the counterfactual algorithmic allocation. Relative to the algorithm, caseworkers reduce alignment with an assignment rule based on unemployment risk and they lower overall employment rates. On average, they assign active labor market programs to jobseekers with both lower unemployment risk and lower treatment effects. Even when the algorithm performs poorly, caseworkers fail to improve outcomes on any dimension—highlighting that keeping humans in the loop does not necessarily correct algorithmic mistakes.

Essay 2 finds that stricter monitoring of job-search requirements shortens unemployment spells, without detectable negative effects on job quality. The mechanism runs through a sequence of responses: a rejected activity report raises the probability of an unemployment insurance sanction and triggers an

[1] `https://www.socialscienceregistry.org/versions/54056/docs/version`

additional call from the PES, strengthening both the incentive to search and the support available to do so. Those most affected by stricter monitoring look broadly similar to other jobseekers, except that they submit very few job applications before the first evaluation. This pattern is consistent with moral hazard rather than genuine barriers to employment. In line with this, the effect is initially larger among jobseekers with low individual unemployment risk, but fades after the first few months. Among jobseekers at high risk of unemployment, and among those previously employed in declining industries, the effect is more persistent, suggesting that early monitoring may also help prevent vulnerable jobseekers from becoming trapped in long-term unemployment.

Essay 3 evaluates a transparency-for-accountability intervention in the form of a multi-stakeholder community workshop. The analysis traces the causal chain from changes in knowledge and institutions, to the quality of WASH facilities as well as their use, and finally to school attendance and exam performance. While the intervention improved knowledge and institutions, it paradoxically had negative effects on the quality and use of WASH facilities for female students—not because treated schools deteriorated, but because control schools improved faster. No effects were found on attendance or exam results.

Essay 3 draws on rich qualitative data and insights from the broader literature to explore several possible explanations. The one-off workshop appears to have generated agreement on the existence of problems, but was insufficient to build consensus around a concrete plan of action. Beyond this, stakeholders who attended the workshop had limited capacity to influence WASH infrastructure alone, where improvements may require substantial resources. In addition, the intervention may have created misleading expectations about future funding—causing local actors to delay action rather than initiate it. The results suggest that information can change awareness and institutions, but may not be sufficient to improve service delivery when local actors face barriers to action.

## Conclusions and Policy Implications

What do the essays teach us about improving the efficiency of public services? Despite spanning two distinct settings and studying three different policy instruments, the essays point to two shared lessons about why interventions succeed or fail.

*1. Aligning Interventions with Objectives and Constraints*

A recurring tension across the essays is between interventions that are easy to scale and those that are well-adapted to the problem at hand. Standardized interventions are attractive because they are less costly, easier to replicate, and may improve consistency in decision-making. Yet standardization can also

be a weakness when the intervention is misaligned with policy objectives or poorly suited to the constraints of the setting.

This tension manifests differently across the three essays. In Essay 1, algorithmic targeting offers a consistent and easily administered decision tool, but the algorithm is designed to predict unemployment risk rather than treatment impact—leaving unresolved the central question of who should be prioritized. In Essay 3, the transparency intervention is similarly low-cost and scalable, but insufficiently adapted to a setting where improvements in WASH provision require resources, coordination, and authority that local stakeholders may not hold. Essay 2 offers a contrasting example: monitoring is less standardized and relies on individual caseworker assessments, offering the flexibility to respond to individual circumstances while still strengthening incentives. The broader lesson is not that standardized interventions are ineffective, but that their success hinges on how well they are matched to the objectives and constraints of the setting.

*2. Accounting for the Full Chain of Responses*

A second lesson concerns the chain of responses that an intervention sets in motion. All three essays study interventions that operate early in such a chain—an algorithmic recommendation, an evaluation of job-search effort, a community workshop. But the ultimate effects depend not on the initial intervention alone, but on what follows: whether caseworkers make good use of algorithmic recommendations, whether caseworkers provide adequate follow-up, and whether local stakeholders can translate awareness into action.

The essays show that these responses can unfold in very different ways. In Essay 2, the chain works: stricter evaluation of job-search efforts triggers not only stronger incentives through the risk of sanctions, but also additional contact and follow-up—supporting jobseekers while holding them accountable. In Essays 1 and 3, the chain breaks down at different points—in Essay 1 because caseworkers fail to translate algorithmic recommendations into good decisions, and in Essay 3 because local stakeholders lack the resources and authority to act on what the workshop revealed. Interventions designed as first steps, without a clear plan for what follows, risk not fulfilling their potential

Public provision of goods and services plays a fundamental role in modern economies, but effective provision cannot be taken for granted. The essays in this thesis do not offer a simple verdict on algorithms, incentives, and information as instruments for improving public policy. Instead, they shed light on the conditions under which such tools succeed or fail. Designing effective interventions requires clarity about objectives and constraints, and careful attention to the full chain of responses that follows.

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# Essay I. Prediction versus Discretion: Human-Algorithm Collaboration in Assignment of Unemployed Jobseekers

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# 1 Introduction

Predictive algorithms are expected to reshape fundamental areas of the economy by improving decision-making. They are believed to change occupational structure (Autor and Thompson, 2025), reorganize production (Agrawal et al., 2022) and increase the efficiency of public services (Ludwig et al., 2024). To date, algorithms are most commonly applied in settings like pretrial risk assessment, medical diagnosis, or job-testing - where the prediction task is narrowly defined, and the objective clear. In these domains, a growing literature compares human and algorithmic performance, finding algorithms to often outperform human decision-makers in terms of both accuracy and bias (Kleinberg et al., 2018; Angelova et al., 2025; Mullainathan and Obermeyer, 2022; Hoffman et al., 2018). Yet in many public policy settings, prediction is only one piece of a broader decision problem.

This paper studies the use of a predictive algorithm to improve public policy targeting in an important labor market domain: assistance to unemployed jobseekers. OECD countries spend on average 0.63 % of GDP on active labor market policy, yet their *value* depends critically on *who* is assigned to them. The assignment decision involves a fundamental trade-off — whether to prioritize the most vulnerable jobseekers, at highest risk of long-term unemployment (needs-based allocation), or those likely to have the greatest treatment effects (efficient allocation). These objectives are aligned only under restrictive conditions. To aid caseworkers, the Swedish Public Employment Service (PES) uses an algorithm that predicts unemployment risk and recommends support to those at highest predicted risk. Final assignment decisions, however, remain at the discretion of individual caseworkers.

Similar dilemmas arise also in other domains, such as hospital triage and anti-poverty targeting (Haushofer et al., 2025). Yet algorithms that combine both risk and treatment effect predictions remain rare in practice.[1] Because while machine learning algorithms excel at predicting baseline outcomes, targeting based on impact requires causal inference—a more demanding task where human judgment may still hold a comparative advantage (Kleinberg et al., 2015). This raises a central question: when algorithms predict risk but policymakers care about impact, can human discretion help bridge the gap—or does it merely reintroduce bias or noise?

I examine the effect of caseworker deviations from algorithmic recommendations along both dimensions of the trade-off. The algorithm itself performs only modestly better than random assignment in predicting reemployment probability, with a ROC-AUC of 0.57. Yet caseworker discretion further worsens outcomes, reducing both employment and the targeting of high-risk jobseekers. This suggests that caseworker discretion leads to an allocation that is both less efficient and less needs-based.

[1] There is, however, a growing literature on *policy learning* that focuses on estimating treatment heterogeneity (Zhou et al., 2023; Mascolo et al., 2024).

Public Employment Services worldwide have long used statistical models to guide decisions, with profiling tools dating back to the 1990s in the U.S. and now present in 40 % of OECD countries (OECD, 2019, 2024). The empirical setting in this paper is the Swedish Public Employment Service's (PES) assignment of jobseekers to either Independent Job Search or an Active Labor Market Program (ALMP). ALMPs range from job search assistance—such as career coaching, interview preparation, or job referrals—to more intensive interventions like education or vocational training. Detailed administrative data allows me to observe the algorithmic prediction, final caseworker allocation and subsequent outcomes.

The institutional setting offers ideal conditions to both evaluate the performance of the algorithm and estimate the impact of deviations. First, jobseekers are as-if randomly assigned to central office caseworkers via a meeting booking system: jobseekers choose a time slot, and caseworkers are subsequently allocated to meetings at random. Second, the PES is characterized by a clear division of tasks. Central office caseworkers conduct labor market assessments and make assignment decisions, while support is delivered by private or public contractors, and monitoring is handled by a separate central unit. As a result, any differences in outcomes between the observed allocation and the algorithmic counterfactual can be attributed to deviations in assignment decisions, rather than variation in caseworker support or monitoring.

I begin by presenting a conceptual framework for the policymakerâs problem, characterized by the PES's dual objective and the resulting trade-off between targeting unemployment risk and targeting treatment effects. Assignment based on predicted unemployment risk is welfare-maximizing if and only if the treatment effect is a deterministic and monotonically decreasing function of baseline reemployment probability. This condition is unlikely to hold in practice for two reasons. First, treatment effects are unlikely to be homogeneous conditional on reemployment probability. Second, even the average relationship between treatment effects and baseline reemployment probability need not be monotonically decreasing. In practice, this relationship is likely to depend both on the type of intervention and on the time horizon considered. While some studies find larger treatment effects among jobseekers with higher unemployment risk, they typically rely on relatively coarse categorizations, for example Van den Berg et al. (2025) and Knaus et al. (2022). Instead, the relationship may be non-monotonic, for example inverse U-shaped. Taken together, this points to a fundamental trade-off between targeting risk and targeting treatment effects.

In the empirical analysis, I compare caseworker and algorithmic performance along two dimensions: alignment with a needs-based allocation and alignment with an efficient allocation. I also estimate the population average reemployment probability, that is, the employment rate under the counterfactual in which all jobseekers are assigned to Independent Search. This estimate helps highlight important mechanisms behind the results. Finally, I examine

group fairness by testing whether either decision-maker systematically favors certain demographic groups.

The key empirical challenge is that only outcomes under realized caseworker decisions are observed. This creates a selection problem: employment outcomes are endogenous to assignment decisions, meaning we only observe employment in the absence of ALMP for a non-random subset of jobseekers â commonly referred to as the selective labels problem (Kleinberg et al., 2018). I address this challenge using two different empirical strategies: one to study needs-based alignment, and one to evaluate efficiency effects.

First, I analyze how caseworker discretion affects alignment with a needs-based allocation. To address selection, I employ a cutting-edge extrapolation method from Angelova, Dobbie, and Yang (2025), who study judges discretion over algorithmic recommendations in pretrial hearings, which leverages as-if random assignment of jobseekers to caseworkers and variation in assignment propensities. The performance measure used is the conditional employment rate - that is, the share of employed among those assigned to independent search.[2] Holding assignment rates constant, a higher conditional employment rate indicates better targeting of ALMPs to jobseekers at high risk of unemployment. Using the extrapolation method, I estimate the conditional employment rate under the algorithmic counterfactual and compare it to that achieved by caseworkers. This method also allows me to estimate the population average reemployment potential, i.e. the conditional employment rate if all jobseekers were assigned to Independent Search. Crucially, the method does not rely on assumptions about ALMP effectiveness.[3]

I find that the algorithm obtains a conditional employment rate which is 1.8 percentage points (3 %) higher than under a random assignment. Despite this relatively modest predictive performance, caseworker discretion reduces the conditional employment rate by 1.3 percentage points (2.2%) relative to the algorithmic counterfactual, implying that caseworkers prioritize high-risk jobseekers to a lesser extent and thereby make the allocation less needs-based. 80% of caseworkers underperform the algorithm on this measure.

Could limited prediction accuracy explain why caseworkers deviate from a needs-based allocation? To test this, I leverage a unique feature of the data: when caseworkers override one type of recommendation, they must record a reason. For those citing "high reemployment probability" and assigning to Independent Search, employment outcomes are observable. In this group, the conditional employment rate exceeds that of random allocation by 12%, suggesting that misalignment is not driven solely by poor prediction. Instead, caseworkers appear to pursue objectives beyond risk targeting.

[2]This corresponds to the share of true positives among predicted positives, i.e. analogous to a measure of precision in machine learning.

[3]Note that if caseworkers were purely targeting unemployment risk, differences in conditional employment rates could be interpreted as differences in prediction skill.

Third, I analyze the impact of caseworker discretion on efficiency by exploiting variation in caseworkers' propensity to deviate from the algorithm. Specifically, I examine the relationship between leave-out deviation rates and employment outcomes. The approach follows a standard 'judge design' (Chyn et al., 2025), and is similar to Hoffman et al. (2018), who study managerial deviations from an algorithmic hiring recommendation. The results show that the average caseworker reduces employment rates within 6 months by 1 percentage point (1.9% ) relative to strict adherence to the algorithm.

These results can be explained by a key finding: caseworker targeting leads to a lower employment rate than would arise if all jobseekers were assigned to Independent Search. This is partly driven by the fact that, among jobseekers assigned to an ALMP, I estimate that roughly half would have found employment on their own. By definition, these jobseekers can have at best zero treatment effects, and potentially negative effects due to lock-in. However, it remains unclear to what extent caseworkers select lower-treatment-effect jobseekers over higher-treatment-effect jobseekers within the group at high risk of unemployment.

Next, I ask why the PES would use an algorithm that appears to perform so poorly. By replicating one of the PES own evaluations of the model's ranking ability, I find that the PES is likely to substantially overestimate its performance. First, it relies on administrative outcome data that are subject to reporting error. Second, it uses realized jobseeker outcomes without accounting for endogeneity. In my estimates, correcting for these two issues reduces the ROC-AUC from 0.67 to 0.59, which is a sizeable difference.[4] The choice of evaluation method matters even more for the assessment of caseworker performance: while calculations that follow the PES approach suggest that caseworkers outperform the algorithm, my preferred estimates imply the opposite.

Finally, the fairness analysis shows that caseworkers reduce the risk-based targeting in all protected groups: gender, country of birth and age. However, the negative effects of deviation rate on employment are concentrated among male, native and young jobseekers.

Existing research on human-algorithm collaboration spans a wide range of settings, and the evidence is mixed. In pretrial hearings, Angelova et al. (2025) find that judicial discretion over algorithmic recommendations reduces prediction accuracy. In hiring, Hoffman et al. (2018) study the use of job-testing technology in the low-skill service sector and find that managers who override recommendations tend to make worse hires. Yet context and implementation matter. Grimon and Mills (2025) find that the use of an algorithmic risk assessment tool improved the allocation of investigations within child protective services by allowing workers to shift their attention from simpler tasks and cases to more complex ones. Agarwal et al. (2023, 2025) study human-AI col-

[4]ROC-AUC is a measure of ranking ability that ranges from 0 to 1, where random assignment corresponds to 0.5 and perfect targeting to 1.

laboration in radiology and fact-checking, and find that performance is highest when humans and algorithms operate independently.

This paper contributes to the literature on human-algorithm collaboration by studying a setting where improving prediction does not automatically advance policymaker goals. Unlike prior work that focuses primarily on prediction accuracy and fairness, I examine a trade-off between targeting risk and targeting impact - making treatment effects a central outcome. As algorithms are introduced into more complex policy domains, understanding the role of human discretion in such trade-offs becomes increasingly important.

Conceptually, my paper builds on the distinction between prediction problems and causal inference problems (Kleinberg et al., 2015). In many algorithmic settings, welfare depends solely on prediction because the decision itself has no causal impact (e.g., diagnosing a disease), or the causal effect is deterministic and monotonic in the predicted outcome (e.g., the treatment effect of detention is trivially largest for the most risky defendants). In such cases, prediction and decision objectives are naturally aligned.

This study also provides the first empirical evidence on the consequences of caseworker discretion over algorithmic recommendations in the Public Employment Service (PES) context. Today, roughly 40 % of OECD countries use statistical profiling tools at their PES (OECD, 2019, 2024). Making this an important setting in which to study human-algorithm interaction.

The early literature on the use of algorithms in the PES focused primarily on model development and predictive performance (Rosholm et al., 2004; O'Connell et al., 2012), and on comparing statistical models with historical caseworker decisions (Frölich et al., 2003; Lechner and Smith, 2007). More recent work examines the contemporary or prospective use of profiling tools. Ernst et al. (2024) study how algorithmic risk scores influence access to job search assistance in the Flemish PES, while van den Berg et al. (2023) compare caseworker and algorithmic predictions of reemployment probabilities using German data. Both studies highlight the value of algorithmic predictions but do not address the endogeneity of jobseeker outcomes to caseworker decisions. This paper contributes to this literature by linking traditional work on statistical profiling to the modern literature on algorithmic decision-making. It does so by examining human discretion over algorithmic recommendations and employing recent econometric methods that explicitly account for selection.

The remainder of the paper is structured as follows: Section 2 provides an overview of the institutional context and Section 3 introduces the data. Section 4 sets up the conceptual framework, which is mapped to the empirical strategy in Section 5. Section 6 and 7 go through identifying assumptions and descriptive statistics. Results are presented in Section 8 and, finally, Section 9 concludes.

# 2 Institutions - The Swedish Public Employment Service

The Swedish Public Employment Service (PES) operates under a dual mandate set by the government: to prioritize jobseekers who are, or are at risk of becoming, long-term unemployed, while also promoting overall employment, matching efficiency, and functioning of the Swedish labor market (Regeringen, 2025).

Since 2019, the Swedish PES has undergone major reforms aimed at improving efficiency through digitization, automation, and privatization. The agency's role has shifted towards labor market assessments, oversight of private providers, data analysis, and digital infrastructure, while Job Search Assistance is delegated to private contractors in a competitive market. As a result, PES caseworkers now focus primarily on assessing eligibility for ALMPs. The reform coincided with growing political attention to long-term unemployment, with the introduction of a statistical profiling tool to target programs toward those most at risk of long-term unemployment.

This section first outlines the standard process for newly registered jobseekers, followed by a detailed description of the algorithmic profiling tool. The information in this section is based on interviews with PES staff and internal documents.[5]

## 2.1 Process for Newly Registered Jobseekers

The process for a typical newly registered jobseeker can be summarized in three main steps. After the summary follows a detailed description of each step of the process.

1. **Registration and as-if random assignment of jobseekers to caseworkers at remote offices.**
   A jobseeker registers online with the Public Employment Service (PES) on the first day of their unemployment spell. They fill in information about themselves and book a first remote meeting with a caseworker at a central office.
2. **Meeting and assignment decision.**
   A first remote meeting is held within two weeks. The caseworker uses the algorithmic recommendation together with private information to make a labor market assessment and assignment decision.
3. **Follow-up and conclusion.**
   If a jobseeker is assigned to an ALMP, the responsibility for the jobseeker and follow-up is transferred to a local office. An assignment to Independent Job Search is assessed every six months at a central office.

[5] Memo Gemensam process för beslutshandläggare – KROM och STOM (2020); Handläggarstöd Placeringsprocessen (2020); Handläggarstöd Innehåll och Avrop (2020); Presentation - Så funkar det AMPB 1.0 (2020).

There are no personal caseworkers. The as-if random assignment is in place for each meeting. At each follow-up, the algorithm is rerun and the algorithmic recommendation updated. Finally, the case is closed when the jobseeker secures employment or enrolls in education.

**Registration and As-If Random Assignment**
**Standard Central-Office Pathway**
An unemployed jobseeker registers online with the Public Employment Service (PES) on the first day of their unemployment spell. Registering is a requirement to apply for unemployment insurance benefits. At registration, jobseekers immediately book a first remote meeting by selecting a free slot. Slots are then allocated across caseworkers in a non-systematic manner, creating an as-if random assignment of jobseekers to central-office caseworkers. Meaning that jobseekers do not have any information about who they will meet, and have no way of selecting a preferred caseworker. All central-office meetings occur during regular working hours; there are no evening or weekend slots. And all caseworkers rotate across all types of slots. The central offices are located throughout Sweden and organized in 7 administrative offices. Central offices are not tied to specific geographic regions but instead serve jobseekers across the entire country. In this paper, I focus exclusively on assignment decisions made at central offices.

**Exemptions & Local-Office Pathway**
If a jobseeker cannot register online or attend a remote meeting: e.g., because they lack an electronic ID or require a translator, they are directed to a local PES office. Alternatively, they may visit a local office in person to register. Local offices serve a restricted geographical region and hold meetings either in person or remotely. Jobseekers assisted at local offices form a selected group with generally lower reemployment probabilities. Since the local-office slot-booking system does not guarantee as-if random assignment, I exclude assignment decisions made there from my analysis.

**The Role of the Caseworker**
The main task of PES caseworkers is to make labor market assessments and determine eligibility to ALMPs. The caseworker sets up an activity plan, with activities based on the assignment decision. Caseworkers also assist jobseekers with practical and administrative tasks, such as choosing a private provider. They do not provide any assistance themselves. The monitoring of jobseekers' job search activities is handled by a separate unit at the PES.

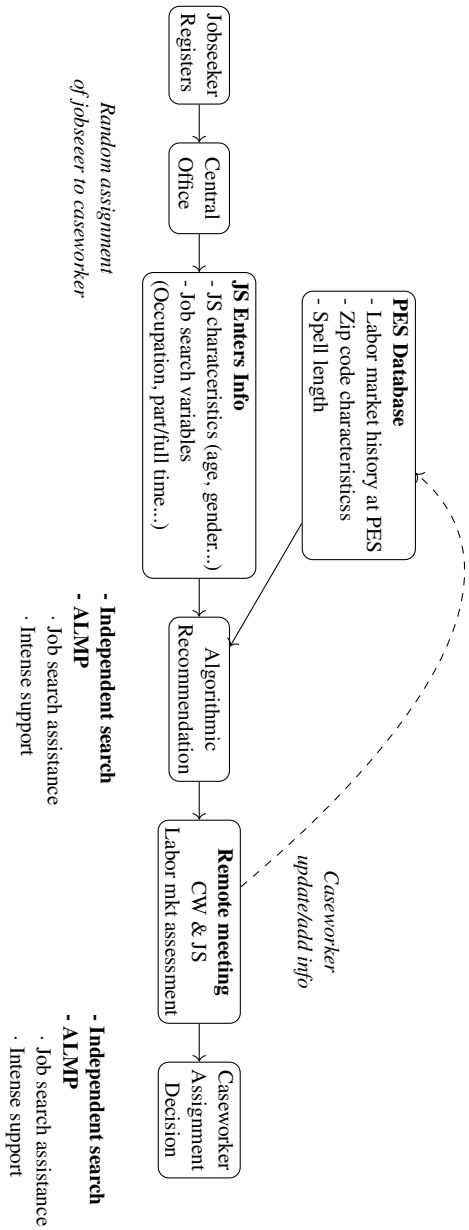



*Figure 1.* Process newly registered jobseekers

Figure 1 illustrates the procedure from registration to the first meeting and assignment decision.

**Active Labor Market Programs**

Active labor market programs at the Swedish PES range from job search assistance, including to more intense support. Job search assistance is a six-month program which is privately provided and include coaching and guidance, CV writing, interview training, and job referrals. The exact activities included are decided on by the private providers together with the jobseeker. There are also intense programs, such as labor market training or educational program lasting 6-12 months. Another alternative is regular education, which can be substantially longer.

## 2.2 Algorithmic Profiling Tool

The algorithmic profiling tool aims to predict the probability that a jobseeker will secure employment independently within six months from the profiling date.

**Input Variables and Data**

The algorithm is trained on the inflow of unemployed jobseekers and incorporates four main groups of input variables:

- **Self-reported characteristics:** age, gender, country of birth, disability status, and education.
- **Job search information:** desired occupation, relevant experience in that occupation, and preferred number of working hours (full-time or part-time).
- **Administrative data from PES registries:** labor market history including prior interventions, registration periods, current spell length, and ongoing programs.
- **Zip code-level characteristics:** local unemployment rate, median income, educational attainment, and share of foreign citizens.

In total, 26 input variables feed into the prediction. The full list appears in Table A1.

The data infrastructure at the PES posed a major obstacle to the design and training of the machine learning algorithm, both in terms of data quality and availability. Input variables were limited to those available in a single database, and self-reported jobseeker characteristics could not be verified.

To ensure high quality of self-reported characteristics, the jobseeker was informed that they were used for the statistical profiling and served as a key input for the caseworker's assignment decision. The lack of individual-level data made the algorithm heavily reliant on zip codeâlevel characteristics, limiting its ability to differentiate between jobseekers within the same area. (Arbetsförmedlingen, 2020).

The algorithm is trained on binary outcome data (0/1) for several events related to employment or full-time studies within a specified time frame. The

two primary outcomes are: (i) full-time employment and (ii) part-time employment within six months. The algorithmâs output is a profiling score, calculated as the sum of the predicted probabilities for these two outcomes. This score ranges from 0 to 1. Thus, the score itself should not be interpreted as a probability (Arbetsförmedlingen, 2020).

The outcomes data used to train the algorithm come from PES registries, which contain substantial measurement error. A manual quality check conducted by the PES revealed that 20-25% of case files included notes about positive jobseeker outcomes (employment or full-time studies) that were never recorded in the registries (Arbetsförmedlingen, 2020).

According to PES (Arbetsförmedlingen, 2023), the algorithm's accuracy (share of correctly classified outcomes) is 68 %, compared to a baseline random-classification accuracy of 57 %. Due to poor performance the PES updated the algorithm in April 2023.

**Score to Recommendation Mapping**

The algorithm produces a continuous profiling score for each jobseeker, which is a number ranging from 0 to 1 where a higher score implies a higher reemployment probability. The score is mapped to a categorical recommendation, which is shown to the caseworkers - i.e. the caseworkers see only the recommendation and not the underlying score.

The output has three main values (from highest to lowest predicted reemployment probability). For the purpose of this paper, I will group job search assistance and intense support into simply 'active labor market programs' (ALMP).

1. **Independent Search**
   Searching for a job mostly independently, with the help of digital self-service tools.

2. **Job Search Assistance**

3. **Intensive support**
   Including e.g. labor market training, labor market education, language training.

Rather than including the current unemployment duration as an input, the system applies time-varying cutoffs to the profiling score. Specifically:

- Cutoffs vary by the number of days since the start of the unemployment spell; jobseekers with shorter duration face lower thresholds than jobseekers with longer duration. Such that at a given score, the probability to be recommended an ALMP increases with the length of the spell.
- Jobseekers with profiling scores above the cutoff associated with their duration are recommended Independent Search. Those whose scores fall below the cutoff are recommended some form of ALMP.

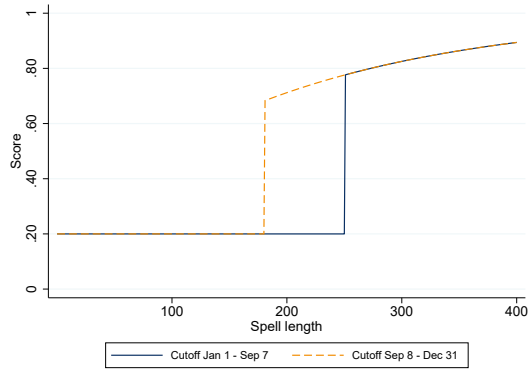



*Figure 2.* Cutoffs based on days unemployed

*Note:* Cutoffs by days unemployed. Jobseekers with a score above the cutoff were recommended Independent Search. From January 1 to September 7, there was a flat cutoff at 0.2 up until 250 days of registration. From September 8 to December 31, there was a flat cutoff at 0.2 up until 180 days of registration. After that the cutoff increased with the length of the spell. Up until 250/180 days the jobseekers could only be recommended Intense Support. But after that the jobseekers could also be recommended Job Search Assistance.

Figure 2 shows the time-varying cutoffs used in 2022 to decide whether a jobseeker is recommended Independent Search or an ALMP track. From January 1 through September 7, the cutoff remained fixed at a profiling score of 0.2 for all jobseekers up to 250 days of unemployment. Beginning September 8, the cutoff stayed at 0.2 only until 180 days, after which the required score increased with the spell length. Consequently, jobseekers needed at least 250 days of unemployment under the January 1 - September 7 rules—and 180 days under the September 8 onward rules—to become eligible for Job Search Assistance. Any jobseeker assigned to an ALMP earlier in the spell could only be placed into the Intense Support category.[6]

## 2.3 Labor Market Assessment & Assignment Decision

Assignment to ALMPs is determined primarily by the caseworkers' labor market assessment. The first meeting is typically held within two weeks after registration, where the caseworker conducts a holistic assessment by combining the algorithmic recommendation with private information elicited during the meeting. If needed, the caseworker can update the inputs used in the profiling tool and rerun the algorithm. In 5 % of first meetings such updated lead to a change in the algorithmic recommendation change relative to the registration, see subsection F5 for details.

[6]Certain groups with generally weak labor market attachment — such as long-term unemployed youth, long-term unemployed adults, and newly arrived immigrants — face less restrictive time-varying cutoffs. Those groups are excluded from the analysis.

Based on the assessment, the caseworker makes an assignment decision, which mirrors the algorithmic recommendation. In case of an assignment to Job Search Assistance, the private provider takes over the responsibility to decide on specific activities. While an assignment to any Intense Support is more specific, for example labor market training, labor market education, or language training. In this paper I focus on the decision to assign either to Independent Search or any ALMP, abstracting from the choice between job search assistance and other programs. If the caseworker determines that an in-person evaluation is necessary, they may refer the jobseeker to a local office. In this paper, such referrals are categorized as assignments to an ALMP, as they indicate that the jobseeker's labor market attachment is too weak to justify Independent Job Search.

The caseworkers always see the recommendation of the algorithm, but are only instructed to strictly follow it under certain circumstances. The algorithm is primarily used to determine eligibility to privately provided Job Search Assistance. Thus, caseworkers are strongly advised against deviating from a recommendation to Job Search Assistance. And during the period studied caseworker discretion was limited such that any deviation from Independent Search or Intense Support to Job Search Assistance required approval from a committee. Otherwise they had free discretion.

In addition, there are some ineligibility criteria such as part-time unemployment or start of employment or studies within 90 days, or other reasons that hinders the jobseeker from participating in an ALMP. To elicit whether the jobseeker is ineligible for ALMP and to facilitate the labor market assessment, the jobseeker is asked to answer 11 standardized questions. Appendix subsection A2 contains the full list of criteria and standardized questions.

Jobseeker preferences should be considered in the assignment decision, and jobseekers must be given the opportunity to express and justify any disagreement. However, caseworkers retain final discretion, and if a jobseeker refuses to participate the case should be referred to the PES monitoring unit, which will decide on a potential sanction.

**Follow-Up and Case Closure**

If a jobseeker is assigned to an ALMP, the responsibility for the jobseeker and follow-up is transferred to a local office. An assignment to Independent Job Search is assessed every six months at a central office. No jobseeker is paired with a single, dedicated caseworker; instead, they are as-if randomly assigned to a caseworker each meeting by simply signing up for a remote meeting slot online. Jobseekers may also request a meeting at any time by calling or booking online.

At each follow-up, the algorithm is rerun and the algorithmic recommendation is updated. Based on the new recommendation and information gathered during the meeting, the caseworker may initiate an ALMP assignment. A case is closed once the jobseeker has found part- or full-time employment (depend-

ing on their stated preference) or enrolled in an education program eligible for student aid. This includes adult education organized by the municipality *(kommunal vuxenutbildning)*, non-formal adult education *(folkhögskola)* at the primary or secondary level, or tertiary education such as higher vocational training or university studies.

# 3 Data

## 3.1 Data Sources and Coverage

I observe the universe of PES profiling events from the introduction in early 2020 through December 2022. On the jobseeker side, data include:

- Detailed PES administrative records (profiling scores, appointment dates, program starts, etc.).
- On the jobseeker side I have linked Swedish registry data (demographics, monthly earnings, income, education.).
- On the caseworker side, I have limited personnel data (tenure, wage and office) from the PES human-resources registry.

The data also contains a reason for deviating, whenever the caseworker has deviated from Job Search Assistance to either Independent Search or Intense Support.

I restrict the analysis to jobseekers starting their unemployment spell between January 1 and December 31, 2022, excluding the roll-out period of 2020 to 2021.[7]

## 3.2 Sample Selection

The main analysis focuses on the inflow of jobseekers from 2022.

*Identifying Decision Events:*
Many jobseekers receive multiple profiling events even when no assignment decision occurs (e.g., during routine administrative updates). To isolate profiling events associated with a caseworker decision, I keep the first profiling event and drop duplicates that are *identical* on three dimensions: (1) original score, (2) algorithmic recommendation, and (3) assignment—if it occurs within one month of the previous identical profiling. This rule removes 4 % of profiling events; the maximum number of profilings per jobseeker falls from 12 to 6, while the median remains 1.

*Exclude groups not due to as-if random assignment:*
I only include jobseekers subject to as-if random assignment at central offices. Following the institutional setting, I exclude the following groups.

[7] With the intention to extend to later profiling events and a longer follow-up period once the data become available.

- Jobseekers aged 16-24 or enrolled in the Youth Job Guarantee (24% of the initial pool).
- Jobseekers with prior long-term unemployment who are routed to specialized programs (8%).
- Jobseekers missing a valid search-occupation code, who are routed to specialized caseworkers (2.5%).

These exclusions remove 32% of profiled cases.

*Exclude non-UI beneficiaries:*
Jobseekers without unemployment-insurance (UI) entitlement who receive benefits are excluded (1.5 %).

*Caseworkerâlevel restrictions:*
The analysis will be conducted at the 'regime' level (Jan 1-Sep 7, Sep 8 - Dec 31), see Figure 2 for details. To ensure sufficient volume per evaluator, I limit the analysis to caseworkers who handled at least 50 cases during each regime. This requirement excludes an additional 17 % of observations.

## 3.3 Reason for Deviating

Whenever the caseworker deviates from Job Search Assistance to either Independent Search or Intense Support, they are required to enter a reason for deviating. Caseworkers can choose from predefined categories or enter a free-text explanation. In total, 3,161 deviations were recorded. In 86% of these cases, a free-text reason was entered; in the remaining 14%, one of eight predefined categories was selected. All free-text responses were manually reviewed and classified.

## 3.4 Defining Key Variables

**Defining Caseworker Assignment**
In the raw data, each profiling event includes the algorithmic recommendation. However, there is no single variable which records the actual assignment decision. Thus, to infer the *actual assignment* made by the caseworker, I use the following rules:

*Any ALMP:*
- Jobseekers whose algorithmic recommendation was Job Search Assistance and who did not have a recorded deviation to Independent Search, inferred by the reason for deviating.
- Jobseekers who start any ALMP (job search assistance, training, etc.) within three months of the profiling date.
- Jobseekers referred to a local PES office.

*Independent Search:*

- Jobseekers whose algorithmic recommendation was Job Search Assistance but for whom the caseworker recorded a deviation to Independent Search, as inferred by the reason for deviating.
- Jobseekers who do not start any ALMP within three months of the profiling date.

**Defining Employment**

The main outcome variable used in the paper will be employment rate, defined as the probability of exiting unemployment for full time employment within the considered time frame. Monthly full time employment is defined using payroll tax data and defined as earning at least 75 % of the collective bargain minimum wage in retail, which equals about 17,800 SEK (1,877 USD) per month. Alternative definitions of employment are used for robustness checks, such as data from the PES registry including exiting the PES to start studying. For 66,441 cases (50 % of the sample) there is tax data to follow up 12 months after the profiling date. After that I follow an increasingly smaller sample up to 17 months post profiling (15,234 observations).

**Adjusted Profiling Score.**

Each profiling event yields an *original profiling score*, between 0 and 1 which is unobservable to the caseworker, and a categorical *algorithmic recommendation* (Independent Search or ALMP). Since the cutoff between Independent Search and ALMP varies by the number of days since the start of the unemployment spell, see Figure 2, I construct an *adjusted score* that captures the distance (in profilingâscore units) to the independentâsearch cutoff at each jobseekerâs spell length. By simply subtracting effective cutoff facing the jobseeker, given their spell length, from the profiling score. This allows me to construct the actual PES ranking of jobseekers, and estimate the impact of caseworker deviations. For completeness, I will construct one algorithmic counterfactual using the "PES score", and one using the original algorithm. See Appendix subsection A6 for more details.

## 3.5 Final Sample

After all exclusions and definitions, the 2022 sample contains:

- 130,705 decision events (first and follow-up profilings).
- 892 caseworkers at central offices.

Most jobseekers appear only once or twice in the profiling data. Figure A3 displays the distribution of profiling counts per caseworker and per jobseeker in this final sample.

# 4 Conceptual Framework

In this section, I present a conceptual framework that formalizes the policymakerâs allocation problem. The policymaker balances two potentially competing goals: (i) a needs-based allocation, which prioritizes jobseekers with high unemployment risk, and (ii) an efficient allocation, which targets individuals with the largest treatment effects from ALMPs.

The framework serves two main purposes. First, it clarifies the conditions under which an allocation purely based on unemployment risk is optimal—highlighting the potential trade-off between a needs-based and efficient allocation—and provides a foundation for interpreting the role of both the algorithm and caseworkers in the allocation process.

Second, I use this framework to guide the empirical analysis. In section 5 I connect the framework to the empirical strategy, and in section 6, I discuss and assess the associated identifying assumptions.

## 4.1 Setting up the Policymaker's Assignment Problem

The policymakerâs problem is how to allocate ALMPs across jobseekers. For each jobseeker $i$, the policymaker chooses between assigning an ALMP ($D_i = 1$) or placing the individual in Independent Search ($D_i = 0$). While multiple ALMPs exist in practice, with potentially heterogeneous effects across individuals, I simplify the framework by assuming that each jobseeker faces a single relevant ALMPâfor instance, the program that would yield the highest employment probability for that individual.

Let the employment outcome for jobseeker $i$ be denoted by potential outcomes $Y_i^1 \in \{0,1\}$ under assignment to ALMP, and $Y_i^0 \in \{0,1\}$ under assignment to Independent Search. The observed outcome is then given by $Y_i = Y_i^0 \cdot (1 - D_i) + Y_i^1 \cdot D_i$. Define $y_i^0 = E[Y_i^0]$ and $y_i^1 = E[Y_i^1]$ as the probabilities of employment under the two assignments. Furthermore, let $(1 - y^0)$ denote the *unemployment risk*. *Treatment effect* is denoted $\gamma_i = E[Y^1 - Y^0] = y^1 - y^0$.

The policymaker is facing a budget constraint on the number of slots in the ALMP, which requires that only a share $\phi$ can be assigned to ALMPs. Let N denote the total number of jobseekers.

The dual objective of the Swedish PES can be interpreted for the policymaker here. Two notes. First, payoff is increasing in jobseeker unemployment risk. Second, payoff is increasing in treatment effect. The first comes trivially from the will to prioritize jobseekers at high risk of long term unemployment. The second can be derived from the goal to maximize overall employment. Under the assumptions that (i) the policymaker is risk neutral, and (ii) program assignment has not spillover effects across jobseekers, then

total employment is maximized by assigning jobseekers to ALMPs based on their expected treatment effects.[8]

Thus, the policymaker problem is to pick an assignment, $D \in \{0,1\}$, in order to maximize total payoff. Let the value of assigning a jobseekers to Independent Search be normalized to zero.

$$\begin{aligned} \max_{D_i,..,D_N \in \{0,1\}} & \sum_{i=1}^{N} D_i \cdot V_i(y_i^0, \gamma_i) + (1 - D_i) \cdot 0 \\ s.t. & \sum_{i=1}^{N} D_i \leq N \times \phi \end{aligned} \tag{1}$$

The value function has three important properties. First, it must be weakly increasing in both unemployment risk and treatment effect. Second, for jobseekers with negative treatment effects, it can never be optimal to assign them to an ALMP. And third, there will be a parameter $\alpha$ guarding the weight put on unemployment risk versus treatment effect. For example, such a function could take the following form:

$$V_i(y_i^0, \gamma_i) = \gamma_i (1 - y_i^0)^{\alpha} \tag{2}$$

Where $\alpha \in \{0, \infty\}$. The parameter $\alpha$ determines the weight put on need. Such that as $\alpha \to \infty$, the policymaker put more emphasis on jobseekers with high risk of long term unemployment. However, at a given level of risk, the payoff is increasing in treatment effect. And importantly, if the treatment effect is negative, it is always optimal to assign the jobseeker to independent search. For $\alpha = 0$, the policymaker only cares about maximizing overall treatment effects.

The solution to the problem, is rank all job seekers based on $V_i$ and assign them to the ALMP if it is positive and above some threshold $s$. The threshold parameter $s$, in turn, is set to satisfy the budget constraint.[9] If the value is negative for a set larger than $\phi N$, the budget constraint will not bind and $s = 0$ is optimal. Under the condition that the treatment effect of the ALMP is positive for at least a share $\phi$ of jobseekers, the threshold $s$ should be set in order to assign the share $\phi$ generating the largest payoff to the policymaker.

[8] Using policymaker risk neutrality and SUTVA implies that total employment is maximized by maximizing expected treatment effects. $E\left[\sum_i^N Y_i\right] = \sum_i^N (1 - D_i) y_i^0 + D_i y_i^1 = \sum_i^N y_i^0 + D_i (y_i^1 - y_i^0) = \sum_i^N y_i^0 + D_i \gamma_i$. Since the term $\sum_i^N y_i^0$ is constant across assignments, the maximization problem reduces to assigning ALMPs to those jobseekers with the largest treatment effects $\gamma_i$. This holds as long as there are no sharp thresholds in employment outcomes—i.e., jobseekers do not require a minimum probability of reemployment to transition into work.

[9] For example, suppose $N = 10$, $\alpha = 0$ and $\phi = 0.4$. Then, we would order the jobseekers according to $\gamma_i$ and set $s$ equal to the fourth highest $\gamma_i$.

Generally, the policymaker's expected payoff is maximized under the following decision rule.

$$D_i = 1\left[\gamma_i(1-y_i^0)^\alpha > s\right] \tag{3}$$

## 4.2 The Needs-Based Algorithm

The algorithm predicts jobseekers' baseline reemployment probability, i.e. $a(X_i) \equiv Pr(Y_i^0 = 1|X_i)$, using observable characteristics $X_i \in \mathscr{X}$. It recommends ALMP for all jobseekers whose probability, or profiling score, is below a threshold $s_a$, which is determined by the policymaker. The objective of the algorithm is to obtain a needs-based allocation.

The algorithmic recommendation follows a simple decision rule, where the cutoff $s_a$ is set to satisfy the budget constraint.

$$A_i = 1[a(X_i) \leq s_a] \tag{4}$$

## 4.3 Setting up the Caseworkers' Assignment Problem

In this section I set up the caseworkers' problem and show how it corresponds to the policymaker's problem and the algorithmic recommendation. Let each caseworker $j$ make an assignment decision analog to the policymaker's problem. For each jobseeker there is a vector of characteristics $X_i \in \mathscr{X}$ observable to both the algorithm and the caseworker and a vector of characteristics $V_{i,j} \in \mathscr{V}$ only observable to the caseworker. $\mathscr{X}$ and $\mathscr{V}$ are mutually exclusive.

Caseworkers first conduct a labor market assessment, predicting reemployment probability under assignment to Independent Search $y_i^0$, as well as reemployment probability under the ALMP $y_i^1$. The caseworkers take the algorithmic recommendation $A_i$, the public information $X_i$ and the private information $V_{i,j}$ into account and predicts the reemployment probabilities. Note that caseworkers do not observe the predicted reemployment probability by the algorithm, but only the assignment recommendation.

$$\begin{aligned} \hat{y}_{i,j}^0 &= h_{0,j}(A_i, X_i, V_{i,j}) \\ \hat{y}_{i,j}^1 &= h_{1,j}(A_i, X_i, V_{i,j}) \end{aligned} \tag{5}$$

The private information $V_{i,j}$ contains any information that the caseworker believes is important to predict reemployment probabilities, but that is not a feature for the algorithm. It could be (i) jobseeker preferences or (ii) jobseekers motivation, effort and search ability. I will allow $V_{i,j}$ to vary by caseworker to reflect differential caseworker ability to exert private information in the meeting. Each caseworker has individual evaluation functions of the reemployment probabilities, reflecting differences in the ability to predict reemployment probabilities, given the information at hand.

The caseworker will assign the jobseeker to an ALMP, if the predicted payoff exceeds the caseworker threshold $s_j$. The payoff to an assignment to Independent Search, $D_i = 0$, is set to zero. Again, the value function exert the same properties as the policymaker, and I will use the same functional form to illustrate sources of caseworker heterogeneity.

$$D_{i,j} = 1\left[\hat{\gamma}_i(1-\hat{y}_i^0)^{\alpha_j} \geq s_j\right] \tag{6}$$

Where $\alpha_j \in [0,\infty]$ and $s_j \in [0,1]$. Where the weight on unemployment risk versus treatment effects is allowed to vary by caseworkers.

While program capacity is determined at the aggregate level, caseworkers may differ in the implicit threshold they apply when deciding whether to assign a jobseeker to an ALMP. Lower-threshold caseworkers assign programs more readily, whereas higher-threshold caseworkers reserve treatment for cases with higher perceived payoff. These differences need not reflect different formal budgets at the caseworker level. Rather, they capture decentralized discretion in how caseworkers interpret the value of treatment, the strictness of the budget constraint, or the suitability of ALMPs for marginal jobseekers. This is captured by the caseworker parameter $s_j$

When different caseworkers face identical jobseekers, they may differ in their decision-making for some reasons. First, their ranking of jobseekers might differ - either due to difference in skill in estimating the unemployment risk and treatment effect, or depending on their weighting parameter $\alpha_j$. But they might also differ in the threshold $s_j$. Altogether, this means that caseworkers will differ in their propensity to assign jobseekers to ALMPs and Independent Search. Let $R_j^{ALMP} = E[D_{i,j}]$ denote the *ALMP rate* and $R_j^{IS} = 1 - E[D_{i,j}]$ denote *Independent Search rate*. Such that $R_j^{ALMP} + R_j^{IS} = 1$.

**Aligning Caseworker Decisions with Policymaker Goals**

The policymaker seeks to align caseworker decisions with policy objectives through both formal and informal channels. Informally, the trade-off between a needs-based and efficient allocation is conveyed through government-set goals and strategic steering documents. In the model, this is captured by assuming that caseworkers share the same underlying payoff function as the policymaker.

While individual caseworkers do not operate under a strict budget constraint or binding volume target, the policymaker indirectly influences overall assignment volumes via the algorithmic recommendation $A_i$, by setting the threshold $s_a$.

## 4.4 Optimal Assignment Mechanism

Under the dual objective of the PES and the statistical profiling tools based solely on unemployment risk an important question arises. When is strict adherence to the algorithm optimal for the policymaker, and when should caseworker discretion be allowed? The answer depends on two key factors. (1) the prediction accuracy of the algorithm relative to that of caseworkers. If caseworkers can improve upon the algorithm's estimates of baseline reemployment probabilities, they may increase the policymakerâs overall payoff. (2) whether a completely needs-based ranking aligns with the policymaker's objective.

Assignment based on unemployment risk is optimal if and only if treatment effect is a deterministic and monotonically decreasing function of reemployment probability.

**Proposition 1 (Trade-off between needs-based and efficient allocation.** )
*Suppose the policymaker's payoff from assigning treatment to jobseeker i is given by $V(\gamma_i, y_i^0)$, where it is increasing in $\gamma_i$ and decreasing in $y_i^0$. $\gamma_i$ denotes the treatment effect of ALMP and $y_i^0$ the baseline reemployment probability. An allocation rule based solely on $y_i^0$ is payoff-maximizing if and only if baseline reemployment probability is sufficient to rank policymaker's payoff from treatment assignment; that is, for any two jobseekers i and j,*

$$y_i^0 < y_j^0 \quad \Rightarrow \quad V(\gamma_i, y_i^0) \geq V(\gamma_j, y_j^0) \quad \forall i, j.$$

*Equivalently, treatment effect must be a deterministic and monotonically decreasing function of baseline reemployment probability*

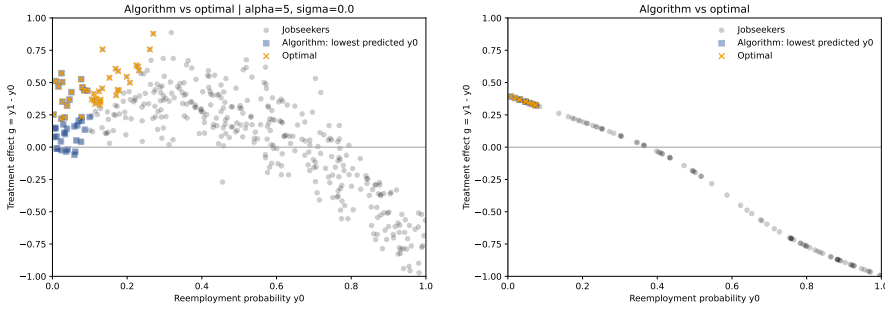
$$y_i^0 < y_j^0 \quad \Rightarrow \quad \gamma(y_i^0) \geq \gamma(y_j^0) \quad \forall i, j.$$

In other words, Proposition 1 implies homogeneous treatment effects conditional on $y_i^0$, which would be a very strong assumption. If treatment effects instead vary across jobseekers with the same baseline risk, then strict adherence to a risk-based ranking cannot generally implement the optimal allocation. Instead, a trade-off arises between assignment based on need and assignment based on treatment effects. And the extent to which the two is valued depend on the policymaker objectives.

Figure 3 shows an illustration of a possible relationship between treatment effect and reemployment probability, together with a selection of jobseekers with the lowest reemployment probability $y^0$ under perfect prediction, and the optimal selection under the value function in Equation 2 and $\alpha = 5$, i.e. a policymaker who values both a needs-based and efficient allocation. It illustrates how monotonically decreasing homogeneous treatment effects is required for the selections to completely overlap.

The underlying relationship is simulated, but it has a both logical and commonly observed feature: treatment effects, on average, decline with baseline reemployment probability. This is intuitive because, by definition, $\gamma = y^1 - y^0$,

so a jobseeker with a high baseline reemployment probability faces a mechanical upper bound on the treatment effect. (Figure G1 shows the distribution when $y^0$ and $y^1$ are drawn at random.) There is also empirical evidence of larger treatment effects among jobseekers with lower reemployment probabilities, at least across relatively coarse categorizations (Van den Berg et al., 2025; Knaus et al., 2022). I have chosen a slight inverted U-shape to reflect that ALMPs may be insufficient to help jobseekers with very weak labor market attachment into work.




(a) Heterogeneous treatment effects (b) Monotonically decreasing treat. effects

*Figure 3.* Simulation: Reemployment Probability and Treatment Effects

*Note:* Simulation of jobseeker reemployment probability and treatment effects. The blue squares represent a selection of jobseekers with the lowest reemployment probability $y^0$ under perfect prediction. The orange crosses represent the optimal selection of jobseekers under the value function in Equation 2 and $\alpha = 5$.

Interestingly, a monotonic and homogeneous relationship between treatment effects and baseline probability or risk is trivially satisfied in many standard algorithmic decision-making contexts involving well-defined prediction tasks—such as pretrial detention. There, the causal mechanism is straightforward: detention reliably prevents pretrial misconduct through incapacitation, meaning that its treatment effect increases with baseline risk. The higher the risk of misconduct, the greater the potential benefit of detention. As a result, assigning detention based on predicted risk is sufficient to minimize misconduct.

In contrast, in the context of active labor market programs (ALMPs), the assumption of monotonic treatment effects is far less likely to hold. If monotonicity does not hold, then a trade-off arises between achieving a needs-based and an efficient allocation. This trade-off creates scope for human decision-makers to improve upon the algorithm, not only by refining predictions of unemployment risk, but also by incorporating broader policy objectives that the algorithm does not account for. The setting thus offers a natural case where human-algorithm collaboration could lead to improved outcomes.

## 4.5 Model Implications

The algorithm assigns ALMPs based on unemployment risk and applies the specific cutoff $s_a$. Any deviation from the algorithm results from (1) a different ranking of jobseekers, (2) a different cutoff, or a combination of the two. Following the conceptual framework, a deviation from the algorithmic recommendation comes from a difference along one of the following three dimensions:

1. Prediction of baseline reemployment probability $\hat{y}^0_{i,j}$.
2. The chosen cutoff $s_j$.
3. Relative weight on unemployment risk versus treatment effect in assignment, represented by $\alpha_j$. And caseworker prediction of treatment effects $\hat{\gamma}_i$.

I will estimate jobseeker outcomes under the caseworker allocation versus the algorithmic counterfactual assignment in order to analyze whether jobseeker discretion improves or reduce policymaker payoff compared to the algorithmic counterfactual. The conceptual framework will help to identify key explanations for the results.

# 5 Empirical approach: Caseworkers versus Algorithm

In this section I will build on the conceptual framework to set up the empirical strategy to evaluate the impact of caseworker discretion in terms of alignment with a (1) needs-based and (2) efficient allocation.

## 5.1 Needs-based Allocation

First, I examine how caseworker discretion affects alignment with a needs-based allocation, where jobseekers at the highest risk of unemployment are prioritized for ALMP assignment. I will compare the realized caseworker allocation to the counterfactual algorithmic allocation. Under the assumption that caseworkers were solely targeting a needs-based allocation, this would reduce to a question about prediction accuracy, since we know that the algorithm is solely predicting unemployment risk. But despite that I will not formally estimate prediction accuracy, standard machine learning classification metrics are useful. Specifically, I will estimate algorithmic and caseworker *precision*, *recall*, and *accuracy*.

**Selective Labels Problem**

The main complicating factor is that all observed outcomes will be endogenous to caseworker decisions. Standard classification metrics require knowledge of which jobseekers would have become employed absent treatment, but this counterfactual is selectively unobserved. This missing data problem is commonly referred to as the selective labels problem (Kleinberg et al., 2018).

There are two main alternatives to deal with this endogeneity problem. The first is to use a *selection on observables* approach, where control variables are used to estimate caseworker outcomes in the absence of treatment. To be unbiased, this approach requires that caseworker assignment decisions are uncorrelated with true employment potential—that is, effectively random—conditional on observables. Such an assumption rules out caseworker usage of private information. Second, one could simply use jobseekers realized employment, despite them being endogenous. Such approach requires assumptions about the causal effects of ALMPs.

This paper instead make use of an estimation of population-level employment probability. Specifically, I combine estimates of average reemployment probabilities in the jobseeker population with the estimated conditional employment rate under Independent Search for the algorithmic counterfactual and the observed conditional employment rate under the caseworker allocation. Together with the assignment propensities under each allocation, these quantities are sufficient to construct unbiased performance measures.

The method builds on the extrapolation approach suggested by Angelova, Dobbie, and Yang (2025) and Arnold et al. (2022), which build on work by Hull (2020). Angelova et al. (2025) study human discretion over an algorithm in the context of pre-trial hearings in the US. The identification strategy relies on random assignment of jobseekers to caseworkers who vary in their propensity to assign to Independent Search.

In contrast to standard approaches, this requires no assumptions about caseworkers' skill and use of private information or ALMP effectiveness.

**Conditional Employment Rate**

The main performance metric used to evaluate the alignment with a needs-based allocation will be *conditional employment rate*, typically referred to as *precision*.

For each caseworker $j$, the conditional employment rate is defined as the employment rate among jobseekers $i$ assigned to Independent Search, $D_{i,j} = 0$.

$$M_j = E[Y_i | D_{i,j} = 0] \tag{7}$$

Next, for each caseworker I define the algorithmic counterfactual employment rate as the employment rate in the pool of jobseekers the algorithm would have assigned to independent search - holding the independent search rate constant. Note that, due to caseworker deviations, the pool of jobseekers assigned to Independent Search under the algorithmic counterfactual can be different from the actual pool of jobseekers assigned to Independent Search by the caseworker. Leading to a different conditional employment rate at the same cutoff.

The algorithm's conditional employment rate at the independent search rate of caseworker j is:

$$M_{a(j)} = E[Y_i | D_{a(j),i} = 0] \tag{8}$$

A higher conditional employment rate, at a given independent search rate, implies a closer alignment with an allocation based on unemployment risk. The intuition behind this metric is that at a given independent search rate, a higher conditional employment rate implies that the decision-maker were more successful in identifying jobseekers likely to find employment independently.

The impact of caseworker discretion on the alignment with a needs-based allocation will be measured by the difference in conditional employment rate:

$$\Delta M_{j,a(j)} = M_j - M_{a(j)} \tag{9}$$

Where $\Delta M_{j,a(j)} > 0$ implies that a caseworker is more aligned with a needs-based allocation than the algorithm. I will quantify the comparison of caseworker versus algorithmic predictive performance using two summary measures. First, I will estimate the share of caseworkers with higher conditional employment rates than the algorithm. Second, I will estimate the total difference in the case-weighted conditional employment rate between caseworkers and algorithm.

Note that a difference in alignment with a needs-based allocation results from (i) difference in skill in predicting reemployment probability, or (ii) that caseworkers are targeting a different objective. Importantly, this metric does not allow me to separate the two channels.

**Estimating the Caseworkers Parameters**

The caseworker-specific independent search rates and employment rates can be estimated using the following specification. I will control for the algorithmic regime (pre/post September 7) nonparametrically by estimating the caseworker parameters separately per regime, such that the indicator $j$ is regime-specific.

$$1 - D_{i,j} = \sum_i \rho_j Z_{i,j} + u_{i,j} \tag{10}$$

$$Y^0_{i,j} = \sum_i \eta_j Z_{i,j} + v_{i,j} \tag{11}$$

Where $(1 - D_{i,j})$ and $Y^0_{i,j}$ are indicators for assignment to Independent Search and employment for jobseeker $i$ conditional on assignment to Independent Search, respectively. $Z_{i,j}$ is an indicator taking the value 1 if jobseeker $i$ was assigned to caseworker $j$. $\rho_j$ provides the estimated independent search rate for for caseworker $j$, and $\eta_j$ provides the estimated conditional employment rate of caseworker $j$.

The caseworker conditional employment rate will be estimated with sampling error, due to a finite number of observations per caseworker. The sampling error will inflate the variance in the caseworker employment rate, which could bias the comparison to the algorithmic counterfactual. To correct for

the sampling error I use linear empirical Bayes shrinkage, as defined by Walters (2024), shrinking the parameters towards the mean. See section A8 for details. I use the shrinked conditional employment rates when calculating the share of caseworkers with higher/lower conditional employment rate than the algorithm as well as the difference in conditional employment rate between the caseworkers and algorithm.

When applying shrinkage to variables that are subsequently used in regression one has to be mindful. While shrinking an independent variable reduces attenuation bias in the slope parameter, shrinkage of a dependent variable will instead induce bias (Walters, 2024). Thus, the empirical Bayes shrinkage is not applied when studying the relationship between caseworker conditional employment rate and independent search rate. The regression is instead weighted by the inverse variance of the estimated conditional employment rate, to put more weight on more precise estimates.

**Estimating Average Employment Potential**

To estimate average employment potential in the population I will use the extrapolation method first suggested by Arnold et al. (2022).

To provide intuition for the extrapolation, consider an idealized setting where there is random assignment of jobseekers to caseworkers, and caseworkers differ in how many jobseekers they assign to Independent Search due to differences in skills or preferences. Let there be one caseworker, caseworker $j^*$, who assigns all jobseekers to Independent Search. Under a large enough case load, the employment rate among the jobseekers assigned to caseworker $j^*$ should be close to the average employment potential in the population of jobseekers.

In the absence of a caseworker $j^*$ the employment potential can be estimated from the relationship between observed conditional employment rate and the Independent Search rate - using extrapolation to independent search rate of 100 %.

**Estimating the Algorithmic Counterfactual Using Extrapolation**

The same extrapolation approach can be used estimate the conditional employment rate under the algorithmic counterfactual. Note that since I can observe all profiling scores, I can observe the complete algorithmic ranking of jobseekers. This ranking can be used to infer which jobseekers would have been assigned to independent search and which would have been assigned to an ALMP under strict adherence to the algorithm, at each potential threshold. And each threshold can be mapped to a specific independent search rate. I want to answer the following question: *If all jobseekers with scores above some threshold $s_a$ were assigned independent search - how many of them would become employed within 6 months?*

As in the full sample of jobseekers, the extrapolation logic is applicable to any score threshold. Consider instead that caseworker $j^*$ assigns all jobseekers

at or above a given threshold $s_a$ to Independent Search, such that the independent search rate at the threshold $s_a$ is equal to one. With random assignment of jobseekers to caseworkers, the employment rate among that subsample of jobseekers should be close to the average employment rate for the subset of jobseekers with profiling scores at or above the threshold.

*Implementation:*
The implementation of the empirical strategy can be summarized in the following steps:

1. **Restrict the sample to jobseekers with scores at or above the given threshold,** $a(x_i) \geq s$

2. **Estimate the caseworker level independent search rates** $R_j^{IS}$ **and conditional employment rates** $M_j$ **within the sub-sample.**

3. **Estimate the average employment potential in the sub-sample**
   The average employment potential within the sub-sample of jobseekers with $a(x_i) \geq s_a$ is equivalent to the employment rate of a caseworker with an independent search rate of 100 % within the sub-sample (due to random assignment of jobseekers to caseworkers). In the absence of such a caseworker, we need to estimate the employment rate using extrapolation to $R_j^{IS} = 1$.

Repeat the procedure for many thresholds such that the whole span of caseworker independent search rates is covered. For robustness, extrapolate using different functional forms. The extrapolation regression is weighted by the inverse variance of the estimated conditional employment rate, to put more weight on more precise estimates. Standard errors and confidence intervals for the algorithmic counterfactual conditional employment rate within each sub-sample are constructed using bootstrapping with random draws of caseworkers.

**Performance: Recall and Accuracy**
Both recall and accuracy can be computed using the estimated average reemployment probability $p$, the estimated conditional employment rates $M$, and independent search rate $q$.

The conditional employment rate—precision—will be the main performance measure used. As complements, I will also compute *recall*, also referred to as true positive rate. Recall is defined as the share of jobseekers who would become employed independently within 6 months who are assigned to Independent Search.

$$Recall^{IS} = E[(1 - D_i)|Y_i^0 = 1] = \frac{M \cdot q}{p} \tag{12}$$

Accuracy is the share of 'correctly classified' cases, i.e. where jobseekers with $Y_i^0 = 0$ is assigned to an ALMP and jobseekers with $Y_i^0 = 1$ is assigned to Independent Search.

$$Accuracy = 1 - p - q + 2 \cdot M \cdot q \tag{13}$$

See Table D1 for a more detailed description of measures used and their calculation.

## 5.2 Efficient Allocation

In this section I present the empirical strategy to study efficiency. An allocation is efficient if it assigns ALMP to jobseekers such that the total employment rate is maximized. Thus, the efficiency can be evaluated by comparing the total employment rate between caseworkers and the algorithmic counterfactual. However, as opposed to studying alignment with a needs-based allocation, we cannot directly estimate the total employment under the algorithmic counterfactual assignment. Instead I will compare the total employment outcomes of caseworkers with different levels of deviation rates, the share of cases for which the caseworker deviate from the algorithmic recommendation. Where a deviation is defined in terms of the choice between Independent Search and some ALMP. The empirical approach relies on a typical 'judge design' (Chyn et al., 2025), exploiting random assignment of jobseekers to caseworkers and variation in caseworker propensity to deviate, and is similar to Hoffman et al. (2018) who study the impact of managerial discretion over an algorithm in hiring.

The employment rate of each caseworker, $E_j = E[Y_{ij}]$, can be written as the weighted sum of the conditional employment rate for jobseekers assigned to Independent Search, $M_j$, and the conditional employment rate for jobseekers assigned to an ALMP.

$$E_j = R_j^{IS} \cdot E[Y_i | D_{i,j} = 0] + R_j^{ALMP} \cdot E[Y_i | D_{i,j} = 1] \tag{14}$$

The propensity to assign to Independent Search $R_j^{IS}$ will depend on the caseworkers' beliefs and predictions regarding treatment effects of ALMPs. Thus, we should let the propensity rate channel also impact the total employment rate. Therefore, I will define the algorithmic counterfactual rate as the zero deviation employment rate.

Define the algorithmic independent search rate for each caseworker as the share of jobseekers assigned to Independent Search under strict adherence to the algorithmic recommendation. $R_j^A$ varies across caseworkers due to sampling variation.

$$R_j^{IS:A} = 1 - E[A_i | cw_j = 1] \tag{15}$$

Then the algorithmic counterfactual employment rate will be a weighted sum of the employment rate conditional on Independent Search, and the employment rate under the ALMP. Note that the algorithmic recommendation does not recommend a specific treatment, but only recommends Independent Search or an ALMP.

$$E_{a(j)} = R_j^{IS;A} \cdot E[Y_i|A_i = 0] + R_j^{ALMP;A} \cdot E[Y_i|A_i = 1] \tag{16}$$

**Estimating the Algorithmic Counterfactual: Deviation rates**

Under random assignment of jobseekers to caseworkers with different deviation rates, a positive association between employment rate and deviation rate implies that caseworker deviation on average improve efficiency. While a negative association implies that caseworkers are reducing efficiency.

This approach is similar to the setup in Hoffman et al. (2018), with two important differences. First, in this paper the institutional setup ensures random assignment of jobseekers to caseworkers, which alleviates worries that jobseekers would systematically differ on unobservable characteristics. Second, caseworkers can impact jobseeker outcomes via the choice of ALMP.

Random assignment of jobseekers to caseworkers ensures that the first term of Equation 16 is comparable across caseworkers with low and high deviation rate. In terms of the second term, a caseworker with zero deviation rate is only a valid estimate of the algorithmic counterfactual if caseworkers are on average equally skilled in picking the employment maximizing ALMP, such that the expected outcomes are equal across caseworkers at the algorithmic ranking of jobseekers. If caseworkers with higher skill are systematically deviating more in terms of choosing between Independent Search and some ALMP, then we will overestimate the effect of caseworker deviations on total employment. Since highly skilled caseworkers would have obtained higher employment rate also under zero deviations. It means that it is not possible to empirically separate the caseworkers ability to identify jobseekers with high treatment effect potential, and their ability to identify the employment maximizing ALMP. I will return to this discussion in the results section.

I will evaluate the impact of caseworker deviation on efficiency holding the algorithmic independent search rate constant. Which allows the caseworkers to impact efficiency through two channels, (i) by adjusting the independent search rate, and (ii) by changing the composition of jobseekers who are assigned to Independent Search or ALMP. I run a regression of jobseeker employment outcomes on the leave-one-out caseworker deviation rate and leave-one-out algorithmic independent search rate at the jobseeker level.

$$y_i = \beta_0 + \beta_1 \text{Deviation rate}_{-i,j} + \beta_2 \text{Alg indep search rate}_{-i,j} + \varepsilon_i \tag{17}$$

$\beta_1$ will recover the causal relationship between caseworker deviation rate and total employment. Unlike with the extrapolation approach, I will not get

an estimate of the distance from the algorithmic counterfactual for each jobseeker.

To understand whether the effect is symmetric, I will complement the reduced form approach with an instrumental variable estimation, where the sample is split by algorithmic recommendation. Asking the questions: For a jobseeker recommended an ALMP - what is the effect of a caseworker deviation to Independent Search?

$$\begin{aligned} ALMP_i &= \alpha_0 + \alpha_1 Z_{j,-i} + \alpha_2 \text{Alg indep search rate}_{j,-i} + \varepsilon_i \quad \forall \quad A_i = 1 \\ y_i &= \eta_0 + \eta_1 Z_{j,-i} + \eta_2 \text{Alg indep search rate}_{j,-i} + \varepsilon_i \quad \forall \quad A_i = 1 \end{aligned} \tag{18}$$

And for a jobseeker recommended Independent Search - what is the effect of a caseworker deviation to an ALMP?

$$\begin{aligned} IS_i &= \alpha_0 + \alpha_1 Z_{j,-i} + \alpha_2 \text{Alg indep search rate}_{j,-i} + \varepsilon_i \quad \forall \quad A_i = 0. \\ y_i &= \eta_0 + \eta_1 Z_{j,-i} + \eta_2 \text{Alg indep search rate}_{j,-i} + \varepsilon_i \quad \forall \quad A_i = 0 \end{aligned} \tag{19}$$

Where the instrument in Equation 18 and Equation 19, $Z_{j,-i}$, is the caseworker leave out deviation rate.[10]

# 6 Identifying Assumptions

The identifying assumptions needed for the two empirical strategies are largely overlapping. Both rely on (i) random assignment of jobseekers to caseworkers, (ii) relevance — caseworker independent search rate (deviation rate) increases the probability of assignment to Independent Search (deviation). (iii) Exclusion restriction — caseworkers only affect outcomes via the decision, assignment or deviation, and not directly. (iv) In the absence of caseworkers with 100 % independent search rates, the extrapolation approach requires that average employment parameters can be accurately extrapolated from the data. (v) The IV approach to examine the impact of caseworker deviations on total employment requires average monotonicity.

## 6.1 Random Assignment of Jobseekers to Caseworkers

The key identifying assumption for the paper is random assignment of jobseekers to caseworkers. The institutional design of assignment of jobseekers at the Swedish PES central offices suggests that the assumption of random assignment holds. Table 1 shows the result of balance tests of both single

[10] Due to limited sample size once the sample is split between jobseekers recommended Independent Search and ALMP, it is not feasible to recalculate deviation rates within subgroups.

and joint significance of jobseekers characteristics. The table show results of single regressions of (1) caseworker leave-out independent search rate , (2) leave-out deviation rate, and (3) profiling score on predetermined jobseeker characteristics. Each row represents single regressions, and coefficients and standard errors are multiplied by 100 to be interpreted as percentage points. I also report the standardized difference: the difference in the mean of the dependent variable, measured in pooled standard deviation units. This provides a unit-free measure of imbalance; for example, a value of 0.1 indicates that the group means differ by one tenth of a standard deviation. As a common rule of thumb, absolute values below 0.1 are typically viewed as indicating negligible imbalance. For 'score' I have instead computed the standardized slope, multiplying the coefficient by the ratio of standard deviation.

The results show that while jobseeker characteristics are strongly correlated with the profiling score, they are largely uncorrelated with the leave-one-out caseworker independent search rate and deviation rate. Even though some single regressions are statistically significant, the magnitudes are economically negligible. Regarding joint significance of jobseeker characteristics, both f-tests on leave-one-out independent search and deviation rates are small and not statistically significant. The balance tests confirm that the institutional setting approximates random assignment of jobseekers to caseworkers.

## 6.2 Relevance

There is a positive and statistically significant relationship between caseworkers' leave-out independent search rate and the probability that a jobseeker is assigned to Independent Search, with a coefficient of 0.374 ($p < 0.01$). Likewise, the leave-out deviation rate is a strong predictor of individual deviation, with a coefficient of 0.326 ($p < 0.01$). The overall deviation rate also predicts deviations within the split samples: 0.274 ($p < 0.01$) among jobseekers recommended IS, and 0.824 ($p < 0.01$) among those recommended ALMP. See Table C2.

It is not obvious ex ante that the overall deviation rate should predict deviations conditional on algorithmic recommendation—a caseworker who frequently overrides IS recommendations need not also override ALMP recommendations. Limited sample sizes make it unfeasible to construct leave-out rates that are recommendation-specific, but the results confirm that the overall rate provides a relevant instrument in both subsamples.

## 6.3 Exclusion Restriction

The institutional setting makes the exclusion restriction a credible assumption. Caseworkers main task is to make labor market assessments and aid jobseekers in administrative matters. Assistance and other active labor market

**Table 1.** *Test random assignment of jobseekers to caseworkers*

| | Assignment Rate | | Deviation Rate | | Profiling Score | |
|---|---|---|---|---|---|---|
| | Coef | Std diff | Coef | Std diff | Coef | Std diff |
| Male | -0.013 | -0.01 | -0.028 | -0.01 | 2.610*** | 0.16 |
| | (0.021) | | (0.021) | | (0.089) | |
| Historical disability | -0.003 | 0.01 | -0.016 | -0.01 | -22.001*** | -1.40 |
| | (0.050) | | (0.049) | | (0.211) | |
| Immigrant | -0.024 | -0.01 | -0.010 | -0.00 | -9.129*** | -0.56 |
| | (0.025) | | (0.024) | | (0.107) | |
| High school | 0.018 | 0.00 | 0.056 | 0.02 | 0.180 | 0.01 |
| | (0.071) | | (0.068) | | (0.317) | |
| Tertiary education | 0.047* | 0.01 | -0.032 | -0.01 | 2.557*** | 0.16 |
| | (0.028) | | (0.027) | | (0.111) | |
| Age 25-34 | 0.037* | 0.01 | -0.034 | -0.01 | 5.246*** | 0.32 |
| | (0.022) | | (0.021) | | (0.094) | |
| Age 35-44 | -0.026 | -0.01 | 0.009 | 0.00 | -1.332*** | -0.08 |
| | (0.024) | | (0.024) | | (0.103) | |
| Age 45-54 | -0.002 | -0.00 | 0.001 | 0.00 | -3.444*** | -0.22 |
| | (0.026) | | (0.026) | | (0.105) | |
| Service and admin | -0.032 | -0.01 | 0.004 | 0.00 | -2.415*** | -0.16 |
| | (0.032) | | (0.031) | | (0.124) | |
| Service, care and shop | -0.032 | -0.01 | 0.028 | 0.01 | -5.757*** | -0.37 |
| | (0.024) | | (0.023) | | (0.094) | |
| Building and manufacturing | 0.073* | 0.01 | -0.072* | -0.02 | 6.920*** | 0.41 |
| | (0.039) | | (0.038) | | (0.180) | |
| Previously registered | 0.037 | 0.05 | -0.035 | -0.02 | -2.014*** | -0.12 |
| | (0.047) | | (0.047) | | (0.210) | |
| Previous treatment | -0.033 | -0.01 | 0.001 | 0.00 | -7.195*** | -0.45 |
| | (0.026) | | (0.025) | | (0.109) | |
| Score | 0.057 | 0.001 | -0.078 | 0.006 | | - |
| | (0.066) | | (0.064) | | - | |
| R-square | 0.0003 | | 0.0001 | | 0.2295 | |
| F-test | 1.20 | | 0.92 | | 2,737 | |
| P-value f-test | 0.269 | | 0.531 | | 0.000 | |
| N | 130,705 | | 130,705 | | 130,705 | |
| N caseworkers | 892 | | 892 | | 892 | |
| Mean depvar | 0.87 | | 0.13 | | 0.50 | |

*Note:* ***, **, * indicate significance at 1%, 5% and 10% respectively. Robust standard errors reported in parentheses. Each row represents single regressions of dependent variables on predetermined jobseekers characteristics. Dependent variables per column: (1) leave-one-out independent search rate, (2) leave-one-out caseworker deviation rate, and (3) jobseeker profiling score. Coefficients and standard errors are multiplied by 100 to be interpreted as percentage points. The table also reports the standardized difference: the difference in the mean of the dependent variable, measured in pooled standard deviation units. As a common rule of thumb, absolute values below 0.1 are typically viewed as indicating negligible imbalance. For 'score' I have instead computed the standardized slope, multiplying the coefficient by the ration of standard deviation. The F-test tests for joint significance of jobseeker characteristics. Regressions control for regime fixed effects and include caseworkers with at least 50 profilings per regime.

programs are carried out by other state or municipal agencies or private companies. Monitoring of job search activities are handled by a separate monitoring unit at the PES. Caseworkers do not provide any practical help with things such as CV, cover letter or job referrals. Altogether, it is therefore unlikely that caseworkers have any large direct effects on the outcomes of jobseekers beyond their assignment decision.

Interviews with caseworkers suggest they can adjust support to jobseekers assigned to Independent Search through the number of contacts and follow-up meetings. I test whether contact frequency correlates with assignment behavior, focusing on jobseekers assigned to Independent Search (Table C1). On average, these jobseekers have virtually no in-person meetings (0.01-0.02) and limited phone contact—0.33 calls before profiling and 0.83 after, of which 0.43 are with the same caseworker. This confirms minimal interaction, though repeat contacts often involve the original caseworker. Column 5 shows that leave-out independent search and deviation rates are negatively correlated with having additional phone contacts. Yet the effects are negligible: a one standard deviation increase in independent search rate (deviation rate) is associated with 0.005 (0.009) fewer calls.

In sum, jobseekers assigned to Independent Search have minimal contact with PES caseworkers beyond the initial assessment. Follow-up meetings, when they occur, are rare and conducted by phone. Given that central office caseworkers serve the entire country and handle a broad range of clients, such calls are likely to involve administrative assistance and general advice rather than personalized support. Combined with the small coefficients reported in Column 5 of Table C1, these findings suggest that assignment decisions are not accompanied by substantial differences in post-assignment assistance — supporting the plausibility of the exclusion restriction.

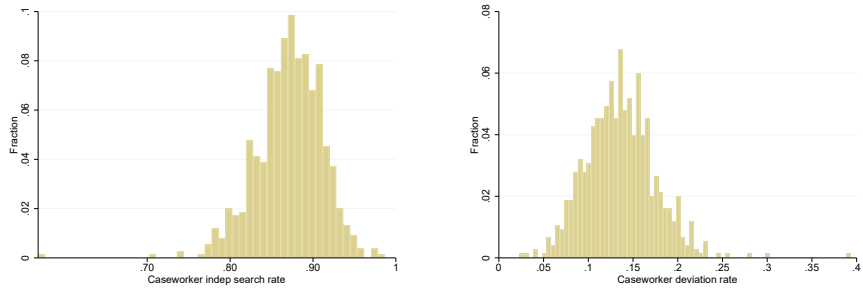
## 6.4 Average Employment Parameters Can Be Accurately Extrapolated

In the absence of caseworkers with 100 % independent search rates, the average employment potential within each subsample is estimated using extrapolation. Whether the extrapolation provides a consistent estimate of the average employment potential will depend on (i) approximating the correct functional form, and (ii) the distance between caseworker observations and the point we are extrapolating to being sufficiently small and that there are sufficiently many caseworkers with high independent search rates.

Figure 4 panel a shows the distribution of independent search rates at the caseworker level. There is significant variation in independent search rates, from 58 % to 100 % in the full sample, which is similar to Angelova, Dobbie, and Yang (2025). Figure C1 shows the distribution of independent search rates at caseworker level for four subsamples, based on different profiling score

cutoffs. The figure confirms that there are caseworkers with sufficiently high independent search rate in a range of sub samples, which indicate that the extrapolation is likely to work well across sub samples.

Regarding approximating the correct functional form, I will use linear extrapolation as the main specification and do robustness checks using quadratic and local linear extrapolation. Results are robust to these alternative specifications.




(a) Independent Search Rate (b) Deviation Rate

*Figure 4.* Independent Search and Deviation Rates

*Note:* Independent search rate and deviation rate at caseworker level. Independent search rate is the caseworkers propensity to assign jobseekers to independent search, ass opposed to some ALMP. Mean=87.4%. Deviation rate is defined as the share of cases where the caseworker deviates from the algorithm. Mean=13.5%. See Table C4 for details on deviations broken down per recommendation category. Includes caseworkers with at least 50 cases per regime

## 6.5 Deviation Rates

In order to estimate the impact of caseworker deviations on employment outcomes, there need to be sufficient variation in their deviation rate. Figure 4 panel b shows the distribution of deviation rates across caseworkers. It ranges from 2.9 % to 37.8 % mean deviation rate is 13.5 %.

## 6.6 Monotonicity

For the IV analysis of the impact of caseworker deviation rate, the Local Average Treatment Effect (LATE) is only identified under *average monotonicity*: for a given jobseeker, the average deviation rate among caseworkers who would deviate must exceed the average rate among those who would follow the recommendation (Chyn et al., 2025). Unlike strict monotonicity, this does not require identical ranking of jobseekers across all caseworkers.

A testable implication of average monotonicity is that the first stage is positive across subgroups. Table C3 reports regressions of the deviation indicator on the leave-out deviation rate for multiple predetermined jobseeker characteristics. In the full sample and among jobseekers recommended Independent

Search, the result confirm that average monotonicity is plausible. Among jobseekers recommended an ALMP, however, 5 of 16 coefficients are insignificant and two are negative — more than would be expected by chance. Small subgroup samples are a likely explanation: the high school education group, for instance, contains only 300 observations. These results nevertheless warrant some caution in interpreting the LATE estimates for this subsample.

# 7 Results

## 7.1 Needs-based Allocation: Baseline Reemployment Probability

First I will examine whether caseworker discretion improve or reduce alignment wiht a needs-based allocation, following the empirical approach presented in section 5. Figure 5 illustrates the main results and compares caseworker and algorithmic performance by plotting the conditional employment rate against the independent search rate. A higher conditional employment rate implies a closer alignment with a needs-based allocation, assigning jobseekers based on unemployment risk. Thus, the fact that the caseworker line is below the algorithmic counterfactuals indicate that caseworkers reduce performance in this dimension: they are assigning jobseekers with lower risk of unemployment relative to the algorithmic counterfactual.

Employment is defined as exiting the unemployment spell for 'full time' employment within 6 months. Where 'full time' employment is defined as earning at least 75 % of the minimum wage in retail, about 17,800 SEK(1,877 USD) per month, for at least one month. The results are robust to alternative definitions (Figure F2).

The solid orange line show average caseworker conditional employment rate for any given independent search rate. The blue short dashed line is the algorithmic counterfactual estimate using the score adjusted for time varying cutoffs. It indicates what the conditional employment rate would be at each independent search rate if the caseworker would have followed the algorithmic recommendation.[11]. The green long dashed line show the algorithmic counterfactual using the original algorithm. Both algorithmic counterfactuals are estimated using extrapolation from linear regression, and the points are combined using local linear interpolation.

The dotted horizontal line shows the conditional employment rate under random assignment — which is equal to the overall population employment potential. The logic is that if caseworkers would randomly assign jobseekers to Independent Search, they would in expectation obtain the average conditional employment rate in the population. The dotted line showing perfect targeting

[11] It shows the algorithmic counterfacutal under the PES ranking. i.e. the impact of caseworkers reranking jobseekers, but holding the caseworker assignment propensities constant

illustrate the potential conditional employment rate if only jobseekers with $Y_i = 0$ were assigned to an ALMP. By construction, all lines coincide at an assignment rate of 1, since assigning everyone to Independent Search yields the same expected employment rate.

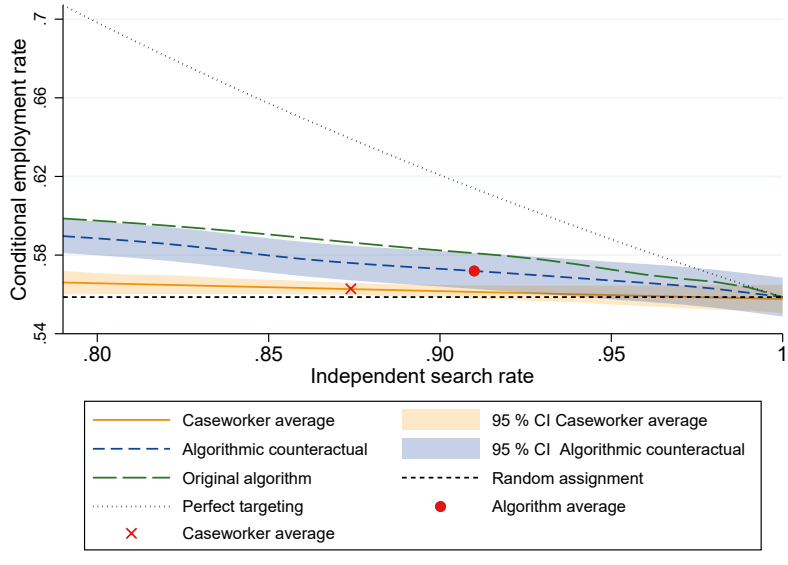



*Figure 5.* Probability of exiting to full time employment within six months

*Note:* The figure shows the conditional employment rate (probability of exiting unemployment for employment within six months among jobseekers assigned to Independent Search) for each level of independent search rate (share assigned to Independent Search). Employment outcomes are based on monthly payroll tax data, as defined in section 3.4. The fitted caseworker average is estimated using linear regression and confidence intervals are constructed using robust standard errors. Algorithmic counterfactuals estimated using linear extrapolation as described in section 5, following Angelova et al. (2025). For the algorithmic counterfactual confidence intervals are constructed using bootstrapping with random draws of caseworkers. Estimates control nonparametrically for regime fixed effects. Includes caseworkers with at least 50 profilings per regime. Table E1 contains details on average performance. Caseworkers obtain on average 1.3 percentage point ($p < 0.001$) lower conditional employment rate relative to the algorithmic counterfactual holding independent search rate constant and 0.5 percentage points ($p < 0.001$) higher conditional employment rate than a random assignment.

The red cross and point indicate the caseworker and algorithm average. Strict adherence to the algorithmic recommendations would have resulted in 91 % assigned to Independent Search and a conditional employment rate of 57.2 %. Caseworker deviations lead to 87.4 % of jobseekers being assigned to Independent Search and conditional employment rate of 56.3 %.

If a decision maker is good at ranking jobseekers by their reemployment probability, we would expect to see a negative relationship between the condi-

tional employment rate and the independent search rate. The intuition is that a decision maker with a low independent search rate, who is more selective, should assign the jobseekers with the highest reemployment probability to Independent Job Search first. So that their pool of jobseekers, for whom the employment rate is calculated, are more likely to become employed on their own. Notably, there is a negative relationship between conditional employment rate and independent search rate for the algorithmic counterfactuals.

Table E1 shows the comparison of algorithm and caseworkers in terms of alignment with a needs-based allocation to ALMPs, using (i) the share of caseworkers achieving a higher conditional employment rate than the algorithm, and (ii) the difference in caseworker conditional employment rate after applying linear empirical Bayes shrinkage. It shows the comparison to both the original algorithm and the resulting recommendation after applying the PES's time-varying cutoffs. Standard errors are constructed using bootstrapping with random draws of caseworkers.

Only 20 % of caseworkers perform better than the PES algorithm and 70.9 % of caseworkers perform better than a random assignment. Deviations lead to a 1.3 percentage point (2.2 %) lower conditional employment rate relative to the algorithm and 0.5 percentage point higher conditional employment rate than a random assignment, conditional on independent search rate.

Compared to a random assignment, using the algorithm increases the conditional employment rate by 1.8 percentage points. However, when comparing the original algorithm (based on the raw profiling score) to the implemented version used by the PES—which incorporates time-varying cutoffs—I find that the implementation distorts the ranking of jobseekers and significantly worsens predictive performance. Specifically, the PES implementation shifts access to ALMPs toward jobseekers with longer current unemployment duration, while reducing access for those at high risk of long-term unemployment but with shorter duration.

**Precision, Recall, and Accuracy**

In the language of standard machine learning classification metrics, the conditional employment rate reported here corresponds to precision. To provide a more complete picture of classification performance, I also report recall, or the true positive rate, defined as the share of jobseekers who would become employed on their own that are in fact assigned to Independent Search: $P(D_{ij} = IndepSearch | Y_i^0 = 1)$. In addition, I report precision and recall on the ALMP side, as well as overall accuracy, defined as the share of correctly classified jobseekers when the objective is a needs-based allocation rule. These measures are evaluated at the realized Independent Search shares of the algorithm and the caseworkers. I also report how much each allocation rule improves over a random assignment, relative to the gain that would be achieved under perfect targeting, and the estimated reemployment probability $p$ and Independent Search rate $q$.

**Table 2.** *Caseworker and Algorithm Performance*

| | Precision | | Recall | | | |
|---|---|---|---|---|---|---|
| | IS | ALMP | IS | ALMP | Accuracy | Gain |
| Algorithm | 0.57 | 0.57 | 0.93 | 0.12 | 0.57 | 0.23 |
| Caseworkers | 0.56 | 0.47 | 0.88 | 0.13 | 0.55 | 0.05 |

*Note:* $Precision_{IS} = P(Y^0 = 1|IS)$, $Precision_{ALMP} = P(Y^0 = 0|ALMP)$, $Recall_{IS} = P(IS|Y^0 = 1)$, $Recall_{ALMP} = P(ALMP|Y^0 = 0)$. Gain is calculated as the gain over a random assignment as a share of performance under perfect targeting. See Appendix Table D1 for details. Includes caseworkers with at least 50 profilings per regime. Controlling for regime fixed effects.

Table 2 reports precision and recall for both the Independent Search and ALMP margins. A first striking result is the large difference in recall between the two sides. This primarily reflects the skewed assignment pattern: only a relatively small share of jobseekers are assigned to an ALMP, even though a substantially larger share remain unemployed after 6 months. I treat this imbalance as given by the budget constraint. Under such a constraint, one would expect both the algorithm and the caseworkers to achieve high precision on the ALMP side: if only a small number of jobseekers can be assigned to an ALMP, those places should ideally go to individuals who are unlikely to find employment on their own. However, the precision on the ALMP side is only 47% for caseworkers and 57% for the algorithmic counterfactual. This implies that a considerable share of jobseekers assigned to ALMPs are individuals who would have found work even without program participation. For these individuals, the treatment effect is at best zero and could potentially be negative over a 6-month horizon. This has important implications for the average realized treatment effects of the programs.

As already shown in Figure 5, the algorithm performs substantially better in terms of overall accuracy. Relative to random assignment, it captures 23% of the feasible gain from perfect targeting. The corresponding number for caseworkers is only 5%. Thus, while neither performs especially well in absolute terms, the algorithm comes considerably closer to a needs-based allocation rule than the observed caseworker assignments.

To assess the algorithm's overall ranking ability, I also use the estimated reemployment potential in the population together with the conditional employment rate to derive the AUC. The algorithm implemented by the PES achieves an AUC of 0.59, which would generally be classified as poor performance.[12] See Appendix Table D1 for details on how these measures are calculated using the estimated population reemployment potential and conditional employment rates.

[12] An AUC of 0.5 corresponds to random assignment. A common rule of thumb is that an AUC between 0.5 and 0.7 is considered poor, 0.7 to 0.8 acceptable, 0.8 to 0.9 excellent, and 0.9 to 1.0 outstanding.

## 7.2 Prediction Accuracy and Deviation Reasons

The estimated reduction in alignment with a needs-based allocation may stem from two sources: poor predictive performance, or caseworkers deliberately targeting treatment effects rather than unemployment risk. To distinguish between these, I exploit a unique feature of the data: caseworkers who deviate from assigning Job Search Assistance (JSA) must report a reason for doing so. Focusing on deviations from JSA to Independent Search, I ask whether realized employment outcomes vary across deviation categories in a way that is consistent with skilled prediction.

Table 3 reports, for each deviation category, the difference between the realized conditional employment rate and the average population reemployment probability. If deviation reasons carried no signal—that is, if caseworkers selected them effectively at random—these differences should be zero.

**Table 3.** *Conditional Employment Rate Deviation to Independent Search*

| | Job | High remp prob | JS declines | Company board | Transition organization | All |
|---|---|---|---|---|---|---|
| CER | 0.235*** | 0.068*** | −0.320*** | −0.175*** | −0.165*** | −0.011 |
| | (0.012) | (0.022) | (0.030) | (0.023) | (0.028) | (0.008) |
| N | 1,164 | 502 | 195 | 448 | 295 | 6,607 |

*Note:* The difference in conditional employment rate of jobseekers recommended job search assistance but where the caseworker deviated to independent search - and the random rate. By reason for deviation. 'Job' indicates that the jobseeker (i) is already working to some degree, or (ii) will start a job within the next few months. 'High reemp' indicates high reemployment probability assessed by the caseworker. 'JS declines' means that the jobseeker has declined the offer. 'Transition org' means that the jobseeker is already receiving help from a transition organization financed by the employer and union. 'company' means that the jobseeker is not eligible due to being in a board of a company. Bootstrapped standard errors in parentheses. *, **, and *** indicate that the conditional employment rate is significantly different from the random rate at a 1 %, 5 %, and 10% level respectively. Controlling for regime fixed effects. Includes caseworkers with at least 50 profilings per regime.

The most common reason is the job category, indicating that the jobseeker reports already working or having a job offer. Here, the conditional employment rate exceeds the random benchmark by 23.5 percentage points (42 %)—a large gap that reflects direct information from the jobseeker rather than caseworker judgment. The more revealing category is high reemployment probability, where the deviation is based on the caseworker's own assessment of the jobseeker's skills and motivation. In this group, the conditional employment rate is 6.8 percentage points (12%) above the benchmark of 55.8%. The improvement is meaningful but substantially smaller, consistent with the greater difficulty of predicting reemployment from soft assessments relative to observing concrete employment status.

At the other end, when caseworkers deviate because the jobseeker declines to participate, the employment rate is very low. These cases reflect jobseeker preferences rather than caseworker assessment: the caseworker judged the jobseeker eligible and suitable for JSA, but the jobseeker refused, risking sanctions on their UI benefits.

The higher conditional employment rate relative to random assignment in the high reemployment probability group suggests that caseworkers possess meaningful prediction skill, which in turn implies that prediction errors alone are unlikely to account for the full reduction in needs-based alignment. There are two caveats to this conclusion, however. First, the random benchmark may not perfectly represent average employment potential within the JSA subsample. Jobseekers recommended JSA have been unemployed for at least 180 or 250 days, and reemployment hazards generally decline with duration, suggesting lower potential than in the full sample. On the other hand, JSA-recommended jobseekers typically have higher profiling scores than those recommended Intense Support. Making the net bias ambiguous.

Second, prediction skill may differ across contexts. Only 47 caseworkers in the sample use the high reemployment probability justification—31 of whom reduce and 16 increase needs-based alignment relative to the algorithm—raising concerns about selection. Moreover, follow-up meetings for JSA deviations are typically longer than initial assessments, and the requirement to justify a deviation may itself induce more careful judgment, making this setting potentially unrepresentative of average caseworker predictive performance.

In sum, the deviation-reason analysis provides suggestive evidence that caseworkers can identify jobseekers with high reemployment probability, but the measure is imperfect. The results make it unlikely that the reduction in needs-based alignment is driven entirely by prediction errors, pointing instead toward caseworkers actively pursuing other objectives. I examine whether that objective is treatment effects in the next section.

## 7.3 Efficient Allocation: Total Employment Outcomes

In this section I analyze the impact of caseworker deviations on efficiency, by examining the relationship between caseworker leave-one-out deviation rate and jobseeker employment outcomes.

Table 4 shows the regression results. The outcome tested is an indicator for exiting to full time employment within 6 and 12 months. Results are robust to controlling for leave-out algorithmic independent search rate. The coefficient in column 2 (-0.071) can be interpreted as when moving from 0 to the mean deviation rate, full time employment is reduced by 1.4 pp or 2.6 % of the mean employment rate. Showing that the average impact of caseworker deviation on employment is small, but negative. Column 4 shows that the negative association between deviation rate and employment shrinks and be-

comes insignificant when I study probability of exiting unemployment within 12 months post profiling.

**Table 4.** *Deviation Rate: Reduced form estimates*

| | Full time employment | | | |
|---|---|---|---|---|
| | 6 months | 6 months | 12 months | 12 months |
| CW deviation rate | -0.114*** | -0.071* | -0.046 | -0.026 |
| | (0.037) | (0.039) | (0.048) | (0.050) |
| Average employment rate | 0.531 | 0.531 | 0.664 | 0.664 |
| Alg. independent search rate | | X | | X |
| Number of jobseekers | 130,705 | 130,705 | 57,894 | 57,894 |
| Number of caseworkers | 892 | 892 | 736 | 736 |

*Note:* ***, **, * indicate significance at 1%, 5% and 10% respectively. Robust standard errors in parentheses. Includes caseworkers with at least 50 profilings per regime. Controlling for regime fixed effects. Mean deviation rate is 0.135 and mean algorithmic independent search rate is 0.906 in the full sample and both measures are leave-one-out means.

To explore mechanisms and robustness I do two more empirical exercises. The first regards *caseworker objectives* and examine whether caseworkers might instead be targeting e.g. job quality rather than fast exit. Table E4 shows that caseworker deviations also negatively affect cumulative earnings. The effect sizes indicate that the average caseworker reduces cumulative earnings by around 2 % of the mean, relative to strict adherence to the algorithm 6 months post profiling. As a proxy for wages I also examine the impact of caseworker deviation rate on earnings in the first month following exit to employment, but find no detectable effects (Table E5).

The second examines whether the effect of deviations to ALMPs and deviations to Independent Search are symmetric. To explore this I estimate a 2SLS, using specifications in Equation 18 and Equation 19. Table 5 reports the results. Coefficients in column 1 and 2 can be interpreted as the local average treatment effect (LATE) of caseworker deviation from Independent Search to an ALMP. And column 3 and 4 show the LATE of caseworker deviation from an ALMP to independent search. The estimates reveal an important asymmetry: while the effect of being assigned to an ALMP when the algorithm recommended IS is negative at -36.2 %, the effect of being assigned to IS while the algorithm recommended an ALMP is positive at 32.6 %.

One interpretation is that caseworkers are more skilled at identifying correct deviations from ALMP to IS than from IS to ALMP. An alternative is that realized treatment effects are on average negative among marginal jobseekers, so that reassignment to IS always improves outcomes. However, while jobseekers are randomly assigned to caseworkers, the set of jobseekers for whom caseworkers choose to deviate is not random. These estimates therefore do not identify average treatment effects of ALMPs—they show only that caseworkers on the margin select jobseekers who, in the short run, do not benefit from

participation. A further consideration is that caseworkers also choose which ALMP to assign, so poor program matching can itself generate negative realized treatment effects.

**Table 5.** *Deviation Rate: IV estimates*

| | Full time employment | | | |
|---|---|---|---|---|
| | IS to ALMP | | ALMP to IS | |
| | 6 months | 6 months | 6 months | 6 months |
| Deviation | -0.477*** | -0.362** | 0.557** | 0.326** |
| | (0.142) | (0.145) | (0.220) | (0.152) |
| Alg. independent search rate | | X | | X |
| Number of jobseekers | 118,618 | 118,618 | 12,087 | 12,087 |

*Note:* ***, **, * indicate significance at 1%, 5% and 10% respectively. Robust standard errors in parentheses. Includes caseworkers with at least 50 profilings per regime. Controlling for regime fixed effects. Deviation is instrumented by caseworker leave-out deviation rate. Column 1 and 2 include jobseekers recommended Independent Search and column 3 and 4 include jobseekers recommended ALMP.

An important question is to what extent the negative effects on reemployment within six months are driven by increased lock-in from caseworkers raising ALMP assignment rates in absolute terms. Several considerations suggest that higher ALMP rates need not mechanically reduce employment. The number of jobseekers who remain unemployed after six months far exceeds the number assigned to ALMPs, so caseworkers could in principle increase ALMP assignment while drawing exclusively from jobseekers who would not have found employment—leaving overall employment unaffected at worse. The negative employment effects arise not from increased ALMP assignment per se, but from caseworkers assigning jobseekers with relatively high reemployment probability, for whom participation generates lock-in without offsetting benefits. I do not decompose this further into the share attributable to assigning jobseekers with zero or negative treatment effects versus selecting, among those with positive effects, jobseekers with smaller gains.

## 7.4 High and Low Performing Caseworkers

The average effects reported so far mask important heterogeneity in caseworker performance. Table 6 and Table 7 compares the characteristics and outcomes of "high-" and "low-performing" caseworkers, defined according to whether they achieve a higher or lower conditional employment rate than their algorithmic counterfactual. To account for sampling variation and avoid overstating differences, Empirical Bayes shrinkage is applied to all variables that are based on average jobseeker outcomes. Table E3 confirms that there are no meaningful differences in observable jobseeker characteristics between the pools assigned to high- and low-performing caseworkers.

First, Panel A shows that high- and low-performing caseworkers are similar along many dimensions. Both groups have substantial experience, similar monthly wages, similar caseloads, and assign a similar share of jobseekers to Independent Search. They also have similar average deviation rate, which, under the assumption that 'skill' is correlated across dimensions, would speak against any worry that deviation rate is systematically correlated with skill in targeting ALMPs. However, high-performing caseworkers have a slightly different deviation pattern, in particular, a much lower rate of deviation to ALMP when the algorithm recommends Independent Search.

Panel B shows that both high- and low-performing caseworkers achieve lower employment rates than had all jobseekers simply been assigned to Independent Search. In other words, the realized allocation implies negative treatment effects on average. This pattern holds both within 6 and 12 months.

Panel C reports additional measures of caseworker performance. Most notably, high-performing caseworkers achieve substantially higher precision in ALMP assignment: among jobseekers assigned to ALMP, a larger share would not have become employed on their own (55.6% compared with 46.2%). High-performing caseworkers also perform significantly better than a random assignment. However, capturing only 19% of the feasible gain would, by most standards, be considered poor performance.

Finally, Panel D reports the observed reemployment rate among jobseekers recommended for Job Search Assistance but reassigned by the caseworker to Independent Search on the basis of a high perceived reemployment probability. High-performing caseworkers achieve a substantially higher rate on this measure as well, suggesting that they are more skilled in predicting reemployment probability.

Altogether, there are no large detectable differences in performance between the "high-" and "low-performing" caseworkers. Rather, the main takeaway is that both groups perform poorly in absolute terms.

**Table 6.** *Caseworker Performance Heterogeneity I*

| | Below algorithm | Above algorithm | P-value diff |
|---|---|---|---|
| **Panel a: Caseworker characteristics** | | | |
| Experience years | 7.387 | 8.294 | [0.000] |
| | (0.027) | (0.055) | |
| Monthly wage (SEK) | 30 207 | 30 016 | [0.000] |
| | (8.572) | (16.272) | |
| Number of cases | 220.995 | 221.379 | [0.026] |
| | (0.400) | (0.823) | |
| Independent Search rate | 0.876 | 0.881 | [0.000] |
| | (0.000) | (0.000) | |
| Deviation rate | 0.133 | 0.127 | [0.000] |
| | (0.000) | (0.000) | |
| Deviation rate indep search | 0.645 | 0.576 | [0.000] |
| | (0.000) | (0.001) | |
| Deviation rate treatment | 0.094 | 0.068 | [0.000] |
| | (0.000) | (0.000) | |
| **Panel b: Jobseeker Outcomes** | | | |
| Total Employment 6m | -0.051 | -0.037 | [0.000] |
| (relative random) | (0.000) | (0.000) | |
| Treatment effect 6m | -0.233 | -0.145 | [0.000] |
| | (0.000) | (0.001) | |
| Total Employment 12m | -0.039 | -0.024 | [0.000] |
| (relative random) | (0.000) | (0.000) | |
| Number of caseworkers 6m | 791 | 266 | |
| Number of caseworkers 12m | 681 | 54 | |

*Note:* Case-weighted means for high and low performing caseworkers. Robust standard errors reported in parentheses. Empirical Bayes shrinkage is applied to all variables that are averages of jobseeker outcomes, besides 'Emprate deviation high re6' for which the sample size is too small. Caseworker experience and wage comes from the personnel registries, in which there is data on 527 low performing and 163 high performing caseworkers. There are deviations due to high reemployment probability in 502 cases, across 154 low performing caseworkers and 99 high performing caseworkers. Treatment effect is $E[Y^1 - Y^0|ALMP]$.

**Table 7.** *Caseworker Performance Heterogeneity II*

| | Below algorithm | Above algorithm | P-value diff |
|---|---|---|---|
| **Panel c: Caseworker Assignment Performance** | | | |
| Recall IS (TPR) | 0.881 | 0.902 | [0.000] |
| | (0.000) | (0.000) | |
| Recall ALMP (TNR) | 0.131 | 0.144 | [0.000] |
| | (0.000) | (0.000) | |
| Precision IS | 0.564 | 0.562 | [0.000] |
| | (0.000) | (0.000) | |
| Precision ALMP | 0.462 | 0.556 | [0.000] |
| | (0.000) | (0.000) | |
| Gain over random | 0.041 | 0.190 | [0.000] |
| | (0.000) | (0.001) | |
| **Panel d: Caseworker Predictive Performance** | | | |
| Emprate deviation high re6 | 0.039 | 0.097 | [0.000] |
| | (0.003) | (0.004) | |
| Number of caseworkers 6m | 791 | 266 | |
| Number of caseworkers 12m | 681 | 54 | |

*Note:* Case-weighted means for high and low performing caseworkers. Robust standard errors reported in parentheses. Empirical Bayes shrinkage is applied to all variables that are averages of jobseeker outcomes, besides 'Emprate deviation high re6' for which the sample size is too small. Caseworker experience and wage comes from the personnel registries, in which there is data on 527 low performing and 163 high performing caseworkers. There are deviations due to high reemployment probability in 502 cases, across 154 low performing caseworkers and 99 high performing caseworkers. Treatment effect is $E[Y^1 - Y^0|ALMP]$.

## 7.5 Bias in Performance Measures

An important question is why the Swedish PES has implemented a seemingly poorly performing algorithm. In this section I benchmark my estimates against those constructed following the approach used in the PES's own evaluation (Arbetsförmedlingen, 2021), which relies on two features that may bias measured performance: measurement errors in the PES's administrative outcomes data, which internal reviews have shown to contain systematic reporting errors, and realized employment outcomes that are endogenous to caseworker decisions. I address the first by using monthly earnings data from Swedish tax records (AGI data), and the second by constructing accuracy measures based on estimated population employment potential and conditional employment rates.

The endogeneity bias causing an overestimate of performance could arise due to negative average treatment effects. Under negative average treatment effects, a caseworker who mistakenly assigns a jobseeker with $Y_i^0 = 1$ to an ALMP will observe $Y_i^1 = 0$, making the assignment appear correct. An algorithm trained on these outcomes will learn to predict not only baseline reemployment potential but also caseworker assignment patterns.

**Table 8.** *Caseworker vs Algorithm: Assignment Accuracy*

| | Original Algorithm | Adjusted Algorithm | Caseworkers |
|---|---|---|---|
| Benchmark estimate | 0.45 | 0.47 | 0.60 |
| (i) Correcting outcomes | 0.31 | 0.32 | 0.42 |
| (ii) Correcting endog. | 0.23 | 0.24 | -0.00 |
| (iii) Both corrections | 0.23 | 0.21 | 0.05 |

*Note:* The table shows classification accuracy, in terms of improvement over a random assignment. Gain = (Accuracy − Random) / (Max − Random). Random accuracy $= pq + (1-p)(1-q)$. The ROC-AUC of the algorithm is 0.67 for the benchmark estimate and 0.59 correcting for both endogeneity and outcome. See Table E2 for raw accuracy, and accuracy under a random and perfectly targeted assignment. Includes caseworkers with at least 50 profilings per regime. Controlling for regime fixed effects. Number of caseworkers is 892, and number of profilings 130,705.

To make estimates comparable, I will express the accuracy rate as a gain over a random assignment, given the potential accuracy under perfect targeting. The random benchmark is calculated for a true reemployment probability $p$ and independent search rate $q$ as $Accuracy^R = p \times q + (1-p) \times (1-q)$. Accuracy under perfect targeting is $Accuracy^P = 1 - abs(p-q)$.

Table 8 presents accuracy estimates under four specifications that progressively correct for measurement error and endogeneity. Here, 'original algorithm' denotes accuracy under the actual algorithmic recommendations, while 'adjusted algorithm' denotes estimated accuracy for the algorithm evaluated at the same Independent Search rate as the average caseworker. Under the benchmark specification—using PES administrative outcomes and endogenous employment—caseworkers appear to outperform the algorithm, capturing 60% of the feasible gain compared to 45% for the algorithm. The estimated algorithmic gain of 45% is close to the estimate of 50% in Arbetsförmedlingen (2021), despite them using a difference test population and during a period where the algorithm was yet not in use. This ranking reverses under corrections: the algorithm captures approximately 23% of the feasible gain, compared to 5% for caseworkers. The endogeneity correction drives the reversal. Under endogenous outcomes, caseworker accuracy is inflated because jobseekers assigned to ALMPs who experience negative treatment effects remain unemployed and are recorded as correctly classified. Correcting for this eliminates the caseworker advantage entirely. The outcome correction alone decreases measured accuracy for all decision makers but does not change the

relative ranking—it is only when endogeneity is addressed that the algorithm's superior targeting is revealed.

The results are robust to the assignment rate at which the algorithm is evaluated: the original and adjusted algorithms achieve nearly identical gains, confirming that the algorithm's advantage reflects ranking quality rather than a more conservative assignment threshold.

These results highlight the importance of taking both measurement errors and endogeneity in outcomes seriously when evaluating algorithmic and caseworker performance. And suggests that the PES's reported accuracy substantially overstates performance.

## 7.6 Fairness in Assignment to ALMP: Gender, Nativity and Age

This section examines *fairness* in assignment to ALMPs. Bias, or discrimination, in machine learning algorithms arises when predictive performance differs across groups. When algorithms are used for decision-making, such differences can translate into discrimination. In the context of ALMP assignment, this means that two jobseekers with the same potential outcomes should receive the same assignment decision. I examine differences in ALMP assignment across protected characteristics, focusing on (1) gender, (2) country of birth, and (3) age.[13]

I begin by examining how well the algorithmic and caseworker allocations align with a risk-based allocation rule within each subgroup. Table 9 reports differences across groups in the Independent Search rate and in accuracy. Importantly, when assessing fairness, it is necessary to account for the fact that groups may differ in their underlying reemployment probabilities. For this reason, differences in accuracy are more informative than simple differences in assignment propensities. Unlike in the previous analysis, however, I do not adjust for Independent Search rates, since one possible mechanism of discrimination is that groups face different access to ALMPs. Using differences in accuracy as measure of bias does not distinguish between the *sources* of bias. It captures both (i) taste based discrimination in assignment decisions and (ii) differences in the ability to predict reemployment probability within each subgroup.

Two findings stand out. First, the algorithm recommends ALMPs to migrant jobseekers to a much greater extent than to natives. For the other groups, assignment rates are very similar. When caseworkers deviate from the algorithm, they increase ALMP assignment for all groups except migrants. Even so, migrants remain more likely than natives to be assigned to ALMPs, although the gap is substantially smaller than under the algorithmic counterfactual. Second, caseworkers reduce accuracy relative to the algorithm in all

[13] Due to limited sample size, I do not analyze fairness with respect to disability status.

groups. The algorithm achieves 7% higher accuracy for women than for men, and likewise for natives relative to migrant jobseekers. Caseworkers widen the gap between natives and migrants to 12%. Figure F1 shows how performance differs in terms of the conditional employment rate.

**Table 9.** *Assignment Propensities and Accuracy Rate*

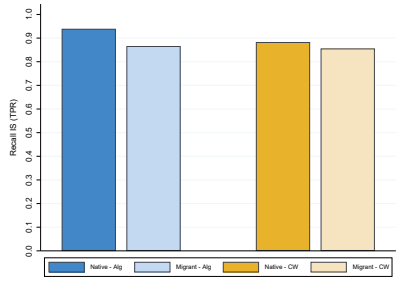
| | Male | Female | Native | Migrant | Old | Young |
|---|---|---|---|---|---|---|
| Panel a: Independent Search Rate | | | | | | |
| Algorithm | 0.91 | 0.90 | 0.93 | 0.83 | 0.9 | 0.91 |
| Caseworkers | 0.87 | 0.88 | 0.88 | 0.86 | 0.87 | 0.88 |
| Panel b: Accuracy | | | | | | |
| Algorithm | 0.55 | 0.59 | 0.58 | 0.54 | 0.58 | 0.57 |
| Caseworkers | 0.53 | 0.57 | 0.56 | 0.50 | 0.55 | 0.54 |
| Number of jobseekers | 64,327 | 66,378 | 98,630 | 32,075 | 63,546 | 67,159 |
| Number of caseworkers | 892 | 892 | 892 | 892 | 892 | 892 |

*Note:* Includes caseworkers with at least 50 profilings per regime. Controlling for regime fixed effects.

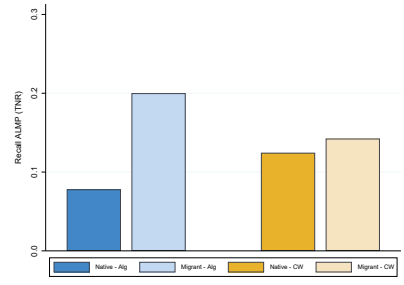
Next, I focus on native and migrant jobseekers and examine more closely how caseworker and algorithmic performance differ. Figure 6 reports recall and precision on both the Independent Search and ALMP margins. The largest differences appear on the ALMP margin. The algorithm achieves more than twice as high recall among migrant jobseekers as among natives; that is, among jobseekers who would remain unemployed, a much larger share of migrants are assigned to an ALMP. This mirrors the fact that migrants are assigned to ALMPs at nearly twice the rate of natives. By narrowing this gap in assignment rates, caseworkers also largely equalize recall across the two groups.

Despite assigning migrants to ALMPs at a much higher rate, the algorithm also achieves higher precision on the ALMP margin for migrants. This suggests that, among jobseekers assigned to ALMPs, a larger share of migrants are correctly identified as having low reemployment potential. By contrast, performance on the Independent Search margin is stronger for natives, with both recall and precision higher in that group. Baseline reemployment potential also differs substantially, at 50% among migrants and 57% among natives. Taken together, these differences help explain why overall accuracy is higher for natives. Although caseworkers reduce disparities in recall and precision on the ALMP margin, they do so at the cost of lower overall accuracy for both natives and migrants.

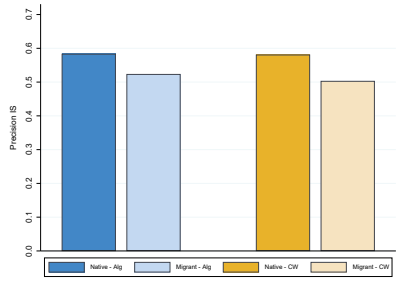
To understand how caseworker deviations affect overall employment rates, I examine the relationship between caseworker deviation rates and employment outcomes within each subsample. I report results only for the 6-month follow-



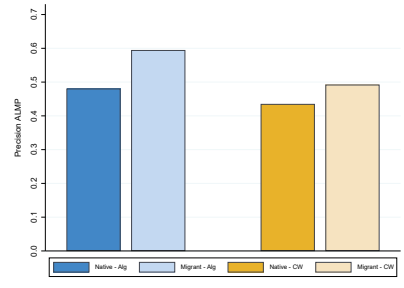

(a) Recall IS (TPR)




(b) Recall ALMP (TNR)




(c) Precision IS




(d) Precision ALMP

*Figure 6.* Fairness in Algorithmic and Caseworker Performance

*Note:* Includes caseworkers with at least 50 profilings per regime. Controlling for regime fixed effects. $Precision_{IS} = P(Y^0 = 1|IS)$, $Precision_{ALMP} = P(Y^0 = 0|ALMP)$, $Recall_{IS} = P(IS|Y^0 = 1)$, $Recall_{ALMP} = P(ALMP|Y^0 = 0)$. Number of caseworkers is 892, and number of profilings 130,705.

up, as the limited sample size makes it infeasible to study outcomes after 12 months

Table 10 shows significant negative effects of caseworker deviation rates on employment among male, native, and younger jobseekers. However, none of the coefficients differ significantly across complementary groups. Interestingly, there is no clear relationship between group-level caseworker accuracy and the employment effects of deviations.

**Table 10.** *Caseworker Deviation Rate: Natives and Migrants*

| | Full time employment, 6 months | | | | | |
|---|---|---|---|---|---|---|
| | Male | Female | Native | Migrant | Old | Young |
| CW deviation rate | -0.096* | -0.073 | -0.096** | -0.046 | -0.032 | -0.133** |
| | (0.055) | (0.055) | (0.045) | (0.078) | (0.056) | (0.054) |
| Alg. assignment rate | X | X | X | X | X | X |
| Deviation rate | 0.135 | 0.134 | 0.120 | 0.180 | 0.141 | 0.129 |
| Number of jobseekers | 64,327 | 66,378 | 98,630 | 32,075 | 63,546 | 67,159 |
| Number of caseworkers | 1,276 | 1,276 | 1,276 | 1,276 | 1,276 | 1,276 |

*Note:* ***, **, * indicate significance at 1%, 5% and 10% respectively. Robust standard errors in parentheses. Includes caseworkers with at least 50 profilings per regime. Controlling for regime fixed effects. P-values of difference between groups: gender $p = 0.796$, country of birth $p = 0.351$, age $p = 0.234$.

## 7.7 Robustness Checks: Targeting Unemployment Risk

**Defining the Outcome**

See subsection F2 for robustness checks with respect to defining the outcome variable. I use both alternatives definition using the payroll tax data, and definitions based on the PES registries. The result that caseworkers reducing alignment with an allocation rule based on unemployment risk is robust to using different definitions of employment. Estimates are overall similar in magnitude for the difference in conditional employment rate, but vary more in terms of the share of caseworkers who achieve a higher conditional employment rate than the algorithm. The difference between the algorithm and caseworkers is smallest using my main measure: exiting unemployment for full time employment within 6 months.

**Controlling for Algorithmic Independent Search Rate**

To control for some of the sampling variation, I run the extrapolation and estimate the algorithmic counterfactuals controlling for the algorithmic independent search rate. See Table F1 for results. The main results are robust.

**Extrapolation: Functional Form**

In the main analysis, I estimate the algorithmic counterfactual conditional employment rate using linear extrapolation. The results are robust to using both quadratic and local linear regression instead. See **??** for results.

**Bounding**

An alternative to extrapolation for estimating the algorithmic counterfactual conditional employment rate is a bounding approach. I apply worst- and best-case bounds, using all observed outcomes of jobseekers assigned to Independent Search at each cutoff. The bounds are calculated by assuming that outcomes for unobserved jobseekers are either all negative (worst case) or all

positive (best case). However, due to substantial caseworker deviations, the resulting bounds are wide and include both the extrapolated algorithmic counterfactual and the caseworker average, rendering them largely uninformative, Figure F4.

**First profiling**
Results are robust to only including the first profiling event per jobseeker, subsection F5.

# 8 Conclusion

This paper asks whether caseworkers can improve on an algorithm that assigns unemployed jobseekers to active labor market programs. The setting is one in which human discretion has clear scope to add value: the algorithm predicts unemployment risk, but the policymaker also cares about treatment effects, and risk-based targeting is optimal only under strong assumptions about the relationship between treatment effects and baseline risk. In principle, caseworkers could use private information and broader judgment to improve on an algorithm that is not designed to solve the full decision problem.

The empirical results show, however, that caseworker discretion reduces alignment with an allocation based on unemployment risk, lowers total employment, and generates negative average realized treatment effects. More than half of the jobseekers caseworkers assign to an ALMP would have found work on their own. At the same time, I find that caseworkers perform relatively well in identifying jobseekers who are likely to become employed on their own, suggesting that the main difficulty may instead lie in assessing treatment effects. These findings are based on a six-month follow-up horizon; whether a longer-run follow-up—particularly for programs lasting 6-12 months—would alter the picture is an important question for future work.

The findings in this paper point to three broader lessons. First, keeping humans in the loop does not guarantee good decision-making. In this setting, the algorithm has limited predictive power, yet caseworkers perform even worse. This challenges the common intuition that human oversight can safeguard against algorithmic mistakes. In practice, such a safeguard exists only when caseworkers are equipped to exercise better judgment.

Second, evaluation methods matter. I show that a standard approach used by both academics and practitioners—one that ignores the endogeneity of observed outcomes to caseworker decisions—can substantially overstate performance and even reverse conclusions about whether caseworkers outperform the algorithm. When an algorithm is both trained and evaluated on biased outcome data, such bias may go undetected. Credible evaluation of both algorithmic and human decision-making is therefore essential to the credibility of the system itself.

Third, the algorithm addresses only one dimension of the policy problem. It predicts unemployment risk, but leaves caseworkers to assess treatment effects on their own—a task at which they appear to perform poorly. In this setting, combining predictions of both unemployment risk and treatment effects into a joint recommendation may do more to improve outcomes than simply adjusting the scope of caseworker discretion.

More broadly, as algorithms are introduced into increasingly complex policy domains—where the prediction target and the policy objective do not fully coincide—the key question is not simply whether humans or algorithms perform better, but whether the overall decision system is designed to support good decisions.

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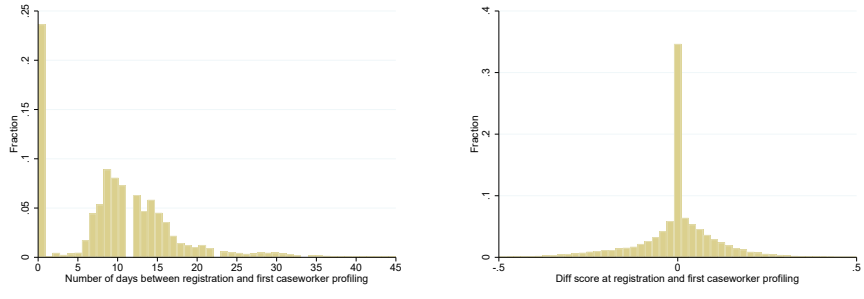
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# PES Institutions and the Profiling Tool

## A1 Timing of First Profiling and Caseworker Updating of the Profiling Score

In the centralâoffice population, 72% of jobseekers receive their first profiling at initial registration (i.e., same day as they sign up online), which occurs between day 0 and 9 of the unemployment spell (median = 0). Among jobseekers whose first profiling occurs during the caseworker meeting, the spell length at profiling ranges from 0 to 287 days on (median = 15). During the meeting, the profiling score is updated only if new information emerges; otherwise, the registration score stands. The median time between registration profiling and the first caseworker profiling is 10 days, and 5% of these firstâmeeting profiling events lead to a change in the algorithmic recommendation.




(a) Number of days between (b) Difference in profiling score

*Figure A1.* Profiling at registration and first caseworker profiling.

*Note:* Sample of jobseekers with their first profiling at registration (72% of total sample). The plot excludes the 5 % most extreme values. The number of days ranges from 0 to 375 days. And the difference in score ranges from -0.76 to 0.80.

## A2 Eligibility Criteria & Standardized Questions

Eligibility for active labor market programs is determined primarily by the jobseeker's predicted reemployment probability. In addition, a jobseeker is ineligible if any of the following conditions apply.

*Eligibility criteria.*

1. Already part-time employed or have secured employment starting within 90 days or enrolled in a full time studies starting within 90 days.
2. Is going on parental leave within 90 days.
3. Is moving abroad.
4. Is a citizen of another EU/EES country, an asylum seeker, or a Ukrainian refugee in Sweden under the EU Temporary Protection Directive.
5. Receiving assistance from a former employer, a transition organization[14], another government agency, or the municipality.
6. Registered on the board of a company.

*Standardized questions.*

The following standardized questions are asked when caseworkers conduct the labor market assessment. The questions are answered either during the online registration or at the first meeting with the caseworker.

1. What did you do before becoming unemployed?
2. Do you know what type of job you would prefer?

2a. Do you know what you would like to study? (If no to question 2)

3. How would you rate your Swedish language skills?
4. Do you know how many jobs you plan to apply for this month?
5. Do you have any obstacles to relocating or commuting long distances?
6. Do you think you will get a job or start studying within three months?
7. What were your average monthly labor earnings during the past year?
8. Do you face any obstacles that prevent you from applying for jobs or getting hired?

9a. Do you have any health issues that may affect your chances of getting a job?

9b. Do you have a disability that may affect your chances of getting a job?

[14]Transition organizations — owned by employer associations and labor unions and regulated through collective bargaining agreements — offer services focused on educational and career counseling and also provide financial support.

## A3 Algorithm Input Variables

**Table A1.** *Input variables algorithm*

| |
|---|
| Date of registration PES |
| Country of birth |
| Disability |
| Number of PES programs or interventions 10 years |
| Number of days in PES programs or interventions 10 years |
| Number of days registered at the PES |
| Current or the number of days registered at the latest registration three years |
| Number of days since the last registration three |
| Age |
| Municipality |
| Zip code |
| Search category |
| Target occupation |
| Education in target occupation |
| Experience in target occupation |
| Searching for work to the extent of (full time, part time) |
| Level of education |
| Field of education |
| Gender |
| Applying for jobs abroad |
| Applying for unemployment insurance |
| Share of unemployed within the zip code |
| Median income zip code |
| Share with secondary education within zip code |
| Share of households types within zip code |
| Share with non-European citizenship within zip code. |

## A4 Spell Durations

Figure A2 shows the distribution of number of days unemployed at the profiling date.

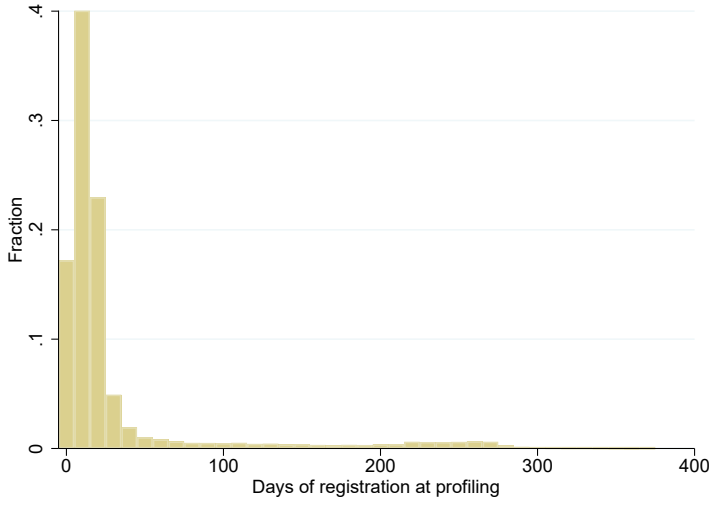
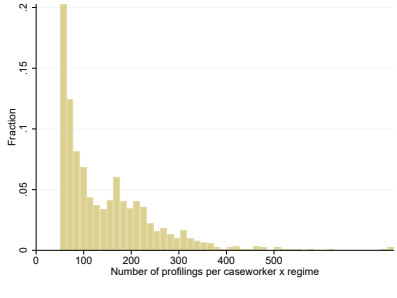


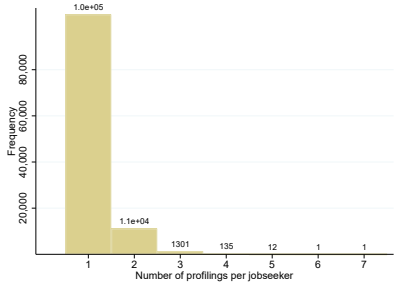

*Figure A2.* Distribution number of days unemployed

*Note:* Length of current unemployment spell at the profiling date. The median length at profiling is 13 days. 10 % of profilings are done at day 80 or later and 5 % of profilings are done at day 196 or later.

## A5 Case Load - Final Data




Number of decision events per caseworker




Number of decision events per jobseeker

*Figure A3.* Distribution of Decision Events Across Caseworkers and Jobseekers

*Note:* Number of decision events in the final sample.

## A6 Adjusted Profiling Score

In order to interpret the results as a result of caseworker deviations from the algorithmic recommendation, I need to adjust the profiling score and take the duration specific cut-offs into account. This section starts with a description of how the adjusted score is computed, followed by descriptive statistics of caseworker deviations.

Each profiling event yields an *original profiling score*, between 0 and 1, and a categorical *algorithmic recommendation* (Independent Search or ALMP). Since the cutoff between Independent Search and ALMP varies by the number of days since the start of the unemployment spell, see Figure 2, I construct an *adjusted score* that captures the distance (in profilingâscore units) to the independentâsearch cutoff at each jobseekerâs spell length.

Formally, let $s_{i,t}$ be the original profiling score for jobseeker $i$ at time $t$, and let $\tau_d$ be the cutoff in place for a jobseeker with $d$ days of unemployment. As described above, $\tau_d$ changed over the study period, why I allow it to vary over time. Define the *adjusted score*:

$$\widetilde{s}_{i,t} \;=\; s_{i,t} \;-\; \tau_{d_{i,t}}\,.$$

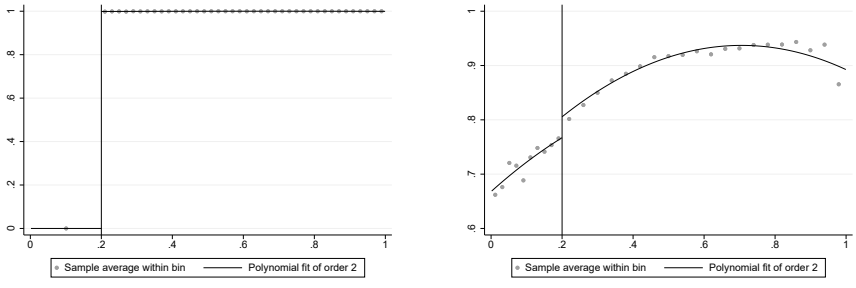
By construction, $\widetilde{s}_{i,t} > 0$ implies "recommended Independent Search", while $\widetilde{s}_{i,t} \leq 0$ implies "recommended ALMP". In the subsample of jobseekers with $\leq 180$ days of unemployment—when the cutoff is flat at 0.2—the original and effective scores yield identical rankings. Almost 95 % of decisions happens before 180 days, see Figure A2 for the full distribution. In the full sample, the correlation between the original and adjusted score is as high as 0.92, indicating that they largely agree on the ordering of jobseekers.

I then normalize the adjusted score to lie in $[0,1]$ for easier interpretation. In the main analysis I will use this normalized adjusted score, instead of the original profiling score. The inferred algorithmic recommendation based on $\widetilde{s}_{i,t}$ matches the recorded recommendation in 99.8% of cases.

## A7 Effect of Algorithmic Recommendation

Figure A4 plots the adjusted score against the probability of being recommended (panel a) or actually assigned (panel b) to Independent Search. In panel (a), the algorithmic recommendation displays a sharp discontinuity at the cutoff: as the score crosses zero, the probability of being recommended jumps from 0 to 0.95. Strikingly, panel (b) shows that caseworker assignment is a smooth, continuous function of the score, with no discrete jump at the cutoff. The smoothness around the cutoff implies that for the marginal cases, the algorithmic recommendation is irrelevant for the caseworkers' assignment decision. Empirically, the caseworkers are deviating more close to the cutoff, which is consistent with the smoothness around the cutoff. See Appendix Fig-

ure A2 for a figure of how the deviation rate changes against adjusted score and the length of the unemployment spell.




(a) Algorithmic recommendation (b) Caseworker assignment

*Figure A4.* Probability of Recommendation and Assignment to Independent Search

*Note:* Probability of being (a) recommended and (b) assigned to independent search using the original profiling score among jobseekers with unemployment duration less than 250 days (Jan 1-Sep 7) and 180 days (Sep 8 to Dec 31) - thus during the windows where the algorithm takes a single cutoff between Independent Search and Intense support (ALMP). RD-plot using 40 bins and a quadratic functional form fit.

## A8 Caseworker overrides: Descriptives

**Table A2.** *Jobseeker Characteristics by Deviations*

| | All | Deviations |
|---|---|---|
| Male | 0.492 | 0.493 |
| Age | 40.990 | 41.482 |
| Historical functional disability | 0.046 | 0.136 |
| Immigrant | 0.245 | 0.328 |
| High school | 0.024 | 0.027 |
| Tertiary education | 0.173 | 0.142 |
| Profiling score | 0.496 | 0.398 |
| Days of registration | 31.802 | 55.608 |
| Service and admin | 0.123 | 0.124 |
| Service, care and shop | 0.278 | 0.313 |
| Agriculture, forestry, fishing | 0.022 | 0.024 |
| Building and manufacturing | 0.082 | 0.087 |
| Previously registered | 0.052 | 0.050 |
| Previous treatment | 0.209 | 0.270 |
| | | |
| N jobseekers | 130,705 | 17,595 |
| N caseworkers | 892 | 892 |

# Appendix B: Linear Empirical Bayes Shrinkage

I use linear empirical Bayes shrinkage, following the three steps lined out by Walters (2024).

## Step 1. Estimation

Estimate the caseworker conditional employment rate, $M_j$.

$$Y_i = M_j + \varepsilon_i \tag{20}$$

Treating that $\hat{M}_j$ is normally distributed with variance $s_j^2$

## Step 2. Deconvolution

Estimate the two parameters $\mu_M$ and $\sigma_M$.

$$\hat{\mu}_M = \frac{1}{J} \sum_{j=1}^{J} \hat{M}_j \tag{21}$$

$$\hat{\sigma}_M^2 = \frac{1}{J} \sum_{j=1}^{J} \left[ (\hat{M}_j - \hat{\mu}_M)^2 - s_j^2 \right] \tag{22}$$

Where subtracting $s_j^2$ corrects for excess noise in the $\hat{theta}_j$ estimates.

## Step 3. Shrinkage

To form the posterior mean for each caseworker estimate $\hat{M}_j$, we take a weighted average of the noisy estimates $\hat{M}_j$ and the prior mean $\mu_M$. How much the caseworker estimates are shrunk towards the prior mean depends on the precision of the estimates relative to the precision of the prior mean.

$$M_j^* = \hat{M}_j \left( \frac{\sigma_M^2}{\sigma_M^2 + s_j^2} \right) + \hat{\mu}_M \left( \frac{s_j^2}{\sigma_M^2 + s_j^2} \right) \tag{23}$$

# Appendix C: Empirical Strategy

## C1 Identifying assumptions

**Table C1.** *Exclusion Restriction: Meetings*

| | All cw In-person Pre Prof | All cw Telephone Pre Prof | All cw In-person Post Prof | All cw Telephone Post Prof | Same cw Telephone Post Prof |
|---|---|---|---|---|---|
| Panel a: Independent Search Rate | | | | | |
| | -0.003 | -0.525*** | 0.008 | 0.117 | -0.139** |
| | (0.007) | (0.059) | (0.013) | (0.107) | (0.064) |
| Std diff | -0.000 | -0.020 | 0.000 | 0.005 | -0.005 |
| Panel a: Deviation Rate | | | | | |
| | 0.009 | 0.558*** | 0.016 | 0.219** | -0.253*** |
| | (0.007) | (0.059) | (0.015) | (0.111) | (0.067) |
| Std diff | 0.000 | 0.021 | 0.001 | 0.008 | -0.009 |
| Mean depvar | 0.01 | 0.33 | 0.02 | 0.83 | 0.43 |
| N | 114,237 | 114,237 | 114,237 | 114,237 | 114,237 |

*Note:* ***, **, * indicate significance at 1%, 5% and 10% respectively. Robust standard errors reported in parentheses. Controlling for regime fixed effects and includes caseworkers with at least 50 profilings per regime. Table shows single regressions of number of meetings/contacts on leave-out independent search rate and leave out deviation rate. "Pre" and "Post" indicate meetings prior to or post the profiling events. Meetings on the day of a profiling event are excluded. Std diff shows the coefficients multiplied by the standard deviation of the independent variable, and can be interpreted as the change in the number of meetings from a 1 sd change in the independent variable. Includes jobseekers assigned to Independent Search.

**Table C2.** *First stage*

| | Independent Job Search | | Deviation All | | Deviation IS to ALMP | | Deviation ALMP to IS | |
|---|---|---|---|---|---|---|---|---|
| CW assignment rate | 0.457*** | 0.374*** | | | | | | |
| | (0.025) | (0.026) | | | | | | |
| CW deviation rate | | | 0.372*** | 0.326*** | 0.270*** | 0.274*** | 0.571*** | 0.824*** |
| | | | (0.026) | (0.027) | (0.023) | (0.025) | (0.119) | (0.127) |
| Alg. IS rate | | X | | X | | X | | X |
| Mean dep var | 0.87 | 0.87 | 0.13 | 0.13 | 0.09 | 0.09 | 0.55 | 0.55 |
| N | 130,705 | 130,705 | 130,705 | 130,705 | 118,618 | 118,618 | 12,087 | 12,087 |

*Note:* First stage regression of assignment to independent search on leave-out assignment rate. It tests the *relevance assumptions*. The positive and significant coefficients indicates that the relevance assumption holds. Robust standard errors reported in parentheses. ***, **, * indicate significance at 1%, 5% and 10% respectively. Dependent variable is the leave-out caseworker independent search rate. Controlling for regime fixed effects and includes caseworkers with at least 50 profilings per regime.

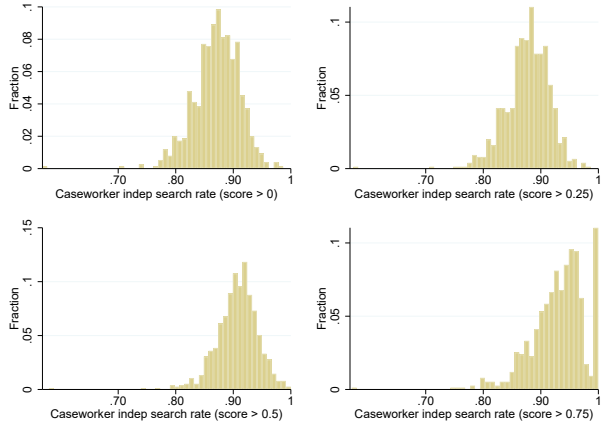



*Figure C1.* Caseworker independent search rates within subsamples

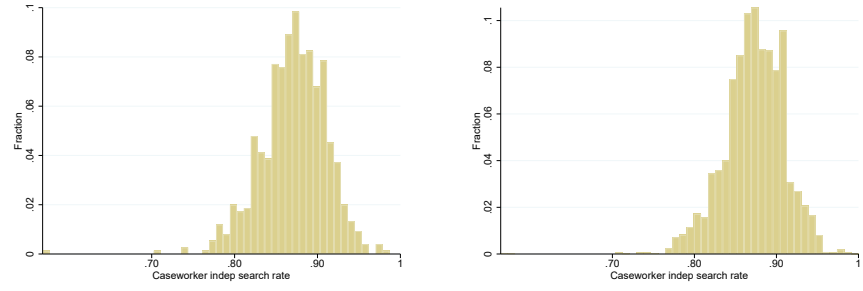
*Note:* Independent search rate is the caseworkers propensity to assign jobseekers to independent search, ass opposed to some ALMP. The figure shows the distribution of caseworker independent search rates across four subsamples. In each figure, the sample is restricted to all jobseekers with a profiling score higher than a cutoff X, and caseworker independent search rates are thereafter calculated. As the sample of jobseekers becomes more restrictive, the distribution of caseworker assignment rates shift to the right. Includes caseworkers with at least 50 cases per regime.

**Table C3.** *Average Monotonicity (deviation rate)*

| | All deviations | | IS to ALMP | | ALMP to IS | |
|---|---|---|---|---|---|---|
| | First stage | N | | | | |
| Male | 0.311*** | 64,327 | 0.282*** | 58,771 | 0.693*** | 5,556 |
| | (0.038) | | (0.036) | | (0.188) | |
| Historical disability | 0.387** | 6,027 | 0.574*** | 3,816 | -0.067 | 2,211 |
| | (0.178) | | (0.208) | | (0.296) | |
| Immigrant | 0.358*** | 32,075 | 0.292*** | 26,723 | 0.499*** | 5,352 |
| | (0.061) | | (0.052) | | (0.180) | |
| High school | 0.317* | 3,077 | 0.238 | 2,777 | 0.146 | 300 |
| | (0.191) | | (0.181) | | (0.874) | |
| Tertiary education | 0.456*** | 22,559 | 0.419*** | 20,853 | 1.258*** | 1,706 |
| | (0.063) | | (0.058) | | (0.341) | |
| Age 25-34 | 0.255*** | 48,621 | 0.230*** | 44,698 | 0.451** | 3,923 |
| | (0.043) | | (0.039) | | (0.216) | |
| Age 35-44 | 0.295*** | 33,648 | 0.274*** | 30,063 | 0.560** | 3,585 |
| | (0.054) | | (0.049) | | (0.228) | |
| Age 45-54 | 0.453*** | 25,925 | 0.347*** | 23,497 | 1.057*** | 2,428 |
| | (0.062) | | (0.057) | | (0.292) | |
| Service and admin | 0.352*** | 16,115 | 0.273*** | 14,515 | 0.870*** | 1,600 |
| | (0.078) | | (0.072) | | (0.336) | |
| Service, care and shop | 0.321*** | 36,301 | 0.258*** | 32,320 | 0.635*** | 3,981 |
| | (0.053) | | (0.048) | | (0.223) | |
| Building and manufacturing | 0.201** | 10,655 | 0.193** | 9,874 | 0.609 | 781 |
| | (0.096) | | (0.091) | | (0.506) | |
| Previously registered | 0.299** | 6,809 | 0.368*** | 6,305 | 0.249 | 504 |
| | (0.117) | | (0.104) | | (0.586) | |
| Not previously registered | 0.327*** | 123,896 | 0.268*** | 112,313 | 0.851*** | 11,583 |
| | (0.028) | | (0.025) | | (0.130) | |
| Previous treatment | 0.274*** | 27,376 | 0.219*** | 23,626 | 0.427* | 3,750 |
| | (0.064) | | (0.058) | | (0.223) | |
| No previous treatment | 0.340*** | 103,329 | 0.288*** | 94,992 | 1.010*** | 8,337 |
| | (0.029) | | (0.027) | | (0.155) | |
| Full time | 0.450*** | 12,181 | 0.550*** | 10,559 | -0.254 | 1,622 |
| | (0.106) | | (0.101) | | (0.332) | |

*Note:* Robust standard errors reported in parentheses. ***, **, * indicate significance at 1%, 5% and 10% respectively. Dependent variable is the leave-out caseworker deviation rate. Controlling for regime fixed effects and includes caseworkers with at least 50 profilings per regime. Controlling for algorithmic independent search rate.

## C2 Descriptives




Caseworker level Case level

*Figure C2.* Caseworker assignment rate

*Note:* Controlling for regime fixed effects and includes caseworkers with at least 50 profilings per regime.

## C3 Deviation Rates

Table C4 shows the matrix of the algorithmic recommendation and caseworker assignment decisions. The table confirms that caseworkers had very limited ability to assign jobseekers to Job Search Assistance if the algorithm recommendation was Independent Search or Intense Support. Instead we observe a substantial number of deviations from independent search to intense support, where caseworkers had free assignment discretion. Likewise, the deviation rate from Intense Support to Independent Search is also high.

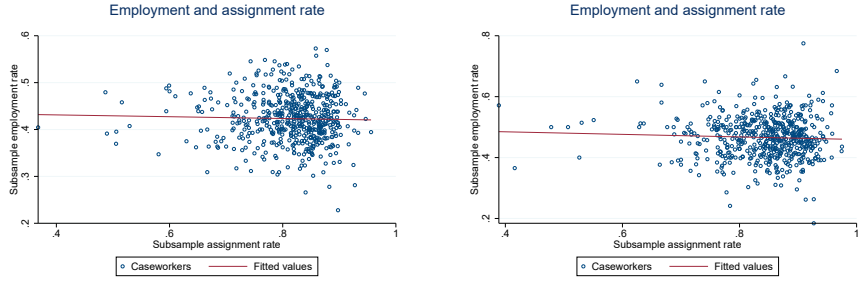
**Table C4.** *Algorithmic recommendation and caseworker assignment*

| | | Caseworker assignment | | |
|---|---|---|---|---|
| | | Independent search | Job search assistance | Intense support |
| Algorithmic recommendation | Independent search | 107,630 | 0 | 10,988 |
| | Job search assistance | 2,604 | 2,138 | 1,915 |
| | Intense support | 4,003 | 0 | 1,427 |

## C4 Implementation

For linear extrapolation, center the independent search rate estimates at $R_j^{IS} = 1$, i.e. by defining a new variable $\hat{R}_j^C = \hat{R}_j^{IS} - 1$. Then run the regression $\hat{M}_j = \beta_0 + \beta_1 \hat{R}_j^C + e_i$. Due to centering around $\hat{R}_j^{IS} = 1$, $\beta_0$ is the estimated employment rate when 100 % of jobseekers are assigned to Independent Search within the sub-sample, which provides the estimate of the threshold-specific average employment rate in the sub-sample of jobseekers with $a(x_i) \geq s$. That is, $\beta_0$ provides the estimate of $M_{a(j)}$. The regression is weighted by the inverse variance of the estimated conditional employment rate, to put more weight on more precise estimates.

Figures below show the extrapolation to independent search rate equal to one for two subsamples. Jobseekers in panel (b) is a subset of jobseekers in panel (a). Note, however, that the caseworker independent search rate and conditional employment rate could differ between the two subset for the same caseworker.




(a) $a(X_i) \geq 0.25$ (b) $a(X_i) \geq 0.49$

*Figure C3.* Subsample assignment rates and conditional employment rates.

*Note:* Extrapolation using linear regression and weighted by the inverse variance of the caseworker estimates. Controlling for regime fixed effects and includes caseworkers with at least 50 profilings per regime.

# Appendix D: Predictive Performance

For the purposes of this paper, any standard machine learning performance measure can be computed from three estimate: (i) The 'random rate' - i.e. the population average reemployment probability, denoted $p$, (ii) the conditional employment rate, i.e. realized employment among jobseekers assigned to independent search, denoted $CER$, and (iii) the assignment propensity (to independent search) denoted $q$.

Recall is the probability of being correctly assigned to IS among jobseekers able to find a job in their own, $p(IS|Y^0 = 1)$. And equivalently for ALMP: $p(ALMP|Y^0 = 0)$.

Precision is the share of IS assigned jobseekers who would truly become employed on their own, that is the conditional employment rate that we observe $p(Y^0 = 1|IS)$ . And the equivalence on the ALMP side which is unobservable: $p(Y^0 = 0|ALMP)$.

Accuracy is simply the share of correctly classified cases.

In this paper, these measures are normalized to show the share captured of the feasible gain over a random assignment:

$$Gain = \frac{Observed - Random}{Perfect - Random} \tag{24}$$

**Table D1.** *Classification metrics under random, perfect, and observed targeting*

| | Observed | Random | Perfect ($q \leq p$) |
|---|---|---|---|
| *Independent Search side (positive = $Y^0 = 1$)* | | | |
| Precision (CER) | CER | $p$ | $p/q$ |
| Recall (TPR) | $\frac{\text{CER} \cdot q}{p}$ | $q$ | 1 |
| *ALMP side (positive = $Y^0 = 0$)* | | | |
| Precision | $1 - \frac{p - \text{CER} \cdot q}{1 - q}$ | $1 - p$ | 1 |
| Recall (TNR) | $1 - \frac{(1 - \text{CER}) \cdot q}{1 - p}$ | $1 - q$ | $\frac{1 - q}{1 - p}$ |
| *Overall* | | | |
| Accuracy | $1 - p - q + 2 \cdot \text{CER} \cdot q$ | $pq + (1 - p)(1 - q)$ | $1 - \lvert p - q \rvert$ |

*Note:* $p$ = population reemployment probability, $q$ = independent search assignment share, CER = observed conditional employment rate under independent search.

The ROC-AUC is a standard metrics for ranking ability of an algorithm. The ROC-AUC measures the probability that a randomly drawn $Y^0 = 1$ job-

seeker is ranked higher (assigned closer to IS) than a randomly $Y^0 = 1$ job-seeker. By construction, the ROC-AUC takes a value between 0 and 1. Where 0.5 implies a random assignment and 1 implies perfect targeting. Since I only estimate the algorithmic CER (TPR) over a specific range of IS rates $q$, I standardize it to lie on the standard ROC-AUC interval.

# Appendix E: Additional Results

## E1 Needs-based Allocation

**Table E1.** *Caseworker vs algorithmic prediction accuracy*

| | (1) Caseworker vs Original alg | (2) Caseworker vs PES alg | (3) Caseworker vs Random | (4) Original Alg vs Random | (5) PES Alg vs Random |
|---|---|---|---|---|---|
| Share CW higher emprate | 0.048*** (0.007) | 0.200*** (0.011 | 0.709*** (0.015) | | |
| Average diff emp. rate | -0.022*** (0.000) | -0.013*** (0.000) | 0.005*** (0.000) | 0.028*** (0.000) | 0.018*** (0.000) |
| N | 1,274 | 1,275 | 1,275 | | |

*Note:* ***, **, * indicate significance at 1%, 5% and 10% respectively. Caseworker vs algorithmic counterfactual, holding caseworker independent search rate constant. Row (1) shows the share of caseworkers with a higher employment rate than (1) the original algorithm, (2) the PES algorithm taking days registered specific cutoffs into account, and (3) a random assignment of jobseekers. Row (2) shows the case-weighted average difference between caseworker conditional employment rate and algorithmic counterfactual and random assignment, as well as the algorithm versus a random assignment. Employment outcomes are based on monthly payroll tax data, as defined in section 3.4. Algorithmic counterfactuals estimated using linear extrapolation as described in section 5, following Angelova et al. (2025). The caseworker conditional employment rate are estimated using linear empirical Bayes shrinkage. Estimates are constructed controlling nonparametrically for regime fixed effects. Includes caseworkers with at least 50 profilings per regime. Standard errors constructed using bootstrapping.

**Table E2.** *Caseworker vs Algorithm: Assignment Accuracy*

| | Actual | Random | Max | Gain |
|---|---|---|---|---|
| *Panel A: Original Algorithm* | | | | |
| Benchmark estimate | 0.41 | 0.38 | 0.44 | 0.45 |
| (i) Correcting outcomes | 0.56 | 0.52 | 0.62 | 0.31 |
| (ii) Correcting endog. | 0.42 | 0.40 | 0.47 | 0.23 |
| (iii) Both corrections | 0.57 | 0.55 | 0.65 | 0.23 |
| *Panel B: Adjusted Algorithm* | | | | |
| Benchmark estimate | 0.43 | 0.39 | 0.48 | 0.47 |
| (i) Correcting outcomes | 0.57 | 0.52 | 0.66 | 0.32 |
| (ii) Correcting endog. | 0.43 | 0.41 | 0.51 | 0.24 |
| (iii) Both corrections | 0.57 | 0.54 | 0.68 | 0.21 |
| *Panel C: Caseworkers* | | | | |
| Benchmark estimate | 0.44 | 0.39 | 0.48 | 0.60 |
| (i) Correcting outcomes | 0.58 | 0.52 | 0.66 | 0.42 |
| (ii) Correcting endog. | 0.41 | 0.41 | 0.51 | -0.00 |
| (iii) Both corrections | 0.55 | 0.54 | 0.68 | 0.05 |

*Note:* Gain = (Actual − Random) / (Max − Random). Random accuracy $= pq + (1-p)(1-q)$. Maximum accuracy $= 1 - |p-q|$. Both benchmarks vary across rows and panels as base rates and assignment rates differ by specification. Controlling for regime fixed effects and including caseworkers with at least 50 profilings per regime. The ROC-AUC of the algorithm is 0.67 for the benchmark estimate and 0.59 correcting for both endogeneity and outcome.

## E2 Low and High Performing Caseworkers

**Table E3.** *Jobseeker Characteristics*

| | Below algorithm | Above algorithm | P-value diff |
|---|---|---|---|
| Male | 0.490 | 0.500 | [0.297] |
| | ( 0.002) | ( 0.003) | |
| Historical functional disability | 0.048 | 0.040 | [0.003] |
| | ( 0.001) | ( 0.001) | |
| Immigrant | 0.248 | 0.236 | [0.000] |
| | ( 0.001) | ( 0.003) | |
| High school | 0.023 | 0.025 | [0.323] |
| | ( 0.000) | ( 0.001) | |
| Tertiary education | 0.172 | 0.173 | [0.971] |
| | ( 0.001) | ( 0.002) | |
| Score | 0.664 | 0.651 | [0.000] |
| | ( 0.000) | ( 0.001) | |
| Service and admin | 0.123 | 0.123 | [0.985] |
| | ( 0.001) | ( 0.002) | |
| Service, care and shop | 0.280 | 0.269 | [0.065] |
| | ( 0.001) | ( 0.003) | |
| Building and manufacturing | 0.081 | 0.085 | [0.984] |
| | ( 0.001) | ( 0.002) | |
| Age 25-34 | 0.375 | 0.362 | [0.870] |
| | ( 0.001) | ( 0.003) | |
| Age 35-44 | 0.258 | 0.253 | [0.309] |
| | ( 0.001) | ( 0.003) | |
| Age 45-54 | 0.197 | 0.202 | [0.321] |
| | ( 0.001) | ( 0.002) | |
| Previously registered | 0.056 | 0.036 | [0.815] |
| | ( 0.001) | ( 0.001) | |
| Days of registration | 30.171 | 38.257 | [0.000] |
| | ( 0.165) | ( 0.448) | |
| Number of jobseekers | 104,545 | 26,070 | |

*Note:* Robust standard errors in parentheses. Includes caseworkers with at least 50 profilings per regime. Controlling for regime fixed effects.

## E3 Efficiency

**Table E4.** *Deviation Rate: Cumulative earnings*

| | Cummulative Earnings | | | |
|---|---|---|---|---|
| | 6 months | 6 months | 12 months | 12 months |
| CW deviation rate | -18,543.920*** | -12,401.709* | -25,616.006 | -19,636.109 |
| | (6,736.726) | (7,063.005) | (17,653.773) | (18,590.393) |
| Alg. assignment rate | | X | | X |
| Number of jobseekers | 130,705 | 130,705 | 57,894 | 57,894 |
| Number of caseworkers | 892 | 892 | 736 | 736 |

*Note:* ***, **, * indicate significance at 1%, 5% and 10% respectively. Robust standard errors in parentheses. Controls for regime fixed effects and include caseworkers with at least 50 profilings per regime. Mean deviation rate is 0.135 and mean algorithmic assignment rate is 0.906 in the full sample and both measures are leave-one-out means.

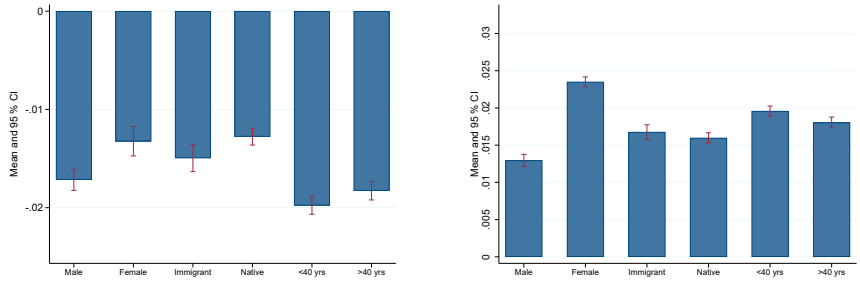
**Table E5.** *Deviation Rate: First earnings post employment*

| | First Earnings | |
|---|---|---|
| CW deviation rate | -1,835.031 | -961.059 |
| | (2,980.637) | (3,180.433) |
| Alg. assignment rate | | X |
| Number of jobseekers | 54,359 | 54,359 |
| Number of caseworkers | 892 | 892 |

*Note:* ***, **, * indicate significance at 1%, 5% and 10% respectively. Robust standard errors in parentheses. Includes balanced panel of caseworkers with at least 75 profilings per year. Mean deviation rate is 0.135 and mean algorithmic assignment rate is 0.906 in the full sample and both measures are leave-one-out means.

# Appendix F: Robustness Checks

## F1 Subgroup results: needs-based allocation



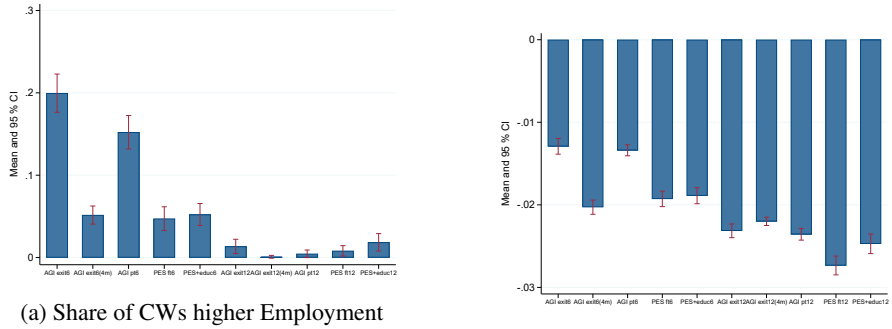

(a) Diff employment rate: caseworkers versus algorithm

(b) Diff employment rate: algorithm versus random assignment

*Figure F1.* Caseworkers versus Algorithmic Counterfactual: subgroups

*Note:* Panel (a) shows the case-weighted difference in employment rate between caseworkers and the algorithmic counterfactual,for the six demographic groups The negative values means that caseworkers obtain lower conditional employment rates than strict adherence to the algorithm. Panel (b) shows the difference in case-weighted conditional employment rate between a random assignment and counterfactual algorithmic allocation. The positive values show that the algorithm performs better than a random allocation in predicting reemployment probability. Controlling for regime fixed effects. 95 % confidence intervals are constructed using bootstrapping.

## F2 Robustness: Outcome Defintion




(a) Share of CWs higher Employment Rate than the Algorithm

(b) Difference in Employment Rate

*Figure F2.* Caseworkers versus Algorithmic Counterfactual

*Note:* Panel (a) shows the share of caseworkers with higher conditional employment rate than the algorithm, and panel (b) shows the difference in employment rate between the caseworkers and algorithmic counteractual. Each bar is using a different definition of 'employment', where indicators with "AGI" are based on payroll tax data, and "PES" are based on PES registries. 6 and 12 indicate the follow-up period. Controlling for regime fixed effects. 95 % confidence intervals are constructed using bootstrapping.

## F3 Robustness: Controlling for algorithmic independent search rate

**Table F1.** *Extrapolation - Controlling for algorithmic independent search rate*

| | (1) Caseworker vs Original alg | (2) Caseworker vs PES alg | (3) Caseworker vs Random | (4) Original Alg | (5) PES Alg vs Random |
|---|---|---|---|---|---|
| Share CW higher emprate | 0.111*** | 0.236*** | 0.909*** | | |
| | (0.009) | (0.013 | (0.011) | | |
| Average difference emprate | -0.015*** | -0.011*** | 0.015*** | 0.030*** | 0.027*** |
| | (0.000) | (0.001) | (0.000) | (0.000) | (0.000) |
| N caseworkers | 892 | 892 | 892 | | |

*Note:* ***, **, * indicate significance at 1%, 5% and 10% respectively. Caseworker vs algorithmic counterfactual, holding caseworker independent search rate constant. Algorithmic counterfactuals estimated using linear extrapolation as described in section 5, following Angelova et al. (2025). The caseworker conditional employment rate are estimated using linear empirical Bayes shrinkage. Estimates are constructed controlling for regime fixed effects. Includes caseworkers with at least 50 profilings per regime. Standard errors constructed using bootstrapping. As opposed to the main results, these estimates additionally control for algorithmic independent search rate.

# F4 Robustness: Extrapolation

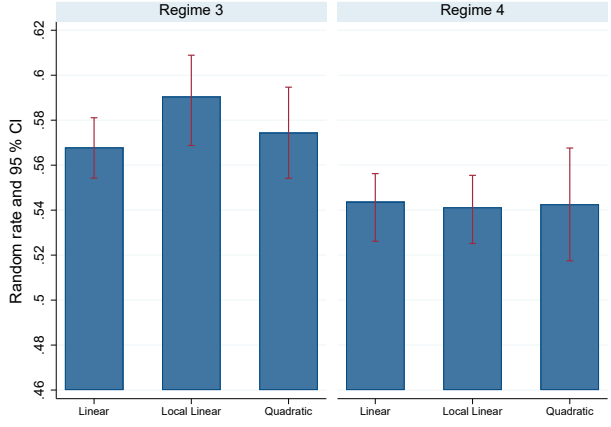



*Figure F3.* Extrapolation Functional Form - Random Rate

*Note:* Figure show how the estimated random rate: the average employment potential, differ across different extrapolation methods and between regimes. Estimates are constructed controlling for regime fixed effects. Includes caseworkers with at least 50 profilings per regime. 95 % confidence interval constructed using bootstrapping.

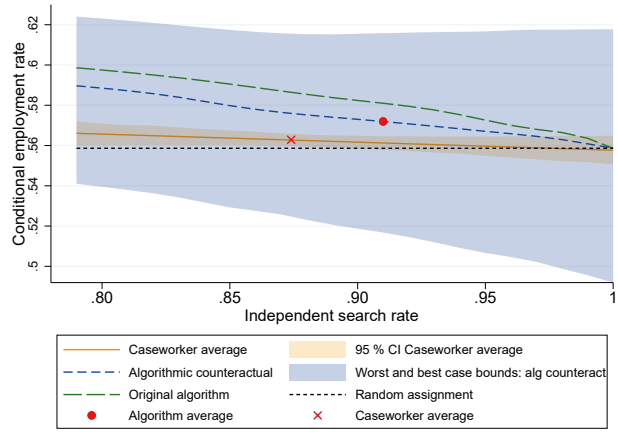



*Figure F4.* Worst- and Best case Bounding

*Note:* Figure show the algorithmic counterfactual estimated using best- and worst case bounding. Includes caseworkers with at least 50 profilings per regime and controlling nonparametrically for regime fixed effects.

**Table F2.** *Extrapolation - Function Form*

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | Caseworkers vs Algorithm | | | Caseworkers vs Random | | |
| | Linear | Local Linear | Quadratic | Linear | Local Linear | Quadratic |
| Share CW higher emprate | 0.200*** | 0.054*** | 0.091*** | 0.709*** | 0.317*** | 0.581*** |
| | (0.012) | (0.007 | (0.010) | (0.014) | (0.014 | (0.016) |
| Average difference emprate | -0.013*** | -0.021*** | -0.019*** | 0.005*** | -0.008*** | 0.002*** |
| | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) |
| N caseworkers | 892 | 892 | 892 | 892 | 892 | 892 |

*Note:* ***, **, * indicate significance at 1%, 5% and 10% respectively. Caseworker vs algorithmic counterfactual, holding caseworker independent search rate constant. Algorithmic counterfactuals estimated using extrapolation as described in section 5, following Angelova et al. (2025) but using three difference extrapolation approaches: liner, local linear and quadratic. The caseworker conditional employment rate are estimated using linear empirical Bayes shrinkage. Estimates are constructed controlling for regime fixed effects. Includes caseworkers with at least 50 profilings per regime. Standard errors constructed using bootstrapping.

## F5 Robustness: First Profiling

This section runs the analysis using only the first profiling per jobseeker.

**Table F3.** *First Profiling: Balance*

| | Assignment Rrate | Deviation Rate | Profiling Score |
|---|---|---|---|
| Male | -0.039* | 0.032 | 2.480*** |
| | (0.020) | (0.021) | (0.101) |
| Historical functional disability | 0.064 | 0.007 | -22.595*** |
| | (0.048) | (0.048) | (0.234) |
| Immigrant | 0.007 | 0.001 | -9.472*** |
| | (0.024) | (0.024) | (0.122) |
| High school | -0.031 | 0.052 | 0.001 |
| | (0.067) | (0.067) | (0.360) |
| Tertiary education | 0.043 | -0.038 | 2.670*** |
| | (0.027) | (0.027) | (0.124) |
| Age 25-34 | -0.026 | 0.019 | 5.227*** |
| | (0.021) | (0.021) | (0.105) |
| Age 35-44 | -0.006 | 0.025 | -1.374*** |
| | (0.023) | (0.024) | (0.116) |
| Age 45-54 | 0.042* | -0.057** | -3.589*** |
| | (0.026) | (0.026) | (0.119) |
| Service and admin | -0.048 | 0.023 | -2.513*** |
| | (0.031) | (0.031) | (0.140) |
| Service, care and shop | -0.031 | 0.020 | -5.824*** |
| | (0.023) | (0.023) | (0.106) |
| Building and manufacturing | 0.038 | -0.034 | 6.701*** |
| | (0.037) | (0.037) | (0.200) |
| Previously registered | -0.075* | 0.024 | -2.052*** |
| | (0.045) | (0.045) | (0.225) |
| Previous treatment | 0.002 | 0.032 | -7.375*** |
| | (0.025) | (0.025) | (0.123) |
| Score | 0.006 | 0.028 | - |
| | (0.063) | (0.063) | - |
| F-test | 1.62 | 1.07 | 2,269 |
| P-value f-test | 0.067 | 0.381 | 0.000 |
| N | 105,354 | 105,354 | 105,354 |
| N caseworkers | 777 | 777 | 777 |
| Mean depvar | 0.90 | 0.11 | 0.50 |

*Note:* ***, **, * indicate significance at 1%, 5% and 10% respectively. Using only first profiling per jobseeker. Robust standard errors in parentheses. Controlling for regime fixed effects. Including caseworkers with 50 cases per regime. Coefficients and standard errors are multiplied by 100 and should be interpreted as percentage points.

**Table F4.** *First Profiling: Results needs-based allocation*

| | (1) Caseworker vs Original alg | (2) Caseworker vs PES alg | (3) Caseworker vs Random | (4) Original Alg | (5) PES Alg vs Random |
|---|---|---|---|---|---|
| Share CW higher emprate | 0.067*** (0.009) | 0.078*** (0.009 | 0.600*** (0.018) | | |
| Average difference emprate | -0.020*** (0.000) | -0.019*** (0.000) | 0.003*** (0.000) | 0.022*** (0.000) | 0.022*** (0.000) |
| N | 777 | 777 | 777 | | |

*Note:* ***, **, * indicate significance at 1%, 5% and 10% respectively. Caseworker vs algorithmic counterfactual, holding caseworker independent search rate constant. Row (1) shows the share of caseworkers with a higher employment rate than (1) the original algorithm, (2) the PES algorithm taking days registered specific cutoffs into account, and (3) a random assignment of jobseekers. Row (2) shows the case-weighted average difference between caseworker conditional employment rate and algorithmic counterfactual and random assignment, as well as the algorithm versus a random assignment. Employment outcomes are based on monthly payroll tax data, as defined in section 3.4. Algorithmic counterfactuals estimated using linear extrapolation as described in section 5, following Angelova et al. (2025). The caseworker conditional employment rate are estimated using linear empirical Bayes shrinkage. Estimates are constructed controlling nonparametrically for regime fixed effects. Includes caseworkers with at least 50 profilings per regime (calculated in the sample of first profilings). Standard errors constructed using bootstrapping.

**Table F5.** *First Profiling: Results efficiency*

| | Full time employment | | | |
|---|---|---|---|---|
| | 6 months | 6 months | 12 months | 12 months |
| CW deviation rate | -0.082* | -0.067 | -0.068 | -0.037 |
| | (0.046) | (0.048) | (0.057) | (0.060) |
| Alg. assignment rate | | X | | X |
| Mean deviation rate | 0.119 | 0.119 | 0.130 | 0.130 |
| Number of caseworkers | 105,354 | 105,354 | 59,594 | 59,594 |
| Number of jobseekers | 777 | 777 | 680 | 680 |

*Note:* ***, **, * indicate significance at 1%, 5% and 10% respectively. Using only first profiling per jobseeker. Robust standard errors in parentheses. Controlling for regime fixed effects. Including caseworkers with 50 cases per regime.

# Appendix G: Theoretical Appendix

## G1 Simulation: Random $y0$ and $y1$

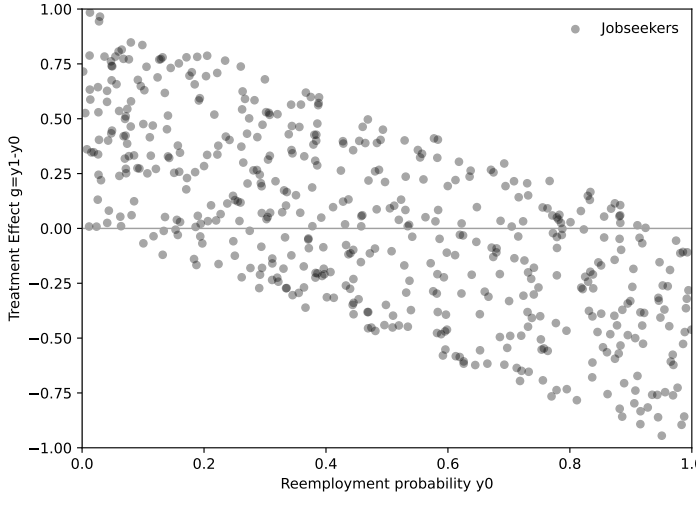



*Figure G1.* Simulation: Random y1 and y0

*Note:* Simulation of jobseeker reemployment probability and treatment effects where $y0$ and $y1$ are randomly drawn.

# Essay II. Strictness in the Evaluation of Job Search Effort and Employment Outcomes

Co-authored with Arne Uhlendorff and Johan Vikström

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# 1 Introduction

Job-search requirements, the monitoring of claimants' search efforts, and sanctions for noncompliance are key features of unemployment insurance (UI) systems in many countries. These measures aim to mitigate moral hazard by encouraging more intensive job search and higher job-finding rates, thereby enabling higher benefit replacement rates and improved consumption smoothing among unemployed job seekers. However, monitoring and job-search requirements may also adversely affect job quality and shift some informal job search toward formal, monitored search. The effects of monitoring may further vary across groups and market conditions.

Beyond supporting job seekers, caseworkers typically play a central role in monitoring job-search effort and enforcing job-search requirements. Caseworkers often differ in how they apply and combine these elements, implying that a job seekerâs employment outcomes may depend on the specific caseworker to whom they are assigned (Le Barbanchon et al., 2024). This paper leverages such caseworker variation. For identification, we exploit a unique institutional setting in Sweden where UI recipients are required to submit monthly reports on their job-search activities. These reports are allocated quasi-randomly to caseworkers in centralized units for assessment. Because caseworkers differ in their strictness and have no direct contact with the job seekers whose reports they review, this setting provides credible evidence on impacts of stricter enforcement of UI requirements.

Using variation in caseworker strictness and detailed administrative data, we estimate the causal effects of stricter monitoring. We study its impact on job-search intensity, job-finding over the unemployment spell, and the quality of the first job obtained, as well as heterogeneity by labor market attachment and prior employment in growing versus declining industries.

Unemployed job seekers in Sweden are required to submit monthly *activity reports*. Caseworkers review these reports and determine whether they meet the required standards. We exploit the fact that during our observation period (2014–2017), these reports were evaluated by caseworkers at a few central offices during the first four months of unemployment, while caseworkers in local offices maintained regular contact with job seekers, providing job-search assistance, counseling, and assigning them to active labor market programs (ALMPs). The allocation of reports to central office caseworkers was effectively quasi-random, as caseworkers pull reports from the top of a common list that is ordered essentially at random. This review process constitutes the only interaction between central caseworkers and job seekers. This setting provides a clear institutional basis for exogenous caseworker variation and reduces the risk of confounding caseworker strategies, addressing two common threats to identification in studies exploiting variation across caseworkers in UI systems (e.g. Arni and Schiprowski, 2019; Humlum et al., 2025; Klaeui, 2025). We

verify empirically that our measure of caseworker strictness is unrelated to observed job seeker characteristics.

We begin by documenting that caseworkers who are stricter in enforcing UI requirements initiate a monitoring chain characterized by significantly more not-approved activity reports, more notifications of violations of the job-search rules, and more benefit sanctions. This stricter monitoring translates into shorter unemployment durations and higher employment rates. Moving from the 10th to the 90th percentile in the distribution of caseworker strictness reduces the probability of remaining unemployed after 120 days by around 1.1 percentage points, corresponding to a relative effect of 3%.

We next shed light on the underlying mechanisms. First, using data on wages and long-run earnings, we find no negative effects on job quality. Second, the positive effects on job finding are reflected in an increase in job applications, both immediately after the assessment of the first report and several months later. Overall, moving from the 10th to the 90th percentile of caseworker strictness increases the number of job applications by around 5.4%. Third, we also observe more phone and in-person meetings for job seekers assigned to stricter caseworkers. This reflects the institutional setting, in which caseworkers contact job seekers following a non-approved report. These contacts are part of the monitoring process and may contribute to the observed increase in job finding.

The effects of stricter enforcement of UI requirements are heterogeneous across job seekers and local labor market conditions. In the short run, the employment effects of a stricter caseworker are more positive for job seekers with a low risk of long-term unemployment, while those with weaker employment prospects experience more positive effects in the longer run. Hence, tighter monitoring can bring job seekers with strong labor market attachment quickly back to work, while providing a more sustained improvement in employment rates for harder-to-place job seekers.

Labor market conditions also play an important role in shaping the effects of monitoring.[1] For job seekers previously employed in high-growth industries, we find a positive short-run effect on job-finding that dissipates over time. In contrast, for job seekers from low-growth industries, the effects are more persistent. This suggests that stricter monitoring is particularly effective in preventing prolonged unemployment among workers in low-growth sectors. This pattern is not driven by higher industry mobility under stricter monitoring, but rather by sustained increases in job-search intensity.

[1] Recent evidence suggests that training programs may be particularly effective for unemployed workers exposed to offshoring or automation risk, who are therefore more likely to experience persistent employment losses (Humlum et al., 2025; Schmidpeter and Winter-Ebmer, 2021). Athey et al. (2026) report that earnings losses after job loss vary substantially by industry and local labor market characteristics, suggesting that enforcing job search requirement may have heterogeneous effects along these dimensions.

There is a broad empirical literature on job-search requirements and monitoring. Several studies find that additional job-search requirements shorten unemployment durations (Arni and Schiprowski, 2019; Bolhaar et al., 2019; Chan et al., 2023; Johnson and Klepinger, 1994). Evidence on the impact of monitoring is more mixed. Based on field experiments, Ashenfelter et al. (2005) show that increased monitoring of job-search effort reduces the duration of UI benefit receipt in the U.S., while Micklewright and Nagy (2010) find similar effects in Hungary, but only for certain subgroups. In contrast, van den Berg and van der Klaauw (2006) find no effect of increased monitoring on re-employment rates, likely due to a shift from informal to formal job search methods.[2] Non-experimental studies, exploiting policy changes over time (Petrongolo, 2009; McVicar, 2008, 2010) or discontinuities due to age (Cockx and Dejemeppe, 2012), generally find positive effects on exits from unemployment and welfare receipt. A related strand of research examines the impact of imposed benefit sanctions, typically using timing-of-events models (Abbring and van den Berg, 2003). These studies consistently document strong positive effects of imposed sanctions on reemployment rates (Abbring et al., 2005; Lalive et al., 2005), though sometimes at the cost of lower subsequent job quality (Arni et al., 2013; van den Berg and Vikström, 2014; van den Berg et al., 2019).[3]

We contribute to the literature on monitoring and sanctions by providing evidence of the causal effects of caseworker strictness in the evaluation of job-search requirements. We combine credible identification, based on the as-if random assignment of caseworkers who differ in strictness, with a non-experimental setting that enhances external validity. Moreover, our large sample and rich administrative data allow us to study how the impact of monitoring varies with job seekers' employment prospects and labor market conditions. We are also able exploit unique longitudinal data on job-search activity, which enable us to observe the dynamics of search intensity over the unemployment spell.

Another strand of the literature focuses more directly on the role of UI caseworkers. Exploiting unplanned caseworker absences, Schiprowski (2020) shows that missing a scheduled meeting with a caseworker reduces re-employment rates on average, with substantial heterogeneity across caseworkers. Cederlöf et al. (forthcoming) estimate caseworker value-added in Swedish local employment offices and document considerable heterogeneity, which affects both unemployment duration and the quality of subsequent jobs. Similarly, Hervelin and Villedieu (2023) analyze French counseling offices and documents considerable variation across caseworkers in training place-

[2] van den Berg et al. (2025b) show that integration agreements, which are written contracts stipulating rights and obligations of UI recipients, increase reemployment rates, especially among job seekers with rather adverse employment prospects.

[3] For structural analyses of the interplay between monitoring and sanctions, see Cockx et al. (2018) and van den Berg et al. (2025a).

ments and employment outcomes.[4] Arni and Schiprowski (2019) use caseworker variation in Switzerland to document that increasing the number of required job applications shortens unemployment durations. Humlum et al. (2025) and Hyman (2018) provide evidence that retraining programs can improve employment outcomes.[5]

Our contribution to this strand of the literature is the focus on the enforcement margin–how variation in caseworker strictness in monitoring job–search requirements affects job seekers' behavior and labor market outcomes. Our setting identifies the impact of stricter enforcement of existing UI requirements, abstracting from broader caseworker, counseling, or training effects. We exploit a clear institutional source of variation: central caseworkers pick reports in random order from a common list. Because these caseworkers have no direct contact with job seekers, we can rule out alternative channels–such as differential access to ALMPs–ensuring that our estimates capture the effects of stricter monitoring.

This paper proceeds as follows. Section 2 describes the institutional background. Section 5 introduces the dataset, details the sample selection criteria and presents some descriptive statistics on activity reports. Section 4 presents the empirical approach and provides evidence on assumptions behind our empirical strategy. Section 5 presents our main results, and Section 6 provides evidence of effect heterogeneity and studies potential mechanisms. Section 7 concludes.

# 2 Institutional Background

We now describe the UI system including, focusing on the rules governing benefit eligibility, monitoring of job-search requirements, and sanctions in place during our observation period from 2014 to 2017. To better understand how these rules have been implemented in practice, we conducted several interviews with individuals who worked as caseworkers during our observation period.

## 2.1 Unemployment Insurance

Sweden is one of the few countries where UI coverage is voluntary. UI benefits are administered by 24 independent UI funds. Workers can choose which fund to join, but since UI funds are linked to labor unions, membership is

[4] Bolhaar et al. (2020) report substantial heterogeneity in how caseworkers assign welfare-to-work programs, and Schiprowski et al. (2024) provide evidence that caseworker meetings are more effective when accompanied by an integration agreement.

[5] Klaeui (2025) show that encouraging broader job search and targeting tight labor market sectors reduces unemployment duration.

typically based on industry, occupation, or a combination of the two.[6] UI fund membership is high in Sweden: approximately 70% of the labor force was enrolled in a UI fund between 2010–2020. Each UI fund sets its own membership contribution, which finances both administrative costs and a fixed government fee based on the number of members.[7]

A job seeker must meet several requirements and conditions to qualify for UI benefits. First, upon becoming unemployed, the individual must have been a member of a UI fund and paid contributions for at least 12 months.[8] Second, the job seeker must have been employed for at least six of the past 12 months. Third, the job seeker must register at the Public Employment Service (PES) and be able and willing to work at least three hours a day and a minimum of 17 hours per week. Fourth, the unemployed must actively search for a job throughout their unemployment spell. Our analysis focuses on this fourth condition.

Job seekers meeting these conditions are entitled to UI benefits for up to 60 weeks. During our observation period, benefits replaced 80% of previous earnings up to a ceiling. Individuals who do not meet the membership requirement may instead qualify for Unemployment Assistance (UA), which is flat-rate and less generous. We restrict our analysis to UI recipients, because monitoring rules were different for UA recipients during our observation period.

## 2.2 Activity Plans and Activity Reports

UI benefit recipients are required to actively search for a job. The Swedish PES is responsible for both monitoring job seekers' job-search activities and for assisting them in their search for suitable jobs. A central element of this process is the *activity plan*, which is set up by a PES caseworker together with the job seeker. The activity plan outlines the support the PES will provide and specifies the activities expected from the job seeker. This is done using eight broad support categories, without any detailed plan.[9] It also does not set a minimum level of required activities. For example, there is no fixed requirement for the minimum number of job applications per month.

[6]Some individuals qualify for more than one fund, but most join the one aligned with their occupation or industry. Selective entry across UI funds is therefore not a concern.

[7]In 2014, the average monthly contribution was around SEK 115 (10.29), with modest variation across funds–from SEK 85 (â¬7.60)*toSEK*160(¬14.31) − −*di f f erencesthatweregenerallynotadecisive f actorin f undchoice*(*IAFDatabase*).

[8]Landais et al. (2021) analyze selection into UI-membership in Sweden. They provide evidence that unemployment risk is higher among workers who buy supplemental coverage, driven partly by risk-based selection and partly by moral hazard.

[9]The categories are: search for jobs, improve your search, guidance to work, education to work, start new business, clarify your qualiï¬¬*cations f orwork*,*adaptyourworksituation*,*andworkpreparatorymeasures*.

To enhance the monitoring of job search efforts, monthly *activity reports* were introduced in 2013. Between the 1st and 14th of each month, all job seekers must submit an activity report detailing their job search activities from the previous month. Most importantly, job seekers must indicate which positions they have applied for. Additionally, the reports can include other activities such as job interviews and job-shadowing visits.

Activity reports serve as the primary tool for monitoring job search behavior. Caseworkers review these reports and decide whether to approve them. Although some general guidelines exist, assessments rely heavily on caseworker discretion. The fact that the activity plan does not specify quantitative targets, such as the number of job applications, leaves substantial room for individual judgment. According to our interviews with PES caseworkers, reviewing a report typically takes about ten minutes for an experienced caseworker. In addition to the number of job applications, caseworkers also assess whether these jobs align with the job seeker's education and occupational skills. They also take into account local labor market conditions relevant to the job seekerâs profile. This combined assessment of quantity and relevance of applications is a key reason why the PES does not impose a fixed minimum number of required applications per month.

## 2.3 Allocation of Activity Reports to Caseworkers

The Swedish PES is organized into two types of offices: *local offices* and *central offices*. At local offices, job seekers register, meet with caseworkers, receive job search assistance, and may be assigned to ALMPs. The central offices were introduced in 2010 to improve efficiency and accessibility by offering information and support to job seekers by phone. During our observation period, there were approximately 200 local offices across the country and 7 central offices. Each central office was responsible for handling activity reports from a specific geographical area.[10]

When activity reports were first introduced in 2013, they were assessed by caseworkers at local PES offices. To reduce the workload of local caseworkers, the responsibility for evaluating most reports was gradually transferred to the central offices. This shift began in 2013 and was fully implemented by February 2014. Central office caseworkers were responsible for assessing the reports but had no direct contact with the job seekers associated with them. Local caseworkers retained responsibility for meetings, job-search assistance, and program assignments.

Activity reports in central offices were organized in an IT system called *The Assistant*. In this system, the *first* activity report of each newly registered

[10]In 2018, the central offices were restructured and reorganized. Caseworkers in central offices now also began holding meetings with job seekers and assigning them to programs, tasks that were previously carried out by local offices.

job seeker appeared on a *common reports-list* accessible to all caseworkers within the office. Reports were ordered by submission date, with the oldest on top, and caseworkers processed them sequentially by selecting reports from the top of the list. The only information available at this stage was a random identification number, and the caseworkers had no prior information about the content of the report or the job seeker. This procedure implies that, conditional on office and time, the allocation of reports to caseworkers is quasi-random.[11]

The common reports list is strictly followed for the first report submitted by each job seeker. Once a caseworker picks the first report, that caseworker is typically responsible for reviewing the same job seekerâs next three reports. After that the reports are checked by the local offices. We therefore focus on the quasi-random allocation of the first report and mainly estimate reduced form effect of being assigned to a stricter caseworker for that initial report.

## 2.4 Benefit Sanctions and Monitoring Chain

Violations of the conditions for retaining UI benefits can lead to a sanction, which are monetary fines corresponding to a temporary suspension of UI benefits. Since UI funds administer UI benefits, they are responsible for imposing sanctions based on information provided by the PES.

The sanction system follows a staircase model, in which penalties increase in severity with each subsequent violation of the rules (IAF, 2014c). Sanctions are categorized into three groups: (1) job search and related misconduct, (2) rejection of job offers, and (3) quitting a job without valid cause. If a caseworker rejects a report due to insufficient job search, it falls into the first category. The staircase model for sanctions due to insufficient job search consists of five steps.[12] The first violation results in a warning. The second leads to a one-day suspension of benefits, the third to a five-day suspension, the fourth to a ten-day suspension, and the fifth to a complete suspension of UI benefits.[13] The job search and related misconduct category also includes other types of misconduct, such as failure to submit activity reports, absence from a scheduled meeting with a PES caseworker, and failure to apply for a referred job vacancy.

[11] Around 15% of the job seekers receive vacancy proposals from central office caseworkers before submitting their first activity report. These proposals are sent without prior contact and, unlike vacancy referrals, are non-binding suggestions rather than mandatory applications. To ensure that we isolate the impact of caseworker strictness, we exclude all job seekers who received a vacancy proposal from central office caseworkers before the assessment of the first report. In robustness analyses, we show that our results are unaffected by this restriction.

[12] Sanctions due to rejection of job offers follows another staircase model that is independent of the staircase for insufficient job search. It involves 5, 10, and 45 days suspension for the first, second, and third violation, followed by complete suspension after the fourth violation. Quitting a job without valid cause leads to a 45-day suspension.

[13] In principle, individuals can appeal a sanction, but this is rare. Of all decisions, only a negligible fraction (2%) is partially or fully reversed.

As described above, the monitoring of job search begins with caseworkers assessing activity reports. On average, about 7% of the reports are rejected. If a report is deemed insufficient, a series of actions take place – a process we refer to as the *monitoring chain*. The local caseworkers are in charge of these events and the central caseworkers are not involved. If a report is rejected by the central caseworker, the job seeker is supposed to be informed promptly, typically through a phone call from a local office caseworker. During this call, the caseworker explains the reasons for the rejection, indicates whether a violation will be registered, and may provide guidance on how to improve the job search. According to our interviews with caseworkers, these feedback calls usually last between 10 and 15 minutes.

In the next step of the monitoring chain, the local caseworker decide whether to report a *violation*. It is not uncommon for an activity report to be rejected without a violation being reported. In fact, only around 30% of all rejected reports result in a registered violation.[14] If a violation is registered, this is usually done on the same day the report is rejected; the median time between rejection and registration is zero days. Once a violation is registered, the job seeker is automatically informed by email. After a violation is registered, a *notification* is automatically sent to the relevant UI fund. In about 95% of cases, this occurs on the same day the violation is registered. Finally, the UI fund decides whether to impose a *sanction* based on the information in the notification. Sanction decisions are usually made within two or three weeks since the notification. The share of notifications that lead to a sanction within the first 120 days of the unemployment spell is 67% in our sample.

In *summary*, job search in Sweden is monitored through monthly activity reports. The centralized review process generates quasi-random variation in report assessments, which can be leveraged using a judges design. If a report is rejected this may have several consequences. First, repeated violations will lead to benefits sanctions. Second, the lack of specific requirements for the number of job applications and the relatively low rate of rejected reports suggest that a rejection may constitute a significant monitoring shock for job seekers. Third, a rejection is typically followed by a phone call with a local caseworker, which may tighten monitoring through questions about planned search activities and provide search advice. Such combinations of consequences are inherent to non-experimental settings, and our analysis captures their combined reduced-form effect. This corresponds to the relevant effect from a policy perspective.

[14]The remaining 70% are split evenly between cases where caseworkers choose not to register a violation after contacting the job seeker and cases where the PES takes no action at all after rejected reports (IAF, 2015a).

# 3 Data and Sample Selection

## 3.1 Data Sources

We use detailed register data from the Swedish PES on employment status, program participation, meetings, and all stages of the monitoring process. These data are complemented with data on labor earnings, wages, and occupational information.

The PES data provide daily information on the unemployment status of all registered job seekers, eligibility for UI benefits, and characteristics recorded at the start of the unemployment spell (such as age, gender, and unemployment history), and information on entries into and exits from ALMPs.[15] It also includes records on interactions between job seekers and caseworkers, including meetings and vacancy referrals. Additionally, the PES data contain information on each activity report, including the submission date, the date it was first accessed, the caseworker who evaluated it, the number of reported job applications and other activities, and the outcome of the assessment (either "approved" or "rejected"). Job applications include responses to advertised vacancies, unsolicited applications, and applications made in response to vacancy referrals and proposals. Other activities include interviews, recruitment meetings, and additional search-related actions.

We also have information on all recorded violations of the UI benefit rules, including the date, reason, reporting caseworker, and whether a notification was sent to the relevant UI fund. The violation reasons allow us to distinguish between cases of insufficient job search and other violations unrelated to rejected activity reports. The data also includes records of all sanction decisions made by the UI funds. These include the date, the type of sanction[16], and the corresponding step on the sanction ladder, indicating its position within the sequence of previous sanctions in the same category.

The PES data are merged with data from population registers and wage statistics from Statistics Sweden. The population register contains additional information, including the country of origin, and the level of education. It also includes yearly information on labor earnings (available up to 2018). The wage statistics contain annual data on wages[17], occupations, and contracted hours. The data cover the entire public sector, all large private firms, and a random sample of smaller firms. In total, the data covers about 50% of all private-sector employees and all public-sector employees. Information is

[15] In our sample, the most common reason for exiting unemployment is transitioning to employment (74.8%). Other exit reasons–such as pursuing education, receiving social assistance, or enrolling in alternative insurance schemes–are less common.

[16] There exist three types of sanctions, related to the reason for the sanction: (1) job search and related misconduct, (2) rejection of job offers, or (3) quitting a job without valid cause.

[17] Wages are recorded as monthly full-time equivalents, and are not derived from annual earnings and hours worked.

collected each fall and refers to all employees working at sampled firms at the time of data collection.

## 3.2 Sample Selection

Our analysis focuses on job seekers who entered as full-time unemployed between 1 February 2014 and 31 December 2017. The centralized assessment was fully implemented in February 2014, and we limit the sample to 2017 because the central offices were restructured in 2018. Our identification strategy relies on the quasi-random allocation of each job seeker's first activity report to caseworkers in central offices. Therefore, we restrict our analysis to job seekers whose first activity reports are reviewed at these offices.[18]

We focus on UI recipients, as the monitoring rules differ for non-recipients. We further restrict our analysis to individuals aged 25 or older, and exclude those with previous sanctions (around 4% of the sample) to focus on job seekers at the first step of the sanction ladder. We impose no restrictions on prior employment or unemployment history. We exclude job seekers who received a vacancy proposal from a central office caseworker before submitting their first activity report. We also exclude job seekers who fail to submit their first report on time. The precision of each caseworkerâs strictness measure depends on the number of cases they handle. To ensure reliable measures, we restrict our main sample to caseworkers who review at least 75 first activity reports during our observation period. This restriction excludes around 5% of the unemployment spells. We will later show that our results are robust to the these sampling restrictions.

The final estimation sample consists of 390 caseworkers across six central offices and 109,866 unemployment spells among 100,703 job seekers. In this sample, the median caseload is 375, and 5% of caseworkers handle more than 804 job seekers. Figure A1 shows the distribution of the number of reports each caseworker reviews during our observation period.

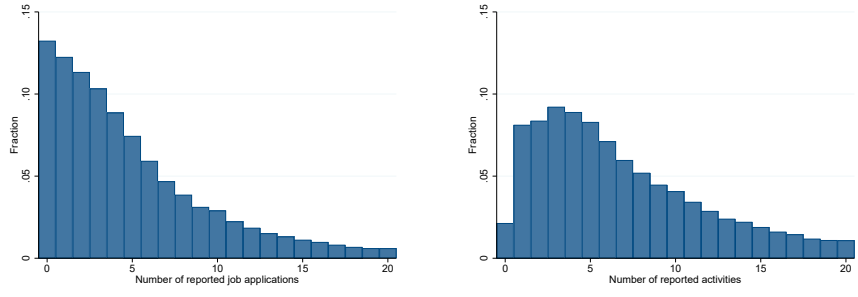
## 3.3 Reported Activities and Approval of Activity Reports

Figure 1 presents descriptive statistics on the number of job applications and the total number of job-search activities reported in the first activity report. Panel A shows that approximately 13% of job seekers do not report any job application, and a majority (56%) report fewer than five applications.[19] Panel

[18] One of the seven offices opened in 2017 and handled very few activity reports prior to September 2017. We exclude cases handled by this office, which account for approximately 1.5% of all reports reviewed by central offices.

[19] The duration of the reporting period for the first report is determined by the date of entry to unemployment, and job seekers entering in the middle of the month will have a reporting period that is less than a month. However, when restricting to job seeker that report for a full month, the share with zero job applications is still substantial (12%).

B indicates that nearly all job seekers report at least one activity. Besides job applications, this includes, for instance, job interviews and job-shadowing visits. Figure A2 in the Appendix shows that the number of reported job applications increases slightly in the second and third activity reports. On average, job seekers report 6 applications in the first report, increasing to 6.6 in the second and 6.7 in the third. The average number of total reported activities remains relatively stable–9.0, 9.4, and 9.2 across the three reports, respectively.

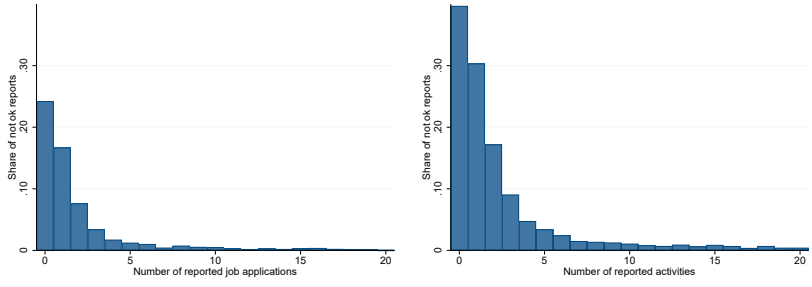



Panel A: Job Applications Report 1 Panel B: Total Activities Report 1

*Figure 1.* Histogram of number of job applications and activities in first activity report
*Notes*: Statistics for first reported report for the analyses sample described in Section 3.2. 109,866 cases and 390 caseworkers. Job applications include standard applications, unsolicited applications, and applications for vacancy referrals or proposals. Total activities additionally include interviews, recruitment meetings, other activities, and activities specified in the action plan. Figure A2 shows equivalent statistics for reports 2 and 3.

Focusing on the first activity report, Figure 2 shows a strong negative relationship between the number of reported job applications and the probability that a report is rejected. The figure also highlights both the comparatively lenient nature of the Swedish monitoring system and the substantial variation in how reports are assessed. Approximately 24% of job seekers who report no job applications receive a rejection, whereas this share falls to about 1% for those reporting five applications (Panel A). When considering all job-search activities rather than job applications alone, we see a similar picture but also that caseworkers take other activities into account. About 40% of the reports with zero activities are rejected, compared a rejection rate of 3% among those with five reported activities.[20]

[20]Figure A3 in the Appendix presents the corresponding distributions for the second and the third report. Although the relationship is somewhat weaker than for the first report, we continue to observe a negative association between both the number of reported job applications and total activities and the likelihood of rejection.




Panel A: Job Applications Report 1 Panel B: Total Activities Report 1

*Figure 2.* Share not-approved reports, by number of reported job applications and activities

*Notes*: Statistics for first reported report for the analyses sample described in Section 3.2. 109,866 cases and 390 caseworkers. Job applications include standard applications, unsolicited applications, and applications for vacancy referrals or proposals. Total activities additionally include interviews, recruitment meetings, other activities, and activities specified in the action plan. Figure A3 shows equivalent statistics for reports 2 and 3.

# 4 Empirical Approach

We are interested in estimating the causal effect of stricter monitoring on employment outcomes of unemployed job seekers. Identification relies on the quasi-random allocation of job seekers' activity reports to caseworkers in central offices. Caseworkers differ systematically in their strictness when evaluating these reports, and they have no direct contact with the job seekers assigned to them. This institutional feature eliminates the risk of confounding caseworker behaviors such as differences in counseling intensity, meeting frequency, or assignment to ALMPs.

## 4.1 Caseworker Strictness

For each unemployed job seeker $i$, we estimate the strictness of the central office caseworker $j$ who reviews the job seeker's first report. Caseworker strictness, $CS_{j(i)}$, is defined as the leave-one-out share of rejected first activity reports among all other first reports assessed by the same caseworker during the observation period:

$$CS_{j(i)} = \frac{\sum_{k \neq i} D_{jk}}{n_j - 1}, \tag{1}$$

with $D_{jk} = 1$ if caseworker $j$ does not approve the first activity report of job seeker $k$, and $D_{jk} = 0$ otherwise. $n_j$ is the total number of first reports reviewed by caseworker $j$.[21]

On average, caseworkers in our sample rejects 7% of first activity reports, but there is substantial heterogeneity in strictness. The 10th and 90th percentiles of the strictness distribution are 0.020 and 0.130, respectively. Figure 3 illustrates this variation, showing that strictness strongly affects the probability that the first activity report is not approved. Caseworker strictness is persistent across reports. Strictness measures based on the first and the second reports are highly correlated. Further, around 75% of the job seekers in our sample are assigned the same caseworker in the first and the second report. Figure A4 in the Appendix shows that caseworkers with above-median strictness exhibit higher rejection rates across all levels of reported job applications.

## 4.2 Empirical Specification

Our main specification is a reduced-form regression of job seeker's employment outcome in period $t$ after entry into unemployment, $Y_{it}$, on caseworker's strictness $CS_{j(i)}$

$$Y_{it} = \beta_1 CS_{j(i)} + \gamma_{q(i)} + \varepsilon_{it}. \tag{2}$$

$\beta_1$ captures the reduced-form effect of caseworker strictness. The term $\gamma_{q(i)}$ denotes central officeâbyâmonth fixed effects, ensuring that we exploit variation in caseworker strictness within a given office and month. This allows us to isolate random variation stemming from the common report lists used at the central offices. In robustness analyses, we also explore adjusting for finer fixed-effects using interacted calendar-day and year fixed-effects for each central office, which does not affect our results. For parsimony, we therefore report estimates using the central officeâbyâmonth fixed effects as our main specification. Given that the allocation of reports to caseworkers is quasi-random conditional on time and central office, standard errors are clustered at the office-by-month level.[22]

We additionally estimate short-run effects of receiving a rejected first activity report. Following the standard judge design approach (Chyn et al., 2025), we use caseworker strictness $CS_{j(i)}$ as an instrument for a rejected activity

[21] We have explored alternatives to this strictness measure. When the model includes covariates, Chyn et al. (2025) shows that it is preferable to use a residualized strictness measure when estimating IV effects. When we calculate IV estimates in Section 5.4, we therefore show that using residualized caseworker strictness (adjusting for calendar month and office fixed effects) does not change the results.

[22] Alternative clustering levels, such as at the office or caseworker level, produce similar results. If anything, clustering at the office-by-month level seems to be the most conservative approach.

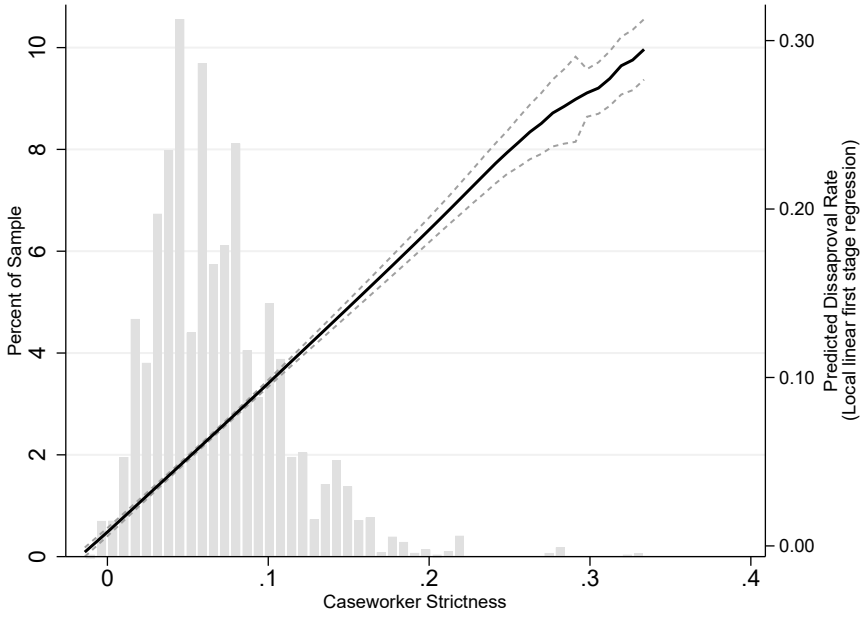



*Figure 3.* Distribution of caseworker strictness and the first stage

*Note*: Statistics for caseworker strictness, defined as the leave-one-out share of activity reports marked not approved for each caseworker, in the analysis sample of first reports described in Section 3. The sample consists of 109,866 cases and 390 caseworkers. The black line plots a local linear regression of the rejection rate on the strictness measure, based on an Epanechnikov kernel (with bandwidth 0.1). Strictness is demeaned using office-month fixed effects, with the global mean added back for interpretability.

report. The first- and second-stage equations are:

$$D_i = \theta_1 CS_{j(i)} + \delta_{q(i)} + \zeta_{it}. \quad (3)$$

$$Y_{it} = \alpha_1 D_i + \eta_{q(i)} + \nu_{it}. \quad (4)$$

Similar to the reduced-form approach, we include central officeâbyâmonth fixed effects $\delta_{q(i)}$ and $\eta_{q(i)}$ and cluster standard errors at the office-by-month level. For this approach to be valid, the caseworker strictness must affect employment outcomes only through its effect on the approval of the first activity report. This condition holds in the short run. However, once the caseworker evaluates the second activity report, the exclusion restriction is violated as it is common that the same caseworker also reviews the second and the third report. For example, job seekers who are facing a stricter caseworker have a higher likelihood that their second activity report is not approved compared to job seekers facing a less strict caseworker. Therefore, we focus on short-

run effects, measured as the probability of leaving unemployment between the first report and the date when the job seeker is expected to submit the second activity report (approximately one month).

## 4.3 Independence: Random Assignment

We next show that the assignment of first activity reports to caseworkers is independent of predetermined job seeker characteristics. To do so, we regress job seeker characteristics while controlling for central officeâbyâmonth fixed effects on (i) a dummy for being unemployed 90 days after entering unemployment, (ii) a dummy for receiving a "not approved" first activity report, and (iii) the strictness of the assigned caseworker. As expected, Table A1 shows that the probabilities of exiting unemployment and receiving a "not approved" report are significantly associated with nearly all characteristics (Columns 1–2). In contrast, and consistent with the quasi-random allocation mechanism based on common report lists, caseworker strictness is largely unrelated to predetermined characteristics–only the coefficients for being married and having less than high school education are statistically significant at the 10% and 5 % level (Column 3). We also perform joint F-tests, regressing the three outcomes on the full set of characteristics, excluding predicted reemployment probability. These tests confirm that caseworker strictness is unrelated to observable job seeker characteristics.

Figure 4 presents regression coefficients of employment states measured at different moments before (and after) entry into unemployment. None of the 23 displayed coefficients related to lagged employment outcomes (observed up to 720 days before the current unemployment spell) is statistically significant at the 5% level. The figure also suggests that caseworker strictness is causally related to job finding, with higher job-finding rates after the first report is assessed.

As further support of randomization, Columns (1)–(2) of Table A2 show that caseworker strictness is uncorrelated with the content of the first activity report, both in terms of the number of reported job applications and the total number of activities. Column (3) shows no association between strictness and the time elapsed between unemployment entry and submission of the first report. However, Column (4) reveals a small positive correlation between caseworker strictness and the time between report submission and evaluation. Moving from the 10th to the 90th percentile of the strictness distribution increases evaluation time by 0.18 days. Given that the average evaluation time is three days, this minor delay is unlikely to meaningfully affect labor market outcomes.

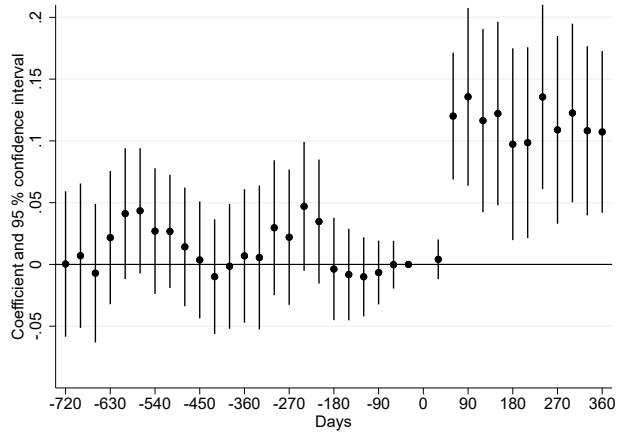



*Figure 4.* Effects of caseworker strictness and the probability of being employed

*Notes:* Regression coefficients from regressions of employment status at t days relative to the start of the unemployment spell on caseworker strictness. All jobseekers are employed at t = -30. Point estimates and 95% confidence intervals shown on vertical axis. Regressions include month-agency fixed effects with standard errors clustered at the month-agency level. N = 109,866; 390 caseworkers.

## 4.4 Caseworker Strictness and Monitoring Decisions

Figure 3 shows that caseworker strictness is positively associated with the probability that the first activity report is rejected. This finding is confirmed by regressing the binary indicator for a "rejected" first activity report on caseworker strictness, while controlling for office-by-month fixed effects (Column 1, Table 1). The coefficient of 0.842 implies that moving from the 10th percentile (0.020) to the 90th percentile (0.130) of the strictness distribution increases the probability of receiving a "rejected" report by 9.3 percentage points—more than doubling the mean probability of 6.9 %.

As described in Section 2.4, a not-approved report represents the first step in the monitoring chain but does not automatically result in a benefit sanction. Columns 2–5 of Table 1 show impacts of caseworker strictness on subsequent monitoring steps. Since it may take time for a rejected report to translate into a registered violation and a benefit sanction, we examine monitoring-chain actions occurring within 120 days after entry into unemployment. We do this to capture the full reduced-form effect on the monitoring chain, although this complicates the comparison of effect estimates across columns since it includes consequences of both the first and subsequent reports.[23]

The probability of a registered job- search-related violation is positively associated with caseworker strictness (Column 2), whereas the probability of

[23]Since time elapses between steps in the monitoring chain, some violations captured in Column 2 toward the end of the window may not yet appear in Columns 4â5.

**Table 1.** *Effects of caseworker strictness on the monitoring chain (rejected reports, registered violations and sanctions)*

| Panel A: Monitoring chain | (1) Not Ok First report | (2) Violation Job search | (3) Violation Other | (4) Notification Sent to UI | (5) Any Sanction |
|---|---|---|---|---|---|
| CW strictness | 0.842*** | 0.209*** | -0.015 | 0.143*** | 0.126*** |
| | (0.028) | (0.016) | (0.024) | (0.026) | (0.021) |
| Mean depvar | 0.069 | 0.022 | 0.122 | 0.124 | 0.083 |
| N | 109,866 | 109,866 | 109,866 | 109,866 | 109,866 |
| Panel B: Sanctions Ladder 1 | (6) Zero-day Sanction 120 days | (7) Zero-day Sanction 240 days | (8) Higher-order Sanction 120 days | (9) Higher-order Sanction 240 days | |
| CW strictness | 0.110*** | 0.076*** | 0.022*** | 0.013 | |
| | (0.020) | (0.023) | (0.008) | (0.011) | |
| Mean depvar | 0.072 | 0.113 | 0.011 | 0.024 | |
| N | 109,866 | 109,866 | 109,866 | 109,866 | |

*Notes:* Standard errors are clustered at the month-agency level and reported in parentheses. ***, **, * indicate significance at 1%, 5% and 10% respectively. We control for month-office fixed effects. Panel A: dependent variables in columns 2–5 are indicators for at least one event within 120 days from the start of the unemployment spell. Column 5 includes sanctions due to both job search violations and other violations. Panel B: columns 6–7 include sanctions at the first step of the ladder, which is a zero day sanction. Columns 8–9 include higher-order sanctions, i.e. 1 day, 5 days, 10 days and suspension. In both cases, within 120/240 days the start of the unemployment spell. Columns (2)–(3) report registered and notified violations. Column (2) shows reason 13 (insufficient job search). Column (3) includes reasons 8â12: failure to set up an activity report, failure to submit an activity report, missed contact or visit, and failure to apply for a referred job. Columns (4) and (5) report notifications sent to the UI funds and the subsequent registered sanctions from the same sanction ladder, covering all reasons 8â13.

violations unrelated to job search remains unaffected (Column 3). Comparing the estimates in Columns 1 and 2, while keeping the caveats mentioned above in mind, suggests that roughly one fourth of rejected reports lead to a registered violation. The likelihood that a notification is sent to the UI fund and that this results in sanction also increases with caseworker strictness (Column 4–5). Moving from the 10th percentile to the 90th percentile of the strictness distribution increases this probability by 1.4 percentage points, corresponding to an increase of about 17% relative to the mean.

All these estimates refer to any type of sanction, including zero-day sanctions. Panel B of Table 1 distinguishes between zero-day sanctions and those involving an actual suspension of benefits. We find that the likelihood of both types of sanction increases with caseworker strictness. This holds both when we look within 120 and 240 days, although the estimated coefficients decline

somewhat in magnitude and become marginally insignificant for sanctions involving benefit suspensions within 240 days.

## 4.5 Exclusion Restriction and Monotonicity

To identify the effect of caseworker strictness and a rejected first report in the short-run, the caseworkers should not affect the job seeker in any other way. Our institutional setting with central caseworkers supports this exclusion restriction assumption.

We consider both the reduced-form effect of being assigned to a stricter caseworker and the IV estimate of the short-run effect of receiving a not-approved first activity report. The IV specification relies on the assumption that our instrument—caseworker strictness—has a monotonic effect on the probability of approval. Following Frandsen et al. (2023), we jointly test the pairwise monotonicity and strict exclusion restrictions. Pairwise monotonicity—the assumption that caseworkers share the same ranking of job seekers—implies that a job seeker whose report is not approved by a more lenient caseworker would also not be approved by any stricter caseworker. We reject the joint null that both pairwise monotonicity and strict exclusion hold.

However, the IV approach still identifies a causal local average treatment effect under the weaker assumption of *average monotonocity*. This assumption requires that, for any given job seeker, the average strictness of caseworkers who would reject a report is weakly greater than that of those who would approve it (Chyn et al., 2025). This assumption has testable implications. First, the first-stage effect—the effect of caseworker strictness on the probability of report rejection—should be non-negative across all observable subpopulations (Autor et al., 2019). Table A3 supports this: we find positive first-stage effects across all subgroups constructed based on observed characteristics in our data.[24] Second, caseworkers who are stricter toward one group—for example, younger individuals—should also be stricter toward the complementary group—in this case, older individuals. Table A4 shows that this holds. Recalculating the strictness measures within each subgroup and correlating them with rejection rates in the complementary group yields consistently positive and statistically significant correlations.

[24]This includes men and women, married and unmarried individuals, younger and older job seekers, those with and without college education, individuals at high and low risk of long-term unemployment, and across regions with low and high unemployment rates.

# 5 Effects of Having a Stricter Caseworker

## 5.1 Employment Probability

Table 2 reports reduced-form effects of caseworker strictness on the probability of leaving unemployment for a part- or full-time job within various time horizons. As expected, we find no effect during the first 30 days in unemployment, which is in line with the institutional set-up and the fact that unemployed job seekers hand in their first activity report after an average of 33 days of unemployment. After 60 days (approximately 30 days after the first report), we find a significantly positive impact on the probability of leaving unemployment. This positive employment effect is the largest at the beginning and then drops somewhat but remains sizable and significant for at least the first 360 days. Quantitatively, moving from the 10th percentile to the 90th percentile of the distribution of caseworker strictness increases the job-finding rate after 120 days by around 1.1 percentage points, corresponding to roughly a 3% increase relative to the mean.[25] These results show that stricter monitoring substantially shortens unemployment durations, and the effect even persists beyond the initial four months, when monitoring is transferred to local offices and no longer depends on the central caseworkerâs strictness.

**Table 2.** *Effects of caseworker strictness on the probability of finding a job*

| Outcome: finding a job within X days | | | | | | |
|---|---|---|---|---|---|---|
| | | After assessment of the first report | | | | |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| | 30 days | 60 days | 90 days | 120 days | 180 days | 360 days |
| CW strictness | 0.008 | 0.114*** | 0.114*** | 0.103*** | 0.076** | 0.064* |
| | (0.009) | (0.027) | (0.035) | (0.038) | (0.038) | (0.033) |
| Mean depvar | 0.011 | 0.131 | 0.269 | 0.380 | 0.528 | 0.704 |
| N | 109,866 | 109,866 | 109,866 | 109,866 | 109,866 | 109,866 |

*Notes:* Outcome is defined as leaving unemployment for a part- or full time job. The models also includes month-agency fixed effects. Standard errors are clustered at the month-agency level and reported in parentheses. ***, **, * indicate significance at 1%, 5% and 10% respectively.

The effects are sizable. There are several potential explanations to this. First, because there is no explicit search requirements and rejections are rare, a rejection likely represents a substantial monitoring shock for job seekers. This is reflected in large effects on job-search intensity (see below). Second, Table A5 reports results from a complier analysis, which show that initial search intensity before the assessment of the first report is low among compliers,

[25] Moving from the 10th to the 90th percentile increases strictness by 0.11 (0.130-0.020). If multiplied with the point estimate 0.104 this gives an effect of 1.1 percentage points.

and monitoring and intensified search may be particularly important for such inactive job seekers.[26]

These effects on exits from unemployment do not take recurring unemployment spells into account, and it could be the case that some individuals who leave unemployment sooner due to stricter caseworkers return faster to unemployment. This is not the case. When examining employment probabilities over time, taking both the current and recurring unemployment spells into account, we observe a similar pattern with positive effects starting from 60 days (Table A7 and Figure 4).

Table 2 relies on job-finding data from the Swedish PES. A potential concern is that these records may not perfectly measure employment as they partly depend on information reported by the job seekers. For instance, some job seekers might report to the PES that they have found a job even if they have not, in order to avoid tighter monitoring or sanctions. This seems unlikely, however, since doing so would imply losing UI benefits. In Table 5, we examine short-run effects on labor earnings, during the calendar when the unemployment spell starts (year 0), one year after (year 1), and so on.[27] The results show a positive earnings effect in the first year after entry to unemployment of similar magnitude as the job-finding estimates based on the PES data. Moving from the 10th to the 90th percentile of the strictness distribution increases job finding within 180 and 360 days by 1.5% and 1.0%, respectively, and raises labor earnings after one year by 1.2%. We return to the longer-run earnings effects in Section 5.5.

## 5.2 Job-search intensity

Table 3 reports effects of caseworker strictness on job-search intensity, using the activity-report data. Report 1 is submitted before job seekers are assigned to a caseworker; accordingly, we find no significant effect for this report. Subsequent reports are submitted after the assessment of the first report: Report 2 roughly corresponds to days 31â60, Report 3 to days 61â90, and so on. Starting with the second report, we find sizable and statistically significant increases in job-search intensity. The effect is strongest during the first four reports and then declines somewhat, coinciding with the point at which report assessments are transferred from central to local offices. These patterns indicate that the positive effects on exits from unemployment are accompa-

[26]Compliers report searching for, on average, 1.5 jobs (3.4 activities), compared to the sample average of 6 jobs (9 activities). Along other dimensions, compliers are similar to the full sample, including in terms of gender, age, unemployment history, unemployment risk, and the growth rate of their previous industry. Complier means are constructed following Goldsmith-Pinkham et al. (2025); details are provided in the note to Table A5.

[27]We have outcome data up to 2018, which means that we lose some observations for each follow-up year: coverage decreases to approximately 77% in the second year and 52% in the third year.

nied by an increased number of job applications, both immediately after the assessment of the first report and in a sustained manner thereafter.

The effect for Report 2 implies that moving from the 10th to the 90th percentile in the distribution of caseworker strictness increases the monthly number of applications by 0.36, or 5.4%. The cumulative increase over the first four reports (first 120 days) corresponds to an increase of around 1.2 applications (4.6%). This increase can be related to a rise in the job-finding rate after 120 days of about 3%.

**Table 3.** *Effects of caseworker strictness on the monthly number of job applications*

| Outcome: Number of job applications report i | | | | | | |
|---|---|---|---|---|---|---|
| | | After assessment of the first report | | | | |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| | Report 1 | Report 2 | Report 3 | Report 4 | Report 5 | Report 6 |
| CW strictness | 0.013 | 3.242*** | 3.462*** | 4.124*** | 2.193** | 2.491* |
| | (0.593) | (0.656) | (0.788) | (0.976) | (1.106) | (1.277) |
| Mean depvar | 6.022 | 6.625 | 6.679 | 6.778 | 6.846 | 6.939 |
| N | 109,866 | 83,123 | 62,422 | 44,484 | 35,365 | 28,220 |

*Notes:* Outcomes are monthly number of job applications based on data from the activity report. The models also includes month-agency fixed effects. Standard errors are clustered at the month-agency level and reported in parentheses. ***, **, * indicate significance at 1%, 5% and 10% respectively.

One potential concern is that we observe job-search intensity only for individuals who remain unemployed, and the probability of remaining unemployed depends on caseworker strictness. This will lead to dynamic selection. Table A8 reports results for a balanced panel of job seekers who submitted the first two reports. This gives a similar pattern as in Table 3, with a positive effect of strictness on the number of job applications in the second but not in the first.

## 5.3 Labor Market Programs and Meetings

Besides affecting job-search intensity, a not-approved report and the ensuing monitoring chain may also influence participation in ALMPs and meetings with caseworkers. For instance, local office caseworkers often contact the job seeker when a report is not approved, which may lead to follow-up meetings or enrollment in ALMPs. Local caseworkers may also interpret a not-approved report as a signal that the job seeker needs additional support.

To examine these channels, we analyze the relationship between caseworker strictness and participation in three most common Swedish ALMPs (work practice, vocational training, and preparatory training) as well as the number

of meetings with caseworkers at the local offices.[28] Note that the job seekers have no meetings with the central caseworkers. Table 4 reveals no significant correlation between caseworker strictness and the probability of participating in any of of the three major ALMPs within 120 days after entry into unemployment (Columns 1–4). The same holds when we consider other time spans.

**Table 4.** *Caseworker strictness, labor market programs and meetings*

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | Labor market programs | | | | Meetings | |
| | Any Program | Work Practice | Vocational Training | Prepatory Training | Face to Face Meetings | Phone Meetings |
| CW strictness | -0.002 | -0.004 | 0.002 | -0.000 | 0.163* | 1.231*** |
| | (0.011) | (0.005) | (0.005) | (0.006) | (0.087) | (0.096) |
| Mean depvar | 0.015 | 0.004 | 0.005 | 0.005 | 1.917 | 1.066 |
| N | 109,866 | 109,866 | 109,866 | 109,866 | 109,866 | 109,866 |

*Notes:* Outcomes are based on participation in labor market programs from the so-called search categories (skat koder) in the PES dataset sokatper, and data on meetings from the PES dataset Histaktso. The models also includes month-agency fixed effects. Standard errors are clustered at the month-agency level and reported in parentheses. ***, **, * indicate significance at 1%, 5% and 10% respectively.

In contrast, we find a positive relationship between the strictness of the caseworker and the number of face-to-face and phone meetings with the local caseworker (Columns 5–6). Especially, the effect on having phone meetings is sizable. The effect of strictness on rejected reports in Table 1 is 0.84. Comparing this with the estimate for phone meetings (1.231) implies that each rejected report generates roughly 1.5 additional phone contacts. This is in line with the institutional set-up as the job-seekers typically discuss a not-approved report with their local caseworker, and this usually happens by phone. Thus, these additional contacts are a natural component of the intensified monitoring process carried out by the Swedish PES, which may contribute to the positive effects on job finding.

## 5.4 Causal Effects of Rejected Activity Reports

Since stricter caseworkers affect the entire monitoring chain and trigger additional meetings, we cannot use caseworker strictness as an instrument for a benefit sanction or for the impact of a rejection on longer run outcomes.

[28] Work practice involves participation in workplace-based activities in both the private and public sectors (Forslund et al., 2013). Vocational training consists of training aimed at improving the skills of unemployed workers, focusing on occupations where skills are (expected to be) in short supply (see, e.g., van den Berg and VikstrÃ¶m, 2022 for a recent evaluation). Preparatory training includes educational and other activities designed to prepare unemployed job seekers for participation in other labor market programs.

However, we can instrument for a not-approved activity report if we focus on short-run outcomes between the first and the second report. The focus on short-run outcomes reflects the fact that the same caseworker typically assesses the first four reports. Consequently, any effects observed after the second report may stem from stricter assessments of both the first and second reports, which cannot be disentangled.

We analyze job finding during the second reporting period, defined as the calendar month immediately following the first reporting period, which ends on the last day of the preceding month. The results in Table A9 indicate a strong positive effect on both job finding and job-search intensity of a not-approved first activity report. The probability of finding a job between the first and the second job increases by 11 percentage points, and the number of job applications increases by 3.9 in the same interval. Under the assumption of average monotonicity, these estimates capture a weighted LATE, reflecting effects for job seekers who would have received a different evaluation of the first activity report if they had been assigned to a stricter caseworker (Chyn et al., 2025). Several factors may contribute to this sizable effect. First, there were no explicit search requirements and non-approvals are rare, implying that a rejection likely constitutes a substantial monitoring shock. Second, initial search intensity prior to the first report assessment is low among job seekers whose reports are not approved, suggesting that a "monitoring push" may be particularly important for these relatively inactive individuals (see Table A5).

## 5.5 Earnings, Job-Quality and Sectoral Mobility

Having a stricter caseworker evaluate the activity reports leads to higher job-finding rates. We have seen that this is reflected in higher job-search intensity, but the increased job finding may also stem from lower wages or shifts in the types of jobs and sectors job seekers enter. If so, stricter caseworkers may come at a cost. We therefore use both wage data and long-run earnings measures to examine potential effects on job quality.

We use the earnings data discussed in Section 5.1, which capture labor earnings $t$ years after entry into unemployment. Because of data availability, the sample size declines as $t$ increases. Note that these earnings effects may reflect both employment effects, and job-quality effects through changes in employment duration and wages. To isolate the effect on wages, we additionally explore data on the wage of the first job after exit from unemployment from the wage statistics. As mentioned above, these wages are recorded as monthly full-time equivalents and are not derived from annual earnings and hours worked. This dataset covers a subsample of all employees, which explains why the sample size is lower for this wage outcome. Since caseworker strictness affect the transition to employment, the corresponding results are based on a selected subsample and have to be interpreted with caution.

The results are in Table 5. Regressing yearly earnings on caseworker strictness yields positive point estimates for all three years, and the coefficient on earnings in the first year is statistically significant at 10 %. The estimate for the wage of the first job is also positive, but statistically insignificant. The positive effects on earnings in the first year after entry into unemployment, along with the positive but insignificant effects during years two and three, and the lack of negative effects on the wage rate, suggests that intensified monitoring does not force job seekers to sacrifice job quality for faster exits from unemployment.

**Table 5.** *Effects of caseworker strictness on labor earnings and wages*

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| | Yearly labor earnings: | | | | |
| | Unempl. Entry Year | Unempl. Entry Year+1 | Unempl. Entry Year+2 | Unempl. Entry Year+3 | Wage first job |
| CW strictness | 16,876 | 18,347* | 17,264 | 13,670 | 989 |
| | (11,655) | (10,333) | (13,071) | (16,025) | (936) |
| Mean depvar | 176,137 | 168,577 | 200,981 | 212,837 | 24,326 |
| N | 109,576 | 109,035 | 83,682 | 56,164 | 44,098 |

*Notes:* Outcomes are yearly labor earnings in SEK based on tax records, and the wage rate of the first job based wage statistics. The models also includes month-agency fixed effects. Standard errors are clustered at the month-agency level and reported in parentheses. ***, **, * indicate significance at 1%, 5% and 10% respectively.

## 5.6 Robustness analyses

Table A11 reports results from several robustness analyses. Our main specification includes month-by-office fixed-effects, assuming that, conditional on the month in which the unemployed job seeker submits their activity report to a central office, the allocation to a caseworker within that central office is quasi-random. Since we observe the exact day at which the unemployed hand in their activity report, we can conduct robustness analyses using calendar day times office fixed-effects. While this increases the number of fixed effects from 276 in the baseline specification to 3,857, this specification is based on the weaker assumption that conditional on the day the unemployed job seeker submits the activity, the allocation to a caseworker is quasi-random. Adding these more detailed fixed effects lead to similar results as our main specification (Panel A of Table A11).

The main specification does not control for observed characteristics of the job seekers. Given our quasi-random research design, the inclusion of background characteristics should not affect the results. This is confirmed by the estimates in Panel B, which adds commonly used observed characteristics. One sample criteria is to only include caseworkers with at least 75 cases, but relaxing this criteria to at least 50 cases, gives very similar results (Panel C).

In Section 2.3, we noted that some job seekers receive vacancy proposals from the central office caseworkers before submitting their first activity report. These proposals are voluntary and sent without prior contact with the job seeker. In the main analyses, we exclude these job seekers, but the estimates in Panel D show that our results are virtually unaffected by this restriction.

For the IV approach, we residualize examiner strictness with respect to month-office fixed effects prior to constructing the instrument. The results are reported in Table A10 and are consistent with our baseline estimates, confirming that our findings are not sensitive to the specific way in which month-office variation is absorbed.

# 6 Heterogeneity by individuals and markets

It is interesting to understand for which types of job seekers and local labor markets stricter monitoring improves employment outcomes. This section, therefore, examines heterogeneity by the individual risk of becoming long-term unemployed and by local labor market situation (growing and declining industries).

## 6.1 Individual Risk of Long-term Unemployment

We begin by constructing a measure of the risk of long-term unemployment in the data. Specifically, we predict the probability that a jobseeker finds a job within 360 days using standard machine learning methods and data on individual characteristics and labor market history.[29] To avoid overfitting, we train the model on observations excluded during sample selection and evaluate its performance on our final analysis sample.[30] The final specification uses Gradient Boosted Decision Trees, and regarding performance it achieves a ROCâAUC of 0.7.[31] We define high unemployment risk as below-median predicted probability of exiting unemployment within 360 days.

[29]We estimate several candidate models (Lasso, ridge regression, random forest and Gradient Boosted Decision Trees) and select the specification with the best out-of-sample predictive performance. The prediction model uses information on demographic characteristics: age, gender, disability, marriage, level of education, region of birth, and employment history: employed t-180 days,employed t-360 days, employed t-540 days, employed t-720 days, number of days of UE last 5 years, number of programs last 5 years, and number of UE spells last 5 years, previous income, and search occupation

[30]Training data include observation dropped due to case load restriction on the caseworker level, and those who received some vacancy proposal.

[31]The ROCâAUC (area under the receiver operating characteristic curve) measures the modelâs ability to rank individuals according to their probability of exiting unemployment. A ROCâAUC of 0.7 implies that the model ranks a randomly selected individual who exits unemployment within 360 days above a randomly selected individual who does not in 70 percent of cases. A random classifier would obtain a ROC-AUC of 0.5.

Table 6 shows the estimated effects of our strictness measure on the probability to find a job.[32] Interestingly, we see different patterns in the beginning and later in the unemployment spell, though the differences between the two groups are not statistically significant. Between 60 and 120 days after entering unemployment, the employment effect is larger for workers with a relatively low risk of long-term unemployment. Beyond 120 days, this pattern reverses, with a larger employment effect for job seekers with a high risk of long-term unemployment. After 360 days, the estimated employment effect is insignificant for the low-risk group, but remains sizeable (job finding up 1.3 percentage point when moving from the 10th to the 90th percentile of caseworker strictness) and significant for the high-risk group.

We have re-estimated the "first-stage" effects on the monitoring chain (Table A12), and programs and meetings (Table A13) for these two groups and find similar estimates, indicating that these differences are not due to differences in the first-stage effects nor in the number of telephone-contacts after a rejected report.

The results indicate that stricter monitoring has a greater impact on job seekers with a low long-term unemployment risk in the short run, while the effect appears larger for high-risk job seekers at longer horizons. This pattern suggests that intensified monitoring and stricter evaluation of active job search are effective tools for rapidly moving individuals with strong labor market attachment back into employment. For job seekers with weaker labor market attachment, intensified monitoring does not generate the same short-run increase in job finding, likely reflecting greater barriers to immediate re-employment. However, for this group, stricter monitoringâand the associated increase in job-search intensityâappears to produce more persistent effects on job finding.

## 6.2 Local Labor Market Situation

We next examine whether the effect of stricter monitoring depends on market conditions, distinguishing between job seekers from declining and growing industries. On the one hand, stricter monitoring may have a larger impact for individuals from growing industries, where more vacancies allow caseworkers to push job seekers toward employment. On the other hand, job seekers from growing industries may find work quickly regardless of monitoring intensity, implying that the monitoring push could be less effective for such industries.

To study this, we use matched employer-employee data and define industry growth as the change in employment between $t-5$ and $t+5$, where $t$ is the year in which the first report is evaluated. Industries are classified as low or high growth depending on whether their growth rate is below or above the

[32] For both groups, we do not see any impact on employment rates after 30 days. Again, this before the assessment of the first report, and in line with the results based on the full sample.

**Table 6.** *Heterogeneous effects by individual risk of long-term unemployment: effects on job-finding*

| Outcome: finding a job within X days | | | | | | |
|---|---|---|---|---|---|---|
| | | After assessment of the first report | | | | |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| | 30 days | 60 days | 90 days | 120 days | 180 days | 360 days |
| Panel a: Low risk | | | | | | |
| CW strictness | 0.015 | 0.152*** | 0.170*** | 0.131** | 0.046 | 0.029 |
| | (0.013) | (0.038) | (0.049) | (0.053) | (0.051) | (0.041) |
| Mean depvar | 0.014 | 0.168 | 0.339 | 0.473 | 0.647 | 0.821 |
| N | 54,933 | 54,933 | 54,933 | 54,933 | 54,933 | 54,933 |
| Panel b: High risk | | | | | | |
| CW strictness | 0.001 | 0.081*** | 0.071 | 0.087* | 0.118** | 0.114** |
| | (0.009) | (0.031) | (0.044) | (0.049) | (0.051) | (0.050) |
| Mean depvar | 0.008 | 0.094 | 0.199 | 0.287 | 0.410 | 0.588 |
| N | 54,933 | 54,933 | 54,933 | 54,933 | 54,933 | 54,933 |
| P-value diff | 0.293 | 0.133 | 0.119 | 0.525 | 0.288 | 0.199 |

Notes: Standard errors are clustered at the month-agency level and reported in parentheses. ***, **, * indicate significance at 1%, 5% and 10% respectively. We control for month-agency fixed effects. High unemployment risk is defined as below median probability of exiting unemployment within 360 days. Low unemployment risk is defined as above median probability of exiting unemployment within 360 days. Prediction of exiting unemployment within 360 days is based on Gradient Boosted Decision Trees, using the following input variables: age, gender, disability, marriage, level of education, region of birth, and employment history: employed t-180 days,employed t-360 days,employed t-540 days, employed t-720 days, number of days of UE last 5 years, number of programs last 5 years, and number of UE spells last 5 years, previous income, and search occupation. P-value diff shows p-values for for tests of differences in coefficients between the high- and low-risk individuals.

median.[33] We do this at the 2-digit industry level but obtain similar results at other levels. Job seekers are assigned to industries based on their last job prior to unemployment. The sample is therefore divided by origin industry, although some individuals may already have shifted their job search toward another industry when the first report is evaluated.

The results are reported in Table 7. For high-growth industries, stricter monitoring substantially increases job finding in the short run, but the effects dissipate after around 120 days. In contrast, for low-growth industries, the initial effects are smaller but persist over a longer horizon, remaining statistically significant even after 360 days.[34] These findings are robust to alternative

[33]The majority of the unemployed workers in our sample have been working in industries with a positive growth rate, but around 20% face a decline in job opportunities in their previous industry.

[34]Table A14 and Table A15 show that the effects on the monitoring chain, and programs and meetings are similar for the two groups, so that these factors cannot explain the industry growth heterogeneity.

**Table 7.** *Heterogeneous effects for low- and high growth industries: effects on job-finding*

| Outcome: finding a job within X days | | | | | | |
|---|---|---|---|---|---|---|
| | | After assessment of the first report | | | | |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| | 30 days | 60 days | 90 days | 120 days | 180 days | 360 days |
| Panel a: Low Growth Industries | | | | | | |
| CW strictness | 0.012 | 0.077** | 0.115** | 0.135** | 0.114** | 0.109** |
| | (0.012) | (0.036) | (0.048) | (0.053) | (0.054) | (0.046) |
| Mean depvar | 0.011 | 0.132 | 0.266 | 0.377 | 0.524 | 0.705 |
| N | 53,021 | 53,021 | 53,021 | 53,021 | 53,021 | 53,021 |
| Panel b: High Growth Industries | | | | | | |
| CW strictness | 0.003 | 0.151*** | 0.129*** | 0.092* | 0.054 | 0.036 |
| | (0.010) | (0.037) | (0.049) | (0.051) | (0.053) | (0.046) |
| Mean depvar | 0.011 | 0.132 | 0.275 | 0.388 | 0.537 | 0.709 |
| N | 55,515 | 55,515 | 55,515 | 55,515 | 55,515 | 55,515 |
| P-value diff | 0.556 | 0.130 | 0.829 | 0.538 | 0.410 | 0.238 |

*Notes:* Standard errors are clustered at the month-agency level and reported in parentheses. ***, **, * indicate significance at 1%, 5% and 10% respectively. We control for month-agency fixed effects. Industry growth rate is defined for 2-digit industry codes, and measures employment change between year t-5 and t+5. P-value diff shows p-values for for tests of differences in coefficients between the high- and low-risk individuals.

measure of industry growth (Table A16). Overall, the results indicate that the employment effects of stricter monitoring are more persistent for job seekers originating from low-growth industries, though the differences are not statistically significant. Stricter enforcement of job-search rules may therefore help reduce prolonged unemployment among workers from low-growth industries. This add to the evidence from Humlum et al. (2025), which shows job seekers exposed to offshoring benefit more from classroom training.

# 7 Conclusions

This paper provides new evidence on how stricter enforcement of job-search requirements affects employment outcomes among unemployed job seekers. We show that stricter caseworker enforcement substantially reduces unemployment durations and increases employment rates. This is because stricter caseworkers initiate a monitoring chain characterized by more rejected activity reports, notifications of violations, and benefit sanctions. This process raises job-search intensity both immediately and persistently. Moreover, because rejected reports trigger additional contact between job seekers and local caseworkers, we also observe more phone and face-to-face meetings, which may contribute to positive employment effects.

We also find interesting heterogeneity across groups and labor markets. Job seekers with strong re-employment prospects benefit more in the short run, while those with weaker labor market attachment experience more persistent gains over time. Importantly, stricter monitoring is particularly effective for job seekers who have been employed in sectors with declining job opportunities. These findings suggest that stricter monitoring plays a crucial role in preventing prolonged unemployment spells, especially for more disadvantaged groups and under worse local labor market conditions.

From a policy perspective, our results imply that the enforcement of job-search requirements can be an effective tool for improving employment outcomes. However, the heterogeneity in effects indicates that a âone-size-fits-allâ approach may not be optimal. Instead, monitoring policies could be designed to balance strict enforcement with targeted support, focusing on individuals and regions with a higher risk of long-term unemployment. Strengthening monitoring in such contexts may yield the greatest returns in terms of reducing unemployment durations and preventing labor market detachment. At the same time, policymakers should be attentive to potential costs associated with stricter enforcement, including administrative burden and the risk of disproportionate hardship for vulnerable job seekers.

Overall, our findings suggest that well-designed monitoring systems, which combine credible enforcement with supportive follow-up, can improve labor market outcomes and enhance the effectiveness of unemployment insurance systems.

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# Appendix A: Additional tables and figures

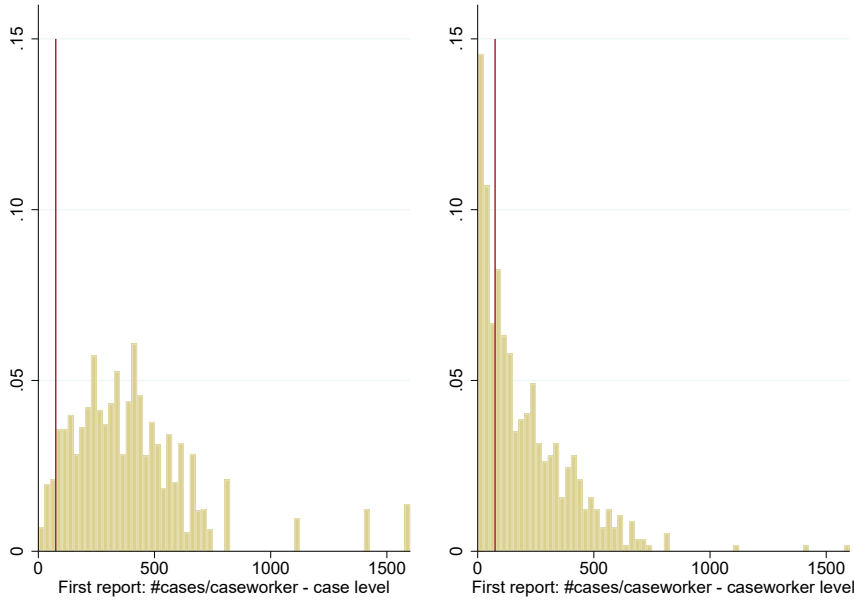
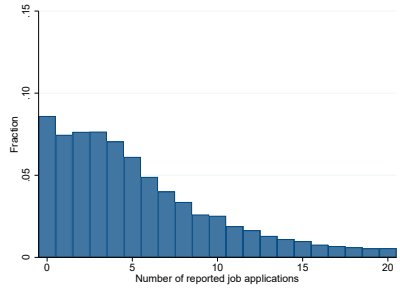
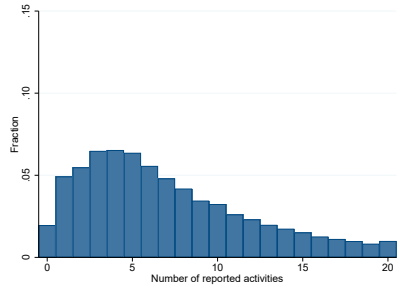



*Figure A1.* Number of checked reports by caseworker at the centralized office

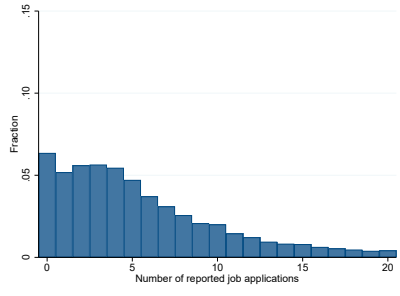
*Notes*: Distribution of number of cases on case level and caseworker level. Number of cases corresponds to the number of unemployed handing in a first activity report during our observation period. Vertical line indicates the threshold of 75 cases per caseworker. In the full sample there are 115,217 cases and 570 caseworkers. After restricting to caseworkers with at least 75 cases, there are 109,866 cases and 390 caseworkers.



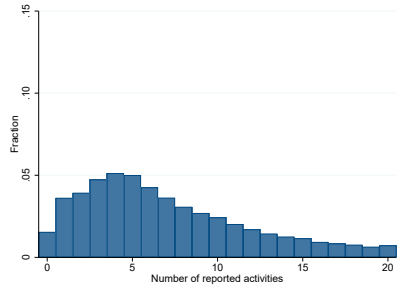

Panel A: Job Applications Report 2




Panel B: Total Activities Report 2



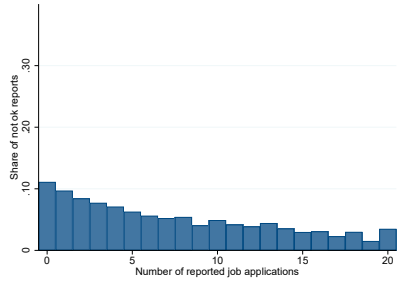

Panel C: Job Applications Report 3



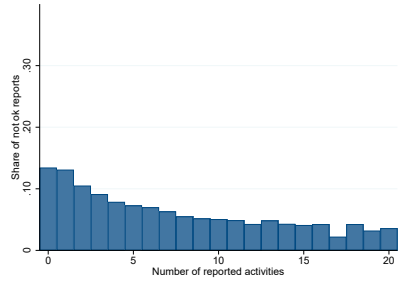

Panel D: Total Activities Report 3

*Figure A2.* Histogram for the number of job applications reported activities in the second and third report

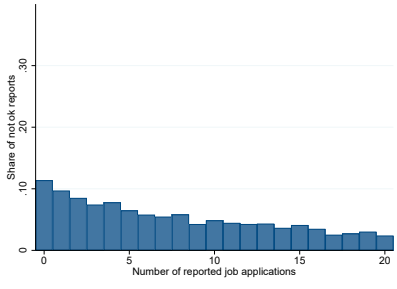
*Notes*: Statistics for second (N= 83,123) and third (N=62,422) report for the analyses sample of described in Section 3. 390 caseworkers. Job applications include standard applications, unsolicited applications, and applications for vacancy referrals or proposals. Total activities additionally include interviews, recruitment meetings, other activities, and activities specified in the action plan.



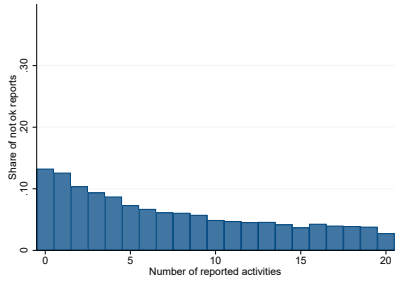

Panel A: Job Applications Report 2




Panel B: Total Activities Report 2



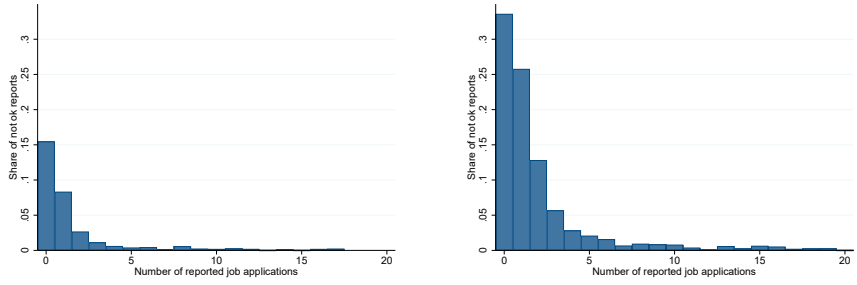

Panel C: Job Applications Report 3




Panel D: Total Activities Report 3

*Figure A3.* Share not-approved reports, by number of reported job applications and activities

*Notes*: Statistics for second and third report for the analyses sample of described in Section 3. Statistics for second (N= 83,123) and third (N=62,422) report for the analyses sample of described in Section 3. 390 caseworkers. Job applications include standard applications, unsolicited applications, and applications for vacancy referrals or proposals. Total activities additionally include interviews, recruitment meetings, other activities, and activities specified in the action plan.




Panel A: Low Caseworker Strictness Panel B: High Caseworker Strictness

*Figure A4.* Share not-approved reports, by caseworker strictness and number of reported job applications

*Notes*: Share of rejected activity reports by low and high caseworker strictness. Low strictness is defined as below median caseworker strictness. High strictness is defined as above median caseworker strictness. Median strictness is 5.8 % rejected activity reports. 109,866 cases and 390 caseworkers.

**Table A1.** *Testing for random assignment of unemployed to caseworkers*

| | Left UE within 90 days | Not ok | Caseworker Strictness |
|---|---|---|---|
| Age | -0.004*** | 0.001*** | -0.000 |
| | (0.000) | (0.000) | (0.000) |
| Male | -0.054*** | 0.011*** | 0.000 |
| | (0.003) | (0.002) | (0.000) |
| Disabled | -0.191*** | 0.037*** | 0.001 |
| | (0.009) | (0.007) | (0.001) |
| Married | -0.026*** | -0.000 | -0.000* |
| | (0.003) | (0.002) | (0.000) |
| *Education level* | | | |
| Less than high school | -0.059*** | 0.033*** | 0.001** |
| | (0.005) | (0.003) | (0.000) |
| High school | 0.011*** | 0.017*** | 0.000 |
| | (0.003) | (0.002) | (0.000) |
| College | 0.018*** | -0.030*** | -0.000 |
| | (0.004) | (0.002) | (0.000) |
| *Region of birth* | | | |
| Sweden | 0.080*** | -0.009*** | -0.000 |
| | (0.003) | (0.002) | (0.000) |
| Nordic countries | -0.002 | 0.013** | -0.001 |
| | (0.009) | (0.005) | (0.001) |
| EU | -0.039*** | -0.010*** | -0.000 |
| | (0.006) | (0.003) | (0.001) |
| Outside of EU | -0.089*** | 0.012*** | 0.000 |
| | (0.004) | (0.002) | (0.000) |
| *Labor market history* | | | |
| Employed t-180 days | 0.028*** | 0.003 | -0.000 |
| | (0.005) | (0.003) | (0.000) |
| Employed t-360 days | -0.006 | -0.003 | 0.000 |
| | (0.006) | (0.002) | (0.000) |
| Employed t-540 days | 0.047*** | -0.001 | 0.000 |
| | (0.004) | (0.002) | (0.000) |
| Employed t-720 days | 0.018*** | -0.004** | 0.000 |
| | (0.005) | (0.002) | (0.000) |
| Number of days UE last 5y | -0.000*** | 0.000*** | -0.000 |
| | (0.000) | (0.000) | (0.000) |
| Number of programs last 5y | 0.012*** | 0.000 | -0.000 |
| | (0.004) | (0.002) | (0.000) |
| Number of UE spells last 5y | 0.015*** | 0.000 | -0.000 |
| | (0.002) | (0.001) | (0.000) |
| Predicted employment within 90 days | 0.957*** | -0.131*** | -0.002 |
| | (0.023) | (0.009) | (0.001) |
| F-test | 144*** | 28*** | 1 |
| Mean depvar | 0.285 | 0.069 | 0.069 |
| N | 109,865 | 109,865 | 109,865 |

*Notes:* Standard errors are clustered at the month-agency level and reported in parentheses. ***, **, * indicate significance at 1%, 5% and 10% respectively. We control for month-agency fixed effects. Each line correspond to a single regression with one predetermined jobseeker characteristic as independent variable. The dependent variable is in column (1) indicator for having left unemployment within 90 days, column (2) indicator for getting a not ok on activity report, and column (3) caseworker strictness. The F-statistic for the joint significance of job seeker characteristics is reported at the bottom of the table. The F-test includes all variables except the predicted probability of exiting unemployment within 90 days.

**Table A2.** *Caseworker strictness, timing of decisions and content of the first report*

| | Number of job applications | Number of activities | Time until unemployed hands in the report | Time untill CW checks the report |
|---|---|---|---|---|
| CW strictness | 0.015 | -0.159 | -0.449 | 1.634** |
| | (0.593) | (0.715) | (1.432) | (0.693) |
| Mean depvar | 6.02 | 8.96 | 33.06 | 3.00 |
| N | 109,865 | 109,865 | 109,865 | 109,836 |

*Notes:* Standard errors are clustered at the month-agency level and reported in parentheses. ***, **, * indicate significance at 1%, 5% and 10% respectively. We control for month-agency fixed effects. The time is measured in days and refers to (1) the time between the entry into UE and the day at which the unemployed hands in the first activity report, and (2) the time between the days the unemployed hands in the report and the moment the caseworker evaluates it.

**Table A3.** *Monotonicity check: job seeker characteristics*

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | Male | Female | Married | Non Married | $<$ 40yrs | $\geq$ 40yrs |
| CW strict | 0.911*** | 0.769*** | 0.860*** | 0.829*** | 0.771*** | 0.910*** |
| | (0.037) | (0.035) | (0.041) | (0.033) | (0.035) | (0.037) |
| Mean strict | 0.075 | 0.063 | 0.069 | 0.069 | 0.063 | 0.075 |
| N | 56,732 | 53,134 | 41,431 | 68,435 | 54,373 | 55,433 |
| | (7) | (8) | (9) | (10) | (11) | (12) |
| | College | High-school or less | Individual Low risk | Individual High risk | Industry Low growth | Industry High growth |
| CW strict | 0.612*** | 0.998*** | 0.803*** | 0.880*** | 0.842*** | 0.830*** |
| | (0.034) | (0.037) | (0.036) | (0.034) | (0.035) | (0.038) |
| Mean strict | 0.052 | 0.081 | 0.061 | 0.076 | 0.068) | 0.069 |
| N | 44,592 | 64,588 | 54,933 | 54,933 | 53,021 | 55,515 |

*Notes:* First stage regressions of caseworker strictness on the probability of getting a rejected activity report within subgroups based on predetermined job seeker characteristics. Standard errors are clustered at the month-agency level and reported in parentheses. ***, **, * indicate significance at 1%, 5% and 10% respectively. We control for month-agency fixed effects.

**Table A4.** *Monotonicity check: across subgroup*

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | Male | Female | Married | Non Married | $<$ 40yrs | $\geq$ 40yrs |
| CW strict | 0.823*** | 0.622*** | 0.832*** | 0.695*** | 0.629*** | 0.842*** |
| | (0.036) | (0.032) | (0.042) | (0.030) | ( (0.031) | 0.039) |
| N | 56,732 | 53,134 | 41,431 | 68,435 | 54,433 | 55,493 |
| | (7) | (8) | (9) | (10) | (11) | (12) |
| | College | High-school or less | Individual Low risk | Individual High risk | Industry Low growth | Industry High growth |
| CW strict | 0.456*** | 0.936*** | 0.697*** | 0.828*** | 0.758*** | 0.752*** |
| | (0.027) | (0.044) | (0.034) | (0.036) | (0.034) | (0.035) |
| N | 65,274 | 45,278 | 54,933 | 54,933 | 55,515 | 53,021 |

*Notes:* First stage regressions of caseworker strictness in subgroup (A), defined as the share of rejected activity reports within the subgroup, on the probability of getting a rejected activity report within subgroup (B). E.g. In column (1) the dependent variable is getting a rejected activity report for males, with strictness among females as the independent variable. Standard errors are clustered at the month-agency level and reported in parentheses. ***, **, * indicate significance at 1%, 5% and 10% respectively. We control for month-agency fixed effects.

**Table A5.** *Complier Characteristics*

| | Sample Mean | Complier Mean | N |
|---|---|---|---|
| Male | 0.516 | 0.551 | 109,866 |
| | | (0.022) | |
| Married | 0.377 | 0.419** | 109,866 |
| | | (0.021) | |
| Age | 41 | 42*** | 109,866 |
| | | (0.463) | |
| Born in Sweden | 0.759 | 0.777 | 109,866 |
| | | (0.019) | |
| Number of days UE last 5y | 284 | 318** | 109,866 |
| | | (15) | |
| Number of programs last 5y | 0.081 | 0.087 | 109,866 |
| | | (0.016) | |
| Number of UE spells last 5y | 1.359 | 1.609*** | 109,866 |
| | | (0.076) | |
| High individual unemployment risk | 0.500 | 0.509 | 109,866 |
| | | (0.022) | |
| High industry growth | 0.511 | 0.511 | 108,536 |
| | | (0.021) | |
| Number of job applications report 1 | 6.022 | 1.479*** | 109,866 |
| | | (0.337) | |
| Number of total activities report 1 | 8.959 | 3.382*** | 109,866 |
| | | (0.423) | |

*Notes:* Mean of predetermined covariates for the whole sample and the complier group. Bootstrap standard errors are reported for the complier group. ***, **, * indicate significant difference in means at significance levels 1%, 5% and 10% respectively. We control for month-agency fixed effects. Complier means are constructed following Goldsmith-Pinkham et al. (2025). First, the means among treated compliers are recovered by regressing the interaction of each covariate $x_i$ with the treatment not ok $D_i$ on the treatment instrumented by caseworker strictness $Z_{j(i)}$ - obtaining a weighted average of the covariate using the same weights as in the main IV specification. Second, to obtain the weighted average among non-treated compliers, the treatment variable is exchanged for $1 - D_i$. The two specifications are pooled using $\tilde{D}_i = 2D_i - 1$ as the treatment indicator. I.e. regressing $x_i \times \tilde{D}_i$ on $\tilde{D}_i$ instrumented by caseworkers strictness, and controlling for month-agency fixed effects.

**Table A6.** *Jobseeker Characteristics: Vacancy Proposals*

| | No vacancy proposal | Vacancy Proposal | p-value diff |
|---|---|---|---|
| Male | 0.517 | 0.493 | 0.000 |
| | (0.001) | (0.003) | |
| Married | 0.377 | 0.391 | 0.036 |
| | (0.001) | (0.003) | |
| Age | 40 | 41 | 0.000 |
| | (0.032) | (0.059) | |
| Born in Sweden | 0.759 | 0.757 | 0.598 |
| | (0.001) | (0.002) | |
| Number of days UE last 5y | 283 | 281 | 0.032 |
| | (1.156) | (2.039) | |
| Number of programs last 5y | 0.081 | 0.080 | 0.254 |
| | (0.001) | (0.002) | |
| Number of UE spells last 5y | 1.358 | 1.296 | 0.091 |
| | (0.005) | (0.009) | |
| Predicted employment within 90 days | 0.265 | 0.263 | 0.000 |
| | (0.000) | (0.000) | |
| N | 111,846 | 35,291 | |

*Notes:* Each line show the difference in means for a predetermined characteristic, between jobseekers with and without any vacancy proposal from a central office caseworker prior to submission of the first activity report. Standard errors are clustered at the month-agency level and reported in parentheses. P-values reported for difference in means between jobseekers with and without vacancy proposals. We control for month-agency fixed effects.

**Table A7.** *Effects of caseworker strictness on employment status*

| | (1) 30 days | (2) 90 days | (3) 180 days | (4) 360 days | (5) 540 days | (6) 720 days |
|---|---|---|---|---|---|---|
| CW strictness | 0.004 | 0.135*** | 0.097** | 0.107*** | 0.105*** | 0.047 |
| | (0.008) | (0.037) | (0.040) | (0.033) | (0.031) | (0.030) |
| Mean depvar | 0.010 | 0.283 | 0.549 | 0.710 | 0.780 | 0.813 |
| N | 109,866 | 109,866 | 109,866 | 109,866 | 109,866 | 109,866 |

*Notes:* Standard errors are clustered at the month-agency level and reported in parentheses. ***, **, * indicate significance at 1%, 5% and 10% respectively. We control for month-agency fixed effects.

**Table A8.** *Robustness analyses for dynamic sample selection*

| | (1) | (2) | (3) |
|---|---|---|---|
| | All reports | Sample two reports | |
| | Report 1 | Report 1 | Report 2 |
| CW Strictness | 0.015 | 0.809 | 3.241*** |
| | (0.593) | (0.724) | (0.657) |
| Mean depvar | 6.022 | 6.133 | 6.625 |
| N | 109,865 | 83,122 | 83,122 |

*Notes:* Standard errors are clustered at the month-agency level and reported in parentheses. ***, **, * indicate significance at 1%, 5% and 10% respectively. We control for month-agency fixed effects. Column 1 reports estimates for the full sample of job seekers, and Columns 2–3 restricts the sample to those who submitted the second report.

**Table A9.** *Effects of caseworker strictness and probability of leaving unemployment within end of second report period*

| | (1) Reduced form | (2) IV estimates |
|---|---|---|
| Panel a: Exits from unemployment | | |
| CW Strictness | 0.093*** | 0.110*** |
| | (0.022) | (0.026) |
| Mean depvar | 0.098 | 0.098 |
| N | 109,866 | 109,866 |
| Panel b: Job-search intensity | | |
| CW Strictness | 3.242*** | 3.893*** |
| | (0.656) | (0.805) |
| Mean depvar | 6.625 | 6.625 |
| N | 83,123 | 83,123 |

*Notes:* Standard errors are clustered at the month-agency level and reported in parentheses. ***, **, * indicate significance at 1%, 5% and 10% respectively. We control for month-agency fixed effects. In the IV estimation the dependent variable not ok on first report is instrumented by caseworker strictness from the first report. The outcome is measured on the inferred last day of the second reporting periodâspecifically, the last day of the month after the first reporting period ends. The number of days from the start of the unemployment spell to this inferred last day of the second reporting period ranges from 40 to 100 days, with an average of 56 days.

**Table A10.** *Robustness: Residual caseworker strictness and probability of leaving unemployment within end of second report period*

| | (1) Reduced form | (2) IV estimates |
|---|---|---|
| Panel a: Exits from unemployment | | |
| CW Strictness | 0.097*** | 0.110*** |
| | (0.023) | (0.026) |
| Mean depvar | 0.098 | 0.098 |
| N | 109,866 | 109,866 |
| Panel b: Job-search intensity | | |
| CW Strictness | 3.481*** | 3.973*** |
| | (0.670) | (0.784) |
| Mean depvar | 6.625 | 6.625 |
| N | 83,123 | 83,123 |

*Notes:* Standard errors are clustered at the month-agency level and reported in parentheses. ***, **, * indicate significance at 1%, 5% and 10% respectively. We control for month-agency fixed effects. In the IV estimation the dependent variable not ok on first report is instrumented by caseworker strictness from the first report. Residual caseworker strictness is demeaned by month times agency fixed effects. Moving from the 10th to the 90th percentile of residual strictness is equivalent to an increase of . The outcome is measured on the inferred last day of the second reporting periodâspecifically, the last day of the month after the first reporting period ends. The number of days from the start of the unemployment spell to this inferred last day of the second reporting period ranges from 40 to 100 days, with an average of 56 days.

**Table A11.** *Robustness analyses*

| | (1) 30 days | (2) 60 days | (3) 90 days | (4) 120 days | (5) 180 days | (6) 360 days |
|---|---|---|---|---|---|---|
| Panel A: Day-agency fixed effects | | | | | | |
| CW strictness | 0.008 | 0.115*** | 0.121*** | 0.111*** | 0.088** | 0.072** |
| | (0.008) | (0.026) | (0.033) | (0.036) | (0.036) | (0.033) |
| Panel B: Controlling for observable jobseeker characteristics | | | | | | |
| CW strictness | 0.008 | 0.116*** | 0.117*** | 0.107*** | 0.082** | 0.072** |
| | (0.009) | (0.027) | (0.035) | (0.038) | (0.038) | (0.033) |
| Mean depvar | 0.011 | 0.131 | 0.269 | 0.380 | 0.528 | 0.704 |
| N | 109,866 | 109,866 | 109,866 | 109,866 | 109,866 | 109,866 |
| Panel C: Caseload min 50 cases | | | | | | |
| CW strictness | 0.007 | 0.115*** | 0.120*** | 0.109*** | 0.087** | 0.086** |
| | (0.008) | (0.025) | (0.034) | (0.037) | (0.038) | (0.033) |
| Mean depvar | 0.011 | 0.131 | 0.269 | 0.380 | 0.528 | 0.704 |
| N | 112,290 | 112,290 | 112,290 | 112,290 | 112,290 | 112,290 |
| Panel D: Including JS with vacancy proposals | | | | | | |
| CW strictness | 0.006 | 0.119*** | 0.120*** | 0.120*** | 0.103*** | 0.069** |
| | (0.007) | (0.023) | (0.031) | (0.034) | (0.033) | (0.029) |
| Mean depvar | 0.011 | 0.127 | 0.264 | 0.376 | 0.524 | 0.702 |
| N | 147,137 | 147,137 | 147,137 | 147,137 | 147,137 | 147,137 |

*Notes:* ***, **, * indicate significance at 1%, 5% and 10% respectively. Panel A: control for day-agency fixed effects. Standard errors are clustered at the day-agency level and reported in parentheses. Panel B: Control for month-agency fixed effects and observable characteristics: Level of education (compulsory school, high-school, college), male, immigrant, married, divorced, disability, days unemployed last 2 years, days unemployed last 5 years, number of unemployment spells last 2 years, number of unemployment spells last 5 years, number of programs last 5 years. Standard errors are clustered at the month-agency level and reported in parentheses. Panel C: Include all caseworkers with at least 50 cases, controlling for month-agency fixed effects. Standard errors are clustered at the day-agency level and reported in parentheses. Panel D: Include both job seekers who did and did not receive any vacancy proposal from a central office caseworker prior to submitting the first activity report. Controlling for month-agency fixed effects. Standard errors are clustered at the day-agency level and reported in parentheses.

**Table A12.** *Heterogeneous effects by individual risk of long-term unemployment: monitoring chain*

| | (1) Not Ok First Report | (2) Violation Job Seach | (3) Violation Other | (4) Notification Sent to UI | (5) Warning/ Sanction |
|---|---|---|---|---|---|
| Panel a: Low risk | | | | | |
| CW strictness | 0.803*** | 0.195*** | 0.020 | 0.146*** | 0.116*** |
| | (0.036) | (0.018) | (0.030) | (0.031) | (0.026) |
| Mean depvar | 0.061 | 0.020 | 0.108 | 0.111 | 0.064 |
| N | 54,933 | 54,933 | 54,933 | 54,933 | 54,933 |
| Panel b: High risk | | | | | |
| CW strictness | 0.880*** | 0.222*** | -0.053 | 0.136*** | 0.103*** |
| | (0.034) | (0.020) | (0.034) | (0.037) | (0.027) |
| Mean depvar | 0.076 | 0.025 | 0.137 | 0.137 | 0.081 |
| N | 54,933 | 54,933 | 54,933 | 54,933 | 54,933 |

Notes: Standard errors are clustered at the month-agency level and reported in parentheses. ***, **, * indicate significance at 1%, 5% and 10% respectively. We control for month-agency fixed effects. High unemployment risk is defined as below median probability of exiting unemployment within 360 days. Low unemployment risk is defined as above median probability of exiting unemployment within 360 days. Prediction of exiting unemployment within 360 days is based on Gradient Boosted Decision Trees, using the following input variables: age, gender, disability, marriage, level of education, region of birth, and employment history: employed t-180 days,employed t-360 days,employed t-540 days, employed t-720 days, number of days of UE last 5 years, number of programs last 5 years, and number of UE spells last 5 years, previous income, and search occupation. P-values for tests of differences in coefficients between the high- and low-risk individuals (by column): (1) p = 0.067, (2) p = 0.238, (3) p = 0.087, (4) p = 0.818, (5) p = 0.713 .

**Table A13.** *Heterogeneous effects by individual risk of long-term unemployment: labor market programs and meetings*

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | Labor market programs | | | | Meetings | |
| | Any Program | Work Practice | Vocational Training | Prepatory Training | Face to Face Meetings | Phone Meetings |
| Panel a: Low risk | | | | | | |
| CW strictness | -0.000 | 0.002 | -0.000 | -0.002 | 0.159 | 1.171*** |
| | (0.013) | (0.006) | (0.007) | (0.006) | (0.116) | (0.132) |
| Mean depvar | 0.012 | 0.003 | 0.005 | 0.004 | 1.793 | 1.051 |
| N | 54,933 | 54,933 | 54,933 | 54,933 | 54,933 | 54,933 |
| Panel b: High risk | | | | | | |
| CW strictness | -0.005 | -0.010 | 0.005 | 0.001 | 0.140 | 1.278*** |
| | (0.016) | (0.007) | (0.008) | (0.009) | (0.129) | (0.130) |
| Mean depvar | 0.018 | 0.005 | 0.005 | 0.007 | 2.041 | 1.082 |
| N | 54,933 | 54,933 | 54,933 | 54,933 | 54,933 | 54,933 |

Notes: Standard errors are clustered at the month-agency level and reported in parentheses. ***, **, * indicate significance at 1%, 5% and 10% respectively. We control for month-agency fixed effects. High unemployment risk is defined as below median probability of exiting unemployment within 360 days. Low unemployment risk is defined as above median probability of exiting unemployment within 360 days. Prediction of exiting unemployment within 360 days is based on Gradient Boosted Decision Trees, using the following input variables: age, gender, disability, marriage, level of education, region of birth, and employment history: employed t-180 days,employed t-360 days,employed t-540 days, employed t-720 days, number of days of UE last 5 years, number of programs last 5 years, and number of UE spells last 5 years, previous income, and search occupation. P-values for tests of differences in coefficients between the high- and low-risk individuals (by column): (1) p = 0.823, (2) p = 0.193, (3) p = 0.614, (4) p = 0.826, (5) p = 0.911, (6) p=0.544 .

**Table A14.** *Heterogeneous effects by industry growth: monitoring chain*

| | (1) Not Ok First Report | (2) Violation Job Seach | (3) Violation Other | (4) Notification Sent to UI | (5) Warning/ Sanction |
|---|---|---|---|---|---|
| Panel a: Low Growth Industries | | | | | |
| CW strictness | 0.842*** | 0.211*** | 0.016 | 0.166*** | 0.109*** |
| | (0.035) | (0.019) | (0.034) | (0.034) | (0.027) |
| Mean depvar | 0.068 | 0.022 | 0.117 | 0.119 | 0.069 |
| N | 53,021 | 53,021 | 53,021 | 53,021 | 53,021 |
| Panel b: High Growth Industries | | | | | |
| CW strictness | 0.830*** | 0.205*** | -0.067* | 0.106*** | 0.102*** |
| | (0.038) | (0.021) | (0.034) | (0.038) | (0.028) |
| Mean depvar | 0.069 | 0.022 | 0.127 | 0.129 | 0.076 |
| N | 55,515 | 55,515 | 55,515 | 55,515 | 55,515 |

Notes: Standard errors are clustered at the month-agency level and reported in parentheses. ***, **, * indicate significance at 1%, 5% and 10% respectively. We control for month-agency fixed effects. Industry growth rate is defined for 2-digit industry codes, and measures employment change between year t-5 and t+5. P-values for tests of differences in coefficients between the high- and low-risk individuals (by column): (1) p = 0.796, (2) p = 0.828, (3) p = 0.098, (4) p = 0.243, (5) p = 0.850 .

**Table A15.** *Heterogeneous effects by industry growth: labor market programs and meetings*

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | Labor market programs | | | | Meetings | |
| | Any Program | Work Practice | Vocational Training | Prepatory Training | Face to Face Meetings | Phone Meetings |
| Panel b: Low Growth Industries | | | | | | |
| CW strictness | 0.008 | -0.002 | 0.009 | 0.000 | 0.234** | 1.210*** |
| | (0.016) | (0.007) | (0.008) | (0.009) | (0.118) | (0.126) |
| Mean depvar | 0.015 | 0.004 | 0.006 | 0.005 | 1.867 | 1.035 |
| N | 53,021 | 53,021 | 53,021 | 53,021 | 53,021 | 53,021 |
| Panel b: Low Growth Industries | | | | | | |
| CW strictness | -0.016 | -0.007 | -0.005 | -0.003 | 0.089 | 1.231*** |
| | (0.013) | (0.006) | (0.006) | (0.007) | (0.128) | (0.125) |
| Mean depvar | 0.014 | 0.004 | 0.004 | 0.005 | 1.960 | 1.099 |
| N | 55,515 | 55,515 | 55,515 | 55,515 | 55,515 | 55,515 |

Notes: Standard errors are clustered at the month-agency level and reported in parentheses. ***, **, * indicate significance at 1%, 5% and 10% respectively. We control for month-agency fixed effects. Industry growth rate is defined for 2-digit industry codes, and measures employment change between year t-5 and t+5. P-values for tests of differences in coefficients between the high- and low-risk individuals (by column): (1) p = 0.236, (2) p = 0.596, (3) p = 0.160, (4) p = 0.724, (5) p = 0.397, (6) p=0.894 .

**Table A16.** *Robustness analyses and the measurement of industry growth*

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | Growth t-5, t+5 | | Growth t-5, t | |
| | 2-digit code | 3-digit code | 2-digit code | 3-digit code |
| Panel a: Low Growth Industries | | | | |
| CW strictness | 0.115** | 0.127** | 0.161*** | 0.120** |
| | (0.048) | (0.049) | (0.048) | (0.048) |
| Mean depvar | 0.266 | 0.268 | 0.260 | 0.266 |
| N | 53,021 | 53,666 | 53,857 | 53,747 |
| Panel b: High Growth Industries | | | | |
| CW strictness | 0.129*** | 0.111** | 0.072 | 0.116** |
| | (0.049) | (0.046) | (0.044) | (0.047) |
| Mean depvar | 0.275 | 0.273 | 0.281 | 0.275 |
| N | 55,515 | 54,863 | 54,679 | 54,782 |

*Notes:* Standard errors are clustered at the month-agency level and reported in parentheses. ***, **, * indicate significance at 1%, 5% and 10% respectively. We control for month-agency fixed effects.

# Essay III. Transparency, governance, and water and sanitation: experimental evidence from schools in rural Bangladesh

Co-authored with Umrbek Allakulov, Serena Cocciolo, Binayak Das, Md. Ahasan Habib, and Anna Tompsett.

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# 1 Introduction

A common belief among policy-makers is that improving water, sanitation, and hygiene—or WASH—services is largely a question of improving governance. For example, the OECD (2015) argues that crises in water management are "primarily" crises in governance, and the UNDP (2011) identifies corruption—a governance failure—as a major impediment to improving WASH service provision. Many policy-makers thus anticipate that interventions to improve governance should improve service provision in the WASH sector. One popular class of such interventions aims to increase transparency in service provision. In practice, however, previous experimental evaluations of similar interventions in other sectors have yielded only mixed results (Kosack and Fung, 2014).

Using a randomized experiment focused on WASH services in 60 schools in rural Bangladesh, we evaluate the impacts of a transparency intervention comprising a deliberative multi-stakeholder workshop initiated with a community scorecard exercise. Following a pre-specified analysis plan, we trace out the full causal path via which the intervention could affect outcomes of interest: changes in knowledge and institutions, improvements to WASH facilities, changes in use patterns in WASH facilities, school attendance, and exam results.[1] We find that the intervention leads to moderate improvements in knowledge and institutions but not to measurable improvements in other outcomes.

Many schoolchildren in Bangladesh lack access to adequate WASH facilities, potentially increasing water-borne disease and absenteeism (UNICEF, 2012). Before our intervention, two schools in this study had no functioning toilets available to students. Those schools that did have functioning toilets had only one for every 226 students, far below the internationally recommended minimum standard of 1:50 (Water Aid, 2016). Only 61% of functioning toilets were clean and could be locked. No school had any toilet meeting the minimum requirements for safe and private menstrual hygiene management (MHM). Students and teachers reported that inadequate sanitation facilities affected female student attendance.

We randomly assigned 30 schools to receive the intervention we study. Developed by the Water Integrity Network (WIN) and the IRC International Water and Sanitation Centre, the Annotated Water Integrity Scan (AWIS) is a participatory assessment tool designed to gather and disseminate information, improve governance, and promote integrity.[2] Integrity is defined as a set of practices that impede corruption and promote respect for the rule of law. During a workshop, participants first anonymously score WASH service

[1] The pre-analysis plan that we follow is publicly available from the AEA RCT registry: `https://www.socialscienceregistry.org/versions/54056/docs/version`.

[2] See `https://www.waterintegritynetwork.net` and `https://www.ircwash.org/home`.

provision across indicators measuring transparency, accountability, and participation. Facilitators then reveal the anonymized scores. When scores differ across participants, facilitators invite them to discuss why, with the goal of understanding differences in opinions rather than defending individual positions. The output of this stage of the discussion is a single, unified set of scores. This process is designed to yield a common understanding of challenges with the goal of identifying concrete and specific priority actions for improvement during the last phase of the workshop. AWIS closely resembles one of a set of "exemplary transparency interventions" identified by Kosack and Fung (2014).[3]

We evaluate the impacts of the AWIS intervention on three families of outcomes, eight to twelve months after implementation. First, we evaluate effects on knowledge about WASH and MHM practices among male and female students, male and female teachers, and headteachers, and institutions for WASH service management. For five out of six outcomes, we see small to medium improvements (between 0.1 and 0.3 standard deviations). Not all the individual effects are statistically significant and none independently survive corrections for multiple hypothesis testing that control the family-wise error rate. However, this pattern of consistently positive results is unlikely to have occurred due to chance ($p = .031$).

Second, we turn to material effects on WASH provision and use. We measure effects on three outcomes: the number of functioning toilets available to students; a toilet quality index, which summarizes measures of functionality, cleanliness, and hygiene; and latrine use. We develop a novel approach to anonymously measuring latrine use, using discreet magnetic door sensors that record when the latrine door is opened or closed. The characteristic signature of a latrine use event in these data is an interval during which the door is closed bookended by two intervals during which the door is open. Door sensors present a potential improvement over other alternatives such as motion sensor cameras, because the entire equipment can be installed outside the cubicle, minimizing intrusion of privacy. We estimate effects on WASH provision and use separately for toilets used by male and female students. We also report the number of functioning toilets available to all students (for which use is not gender-separated), although few schools have them. Three out of seven metrics show statistically insignificant improvements, and four show declines. We thus find no evidence that the intervention improved WASH service provision or increased WASH facility use.

[3] Specifically, Kosack and Fung (2014) identify "citizen report cards or community score cards followed by deliberative community meetings" as the exemplary transparency intervention in contexts in which service providers are willing to cooperate in reforms and the main role of transparency interventions is to feed collaborative problem solving. While it is often uncertain whether or not service providers are willing to cooperate in reforms, and transparency interventions are less likely to be successful when they are not, successful transparency interventions often reveal service providers to be willing to engage in reform (Kosack and Fung, 2014).

Most disappointing, the impacts on WASH provision and use are negative for all metrics for toilets available to female students. We did not design our experiment to detect negative effects of the intervention; we prespecified one-sided hypothesis tests to maximize power, given the limited sample size. However, two out of three of these negative effects are statistically significant at the 10 percent level in two-sided hypothesis tests, a pattern of results with a low likelihood of occurring if the intervention had no effect on school WASH services. WASH provision and use for female students decline in treated schools relative to control schools not because WASH service provision worsens in treated schools but because control schools more successfully improve WASH service provision. Future research should investigate whether similar interventions can have negative unforeseen discouragement effects, for example by creating expectations of future external material support that disincentivize independent local action.

Finally, we turn to effects on school attendance and exam results. We do not find that the intervention leads to measurable changes in attendance or achievement in exams. The 90% confidence intervals exclude increases in female student attendance larger than 4.1 percentage points (p.p) or declines larger than 2.9 p.p. and increases in female student exam results of greater than 0.16 points on an 5 point scale or declines larger than 0.12 points. For female students, baseline mean attendance is 66% and the mean exam score is 2.3.[4] That the intervention does not measurably affect attendance or results may not be surprising, given the mixed results on knowledge and institutions, and WASH provision and use.

Although the number of schools in the study is relatively small, we maximize our ability to detect effects by collecting rich data at a more granular unit or temporal level wherever feasible. Conversely, we minimize the risk of drawing erroneous conclusions by using randomization-based inference and reporting conservative multiple hypothesis tests that control the risk of erroneously rejecting any of the null hypotheses within a family of outcomes. Ex post minimum detectable effects using measures of the variance of the estimated coefficients (that are independent of the realized point estimates themselves) suggest that the study was powered (at significance level 10% and power 80%) to detect increases in indexes of between 0.2 and 0.4 standard deviations, typically characterized as small to medium-sized effects (Cohen, 1988). Thus the reason we do not detect effects on WASH service provision is not solely a consequence of power. Adherence to the study protocol was also high, with all schools assigned to treatment successfully treated.

Our results contribute to the literature on interventions designed to improve public services through greater transparency or accountability. The interven-

[4]For male students, the 90% confidence intervals rule out increases in attendance larger than 5.3 percentage points or declines larger than 3.3 p.p., and increases in student exam results of greater than 0.07 points on an 5 point scale or declines larger than 0.19 points. Baseline mean attendance is 57% and mean exam score is 2.3.

tion we evaluate belongs to a class of interventions called "transparency for accountability" that help users of a public service access information that increases their ability to influence service provision for the better (Kosack and Fung, 2014). These reforms are increasingly "in vogue" (Kosack and Fung, 2014), along with a closely-related group of interventions called "social accountability" interventions that attempt to improve service provision through citizen engagement (Fox, 2015). In practice, transparency for accountability and social accountability interventions commonly share a twin focus on "information"—in our case, the scorecard exercise—and "participation"—in our case, the deliberative multi-stakeholder workshop. Total spending on transparency and accountability is difficult to measure, but it most likely exceeds US$200 million a year.[5] The enthusiasm from policy-makers contrasts, however, with the mixed experimental evidence regarding the efficacy of these interventions (Joshi, 2010; Kosack and Fung, 2014; Fox, 2015).

Our primary contribution to this literature is to provide the first experimental evaluation of a transparency intervention in the water and sanitation sector, a sector in which many policy-makers expected high impact. Our results show that a single, time-limited transparency intervention improved knowledge and institutions but not WASH service provision. While improvements to knowledge and institutions are intrinsically important, they would also, all else equal, would be expected to improve service provision. We draw on rich qualitative and quantitative descriptive data to understand why improvements in knowledge and institutions did not translate into improvements in WASH service provision in this case. This exercise, in turn, sheds light on results from previous literature and suggests ways to improve the design of future interventions. We identify three potential reasons.

First, we study a relatively "light-touch" intervention: a one-off workshop that draws only on information available to the community. Such interventions are naturally of particular policy interest because they are relatively cheap to implement and would thus be highly cost-effective if they led to meaningful changes. However, time-limited interventions may be insufficient to change outcomes. In particular, the AWIS intervention appears to have more successfully created consensus over problem diagnosis than plans for action. A longer time horizon might allow both to receive equal attention. In previous studies, less effective interventions were often characterized by shorter time horizons (e.g., Banerjee et al., 2010; BenYishay et al., 2022) while more effective interventions were characterized by longer time horizons, more extended engagement, or accountability measures to incentivize adherence to agreed-upon action plans (e.g., Björkman and Svensson, 2009; Barr et al., 2012; Mohanan et al., 2020). Such modifications most likely increase implementation costs,

[5]Members of the Transparency and Accountability Initiative, a consortium of eight private and public funders, report spending US$200 million annually on transparency and accountability work in the Global South (Fox, 2015).

but they may be necessary if transparency interventions are to be effectively deployed.

Second, it is possible that the intervention may have inadvertently created misleading expectations of forthcoming financial support, discouraging stakeholders from seeking independent solutions to their problems. While our study is to our knowledge unique in documenting negative effects of a transparency intervention—though we cannot rule out the possibility that negative effects have been encountered in unpublished work—null effects are common and other studies also note that expectations about development projects "work" could influence how participants behave during similar interventions (Arkedis et al., 2021).

Third, the results may reflect the challenges of scale in intervention design. Schools in rural Bangladesh may not be able to solve all their WASH service problems internally. Funding new infrastructure typically requires schools to secure additional funding from local communities, non-governmental organizations, or, on an ad hoc basis, from government bodies. Local government administrators have limited and fragmented responsibility for secondary school WASH service provision and proved unwilling or unable to attend school-level workshops. Transparency interventions may be more effective if they diversify their approaches to engagement with different stakeholders or simultaneously provide financial support that expands the choices available to stakeholders and frees them from dependence on external funding (as, for example, in successful precedents such as Pradhan et al., 2014).

Our results leave open the possibility that the improvements we observe in knowledge and institutions might translate into improvements in WASH service provision over longer time horizons. Few previous studies evaluate how the effects of transparency or accountability interventions vary over time and those that do tend to be following up successful interventions (Nyqvist et al., 2017; Christensen et al., 2021), implying that their results may not necessarily generalize. Our findings also do not speak to whether or not similar interventions might be more effective in different WASH service contexts. In particular, similar interventions might be more effective when WASH service provision is the core objective of the provider, as with a water utility, rather than a secondary or indirect objective, as it is in schools. Future research may wish to investigate impacts over longer time horizons or in different WASH service provision contexts.

The paper proceeds as follows: Section 2 describes WASH service provision in the study schools at baseline. Section 3 provides more details about the AWIS intervention. Section 4 outlines a theory of change for how the intervention might alter outcomes. Section 5 describes the data, and section 6, the experiment. Section 7 outlines our approach to analysis, and section 8 reports the results. Section 9 summarizes descriptive evidence that suggests reasons why the intervention did not improve WASH service provision. Section 10 concludes.

## 2 School WASH service provision in rural Bangladesh

School WASH provision in Bangladesh remains very far from adequate. Although between 2013 and 2018 the provision of toilets for secondary school students increased from one toilet per 200 students (ICRRDB et al., 2014) to one toilet per 115 students (Bangladesh Bureau of Statistics, 2020), the latter figure is still less than half the internationally recommended level. Fewer than half of schools had improved, accessible, gender-separated toilets with soap and water available (Bangladesh Bureau of Statistics, 2020). Rural areas lagged behind urban areas (Bangladesh Bureau of Statistics, 2020).

Secondary school governance and management in Bangladesh is largely decentralized and institutionally fragmented (Water Aid, 2016). Secondary schools are regulated by the Directorate of Secondary and Higher Education (DSHE) but few (around 2% nationwide) are directly managed by the government (Water Aid, 2016). In non-governmental schools, the primary governance structure is the school management committee, a committee formed by local dignitaries, teachers, and parents or guardians. An upazila officer of the DSHE is nominally part of each school management committee but rarely participates actively in school governance, since the upazila-level DSHE offices have only a few staff members that are responsible for overseeing all secondary education in the upazila. The government subsidizes teacher and headteacher wages. Additionally, the government pays tuition for female students as long as they meet attendance and achievement goals, and remain unmarried.

The school management committee is responsible for providing adequate WASH services (Water Aid, 2016) but receives no direct or systematic governmental support to meet this responsibility. The Ministry of Education issued a national circular in 2015 that directs secondary schools to improve WASH services, which in principle is enforceable, but no resources are made available to schools to comply with the circular (Water Aid, 2016). The Department of Public Health and Engineering (DPHE) has a general responsibility for sanitation in rural areas but no specific budget or responsibility for secondary schools. The most decentralized level of local government, the union parishad or council, plays no direct role in governing secondary education. To fund any project, such as investments in construction, repair, or maintenance of school infrastructure, non-governmental schools must primarily raise funds themselves on a case-by-case basis.

We focus on secondary schools, covering grades 6 to 10, in two *upazilas* (subdistricts), Bhola Sadar and Ramgati, in southern Bangladesh. Both upazilas have a large number of unions (local government units) that are classified as "Extremely Hard to Reach", a classification the Bangladeshi government uses to describe areas with poor water and sanitation coverage and particular obstacles to improvement, such as unfavourable hydrogeological conditions,

inadequate communications networks, or frequent natural disasters (see Figure 1).

A local implementing NGO, Development Organization of the Rural Poor (DORP),[6] identified 60 schools with sanitation facilities below national standards to target for the intervention. In practice, almost all schools in the study areas were eligible under these criteria. Since school sanitation facilities may be most important for female adolescent students, we focused on schools with either all female students (11 schools) or both female and male students (49 schools). No school declined to participate in the study, and we did not exclude any schools from the selected pool. All study schools were initially non-governmental schools.[7] Around a third are madrasahs, or Islamic schools. The madrasahs in our study area closely resemble other non-governmental schools in organization and management but offer religious education and Arabic alongside the standard curriculum. At baseline, on average, female-only schools had 424 students and mixed schools had 609 students, of which 318 were female. Schools had on average 12.5 teachers, implying a mean student-to-teacher ratio of 46.5, slightly higher than the national average of 34.[8]

Our data confirm that sanitation facilities in the study schools do not meet Bangladeshi or international standards, either in terms of quantity or quality. At baseline, enumerators identified on average 4.5 toilet cubicles available to students per school, of which only 2.9 were functional, equivalent to one for every 226 students. Only three in four schools had any functioning toilet cubicle available exclusively for female students. Fewer than half of schools had any clean, lockable, functioning toilet cubicle available exclusively for female students. No school had any functioning toilet cubicle meeting the minimum guidelines for menstrual health management: exclusively for female students, clean, lockable, and having water, soap, and an adequate container for disposal of menstrual health products inside the cubicle (UNESCO, 2014). Only ten schools had any functioning toilet cubicle that was accessible to disabled students.

Before the intervention, students reported that school sanitation facilities affected female student attendance. Among female students interviewed, only 19% reported that school attendance is unaffected by menstruation: 73% reported that girls leave school earlier when they have their periods, and 8% reported that girls do not usually come to school at all during their period. The majority of girls interviewed (85%) cited poor or inadequate school sanitation facilities among the reasons for which girls miss school when they have their periods, including lack of private facilities for girls to change (64%), to wash cloth pads (51%), and to dispose of sanitary napkins (34%).

[6] See `https://www.dorpbd.org`.

[7] One study school was nationalized in September 2018, just after our study began, as part of a national program to ensure that every upazila had at least one government-run school.

[8] National data from the UNESCO Institute for Statistics, 2017.

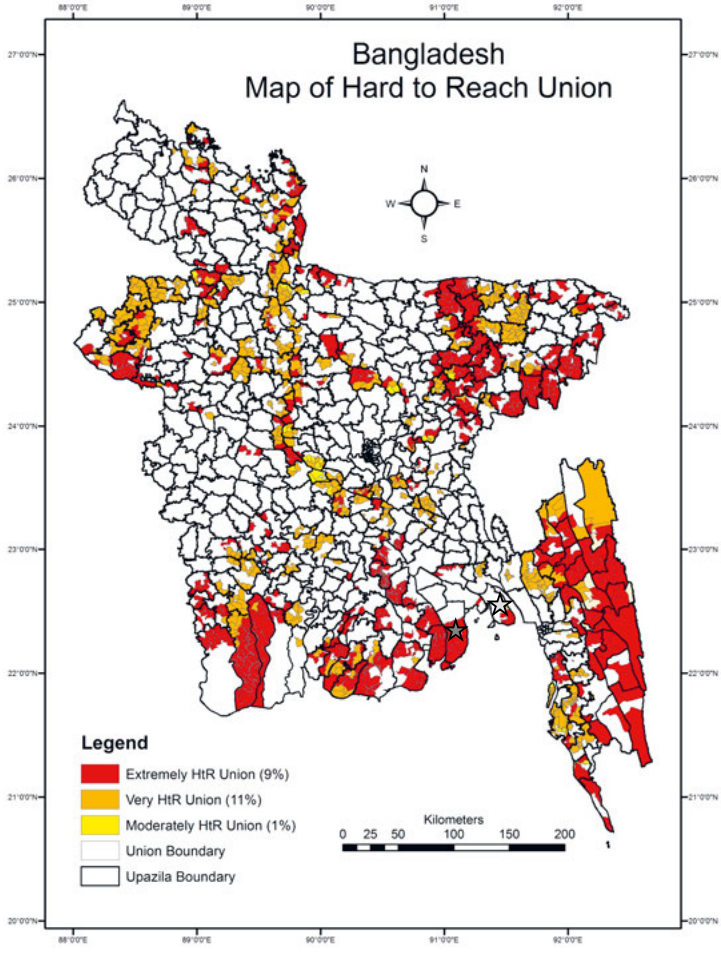



*Figure 1.* Location of study upazilas in Bangladesh

*Notes* Map shows unions in Bangladesh defined as "Hard-to-Reach" in terms of water and sanitation coverage (Government of Bangladesh, 2011). Dark grey star indicates Bhola Sadar upazila; light grey star indicates Ramgati upazila.

What students reported about the consequences of inadequate sanitation facilities contrasts with what school staff reported. When asked about the needs of female students, only 37% of headteachers cited separate toilet cubicles for boys and girls; 23%, a mechanism for the collection and disposal of menstrual health products; and 18%, a space for girls to wash in private. Female teachers were more aware of student needs regarding menstrual health management (MHM) than male teachers.[9] However, women are underrepresented among school staff and management. At baseline, few headteachers (2 out of 60) were female, only one in five school management committee (SMC) members and teachers were female, and ten mixed-gender schools had no female teachers.

## 3 The AWIS intervention

The Annotated Water Integrity Scan, or AWIS, is an intervention designed by the Water Integrity Network (WIN) and the International Water and Sanitation Centre (IRC) to improve governance in the water sector (WIN and IRC, 2011). AWIS is designed to measure integrity, defined as a set of practices that impede corruption and promote respect for the rule of law. The practices that characterize integrity fall into three categories or "pillars": transparency, accountability, and participation, as defined in Table 1. AWIS is designed to be replicable and adaptable to water governance at all jurisdictional and geographical scales. Before this study, WIN adapted and piloted AWIS for implementation in several countries and contexts, including rural WASH services in Benin and school WASH services in India. In this study, we apply and evaluate AWIS in the context of high schools in rural Bangladesh, as described in the previous section.

AWIS is designed to initiate discussions that facilitate the identification of specific and concrete priority steps for improvement. The cornerstone of the intervention is a participatory workshop that brings together stakeholders. The workshop process is designed to facilitate constructive dialogue, to allow discussion of integrity without antagonizing participants with divergent views, to encourage revelation of new information, and to raise awareness of contrasting points of view or unexpected risks. Participating in the AWIS process is intended to help policy-makers and user representatives establish priority actions to enhance water integrity and governance.

The local implementing NGO, Development Organization for the Rural Poor (DORP), carried out the AWIS intervention between September and December 2018 (Figure 2). Local DORP staff facilitated the AWIS workshops and documented the process.

[9] E.g., when asked about the characteristics that a sanitation facility should have for safe MHM, a much higher share of female teachers than male teachers cited the importance of a mechanism for the collection and disposal of sanitary protection materials (56% vs 12%) or of a space to wash privately (62% vs 16%).

**Table 1.** *AWIS "pillars of integrity"*

| Pillar of integrity | Explanation |
|---|---|
| Transparency | The existence of written procedures, agreements and contracts that explain the roles and responsibilities of actors. |
| Accountability | The application of the written procedures and agreements and, where feasible, the potential compliance of actors ("internal accountability"). |
| Participation | The ability of the public, and the users or their representatives (including marginalised and resource-poor groups), to access information, influence decision-making, file complaints effectively and be heard ("external accountability"). |

*Notes* "Pillars of integrity" used to characterize "integrity" (from WIN and IRC, 2011).

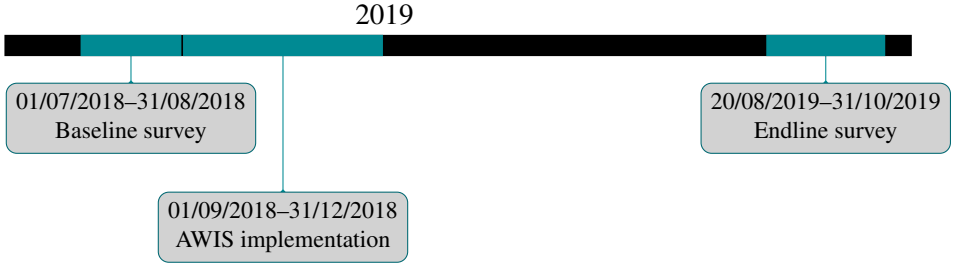



*Figure 2.* Project timeline

*Participants*

The stakeholders AWIS aimed to bring together comprised students, parents, teachers, the school management committee, social leaders, and local government administrators. By design, four representatives from each of the first four groups should have participated in the workshop. In practice, the number of meeting participants from each group varied slightly (Table 2).[10] In some cases, participants represented multiple groups.

The most important deviation from the original project design was that, with one exception, no local government administrators attended any workshops. The exception was one of the first workshops organized by the project, in Bhola, which was attended by a member of the union parishad or council. The absence of local government administrators likely reflects the fragmented nature of the institutional landscape. Government administrators have limited responsibility for improving WASH service provision in schools. Also, many governmental agencies have a limited staff. It may not be surprising that they

[10]In 8 out of 30 cases, the total number of participants was 18 (N = 1), 19 (N = 3), or 21 (N = 4), instead of 20 as per the intervention design.

were unwilling or unable to attend school-level workshops. The DORP facilitators invited social leaders in their place.

**Table 2.** *Participants in AWIS workshops*

| | Mean no. of participants per school | | |
|---|---|---|---|
| | Male | Female | Total |
| Students | 1.60 | 2.47 | 4.07 |
| Parents | 2.37 | 1.57 | 3.93 |
| Teachers | 2.53 | 1.10 | 3.63 |
| SMC members | 3.17 | 0.73 | 3.90 |
| Social leaders | 2.37 | 1.13 | 3.50 |
| Headteachers | 0.50 | 0.03 | 0.53 |
| Local government/administrators | 0.00 | 0.03 | 0.03 |
| Parents/SMC members | 0.10 | 0.13 | 0.23 |
| Teachers/SMC members | 0.10 | 0.03 | 0.13 |

*Notes* Mean number of participants in each category participating across schools. One workshop held in each school. SMC: School Management Committee.

*Process*

The AWIS workshop proceeds in three steps:

Workshop participants anonymously score the quality of WASH services and WASH-related processes with respect to indicators for transparency, accountability and participation. The scorecard covers five areas: (i) quality of sanitation facilities; (ii) gender; (iii) menstrual hygiene; (iv) facilities for disabled students; and (v) budget and expenditure. For each indicator, local facilitators developed a scale from 1 to 3 where each score is associated with a statement describing quality with respect to that indicator. Participants select which statement best described conditions at their school, yielding an individual score for each indicator between 1 to 3 (Appendix A).

Facilitators invite participants to openly discuss the scores for each area, asking participants to share their motivations behind their scores for each item. The objective is not to defend individual opinions but to understand differences in opinions. Based on the outcomes of this discussion, the scores are jointly adjusted and aggregated, until one set of scores is collectively agreed upon.

Each participant identifies priority areas for action and specific actions that can be taken within each priority area. Facilitators disclose the results of this exercise to participants and ensure that priority areas and proposed actions are clearly understood by all participants.

*Scores*

Facilitators recorded the scores agreed upon during steps 1 and 2 of the process (Appendix Figure K1). Consistent with the baseline survey evidence described in section 2, workshop participants hold different views about baseline conditions at the study schools. Two randomly-drawn workshop participants from the same school give, on average, the same scores for only half (0.52) of the items, and the within-school range of scores given by participants includes both the minimum and maximum score for about half (also, coincidentally, 0.52) of all items. Mean differences across schools explain less than half of the variation in scores across respondents, implying that more than half is driven by variation across workshop participants within schools.[11] While some of this residual variation is likely explained by measurement error,[12] mean scores given by participants correlate positively with direct, independent observations by enumerators during school visits (Appendix Table K1), suggesting that the information captured by the AWIS process is meaningful. Taken together, this descriptive evidence suggests that there is scope for the intervention to improve stakeholder understanding of conditions at the study schools.

*Priorities and actions*

Workshop participants identified, on average, a total of 20 priority areas and 20 specific actions that could be taken.[13] Around half of the most commonly-identified priority areas concerned improvements to physical infrastructure (Appendix Figure K2). The most-commonly identified actions included reaching out to possible sources of funding, awareness-raising, organizing meetings, or delegating responsibility for cleaning and monitoring of WASH facilities to students or teachers. Just under half of schools proposed to allocate a greater share of school funds towards WASH facilities (Appendix Figure K3).

*Costs*

A conservative estimate of the costs of implementing the AWIS intervention is around USD$770 per school, equivalent to USD$1.3 per student. This estimate is conservative because it includes costs such as liaising with local government officials that would likely reduce in proportional terms if the program were implemented at scale. If the intervention successfully increased school

[11] The (unadjusted) $R^2$ from a regression of participant scores on school-item fixed effects is 0.46.

[12] Some participants may also not have fully understood all the questions. Also, although scoring is intended to be anonymous, some participants required assistance to complete the scoring process, which might have affected their choices. Note that because scores were recorded anonymously, we cannot evaluate whether participants with different characteristics gave different scores.

[13] Participants listed priorities and actions descriptively. Research assistants categorized responses into discrete categories.

attendance, it would have been among the most cost-effective interventions available (J-PAL, 2018).[14] The per school implementation costs are additional to costs of about USD$23,000 for adapting the AWIS intervention for application to schools in rural Bangladesh and about USD$11,500 to train facilitators.

*Comparison with other previously-evaluated interventions*

Tables 3a and 3b compare AWIS to other closely-related "transparency for accountability" and "social accountability" interventions that have been evaluated in other sectors. Like AWIS, these interventions typically include elements of *information* and *participation*. The *information* elements can be broadly categorized into three types: 1) general, non-specific information, such as information about rights to public services or responsibilities of service providers; 2) specific information about the service provider (e.g., school or clinic) that is observable to the users or service providers; and 3) specific information about the service provider obtained from external sources, such as surveys or administrative data. The information elements in AWIS belong to types 1 and 2, while type 3 is typically associated with higher costs. The *participation* element comprises some type of meeting or workshop that usually brings service providers and users together. Relative to the other interventions in Tables 3a and 3a, the participatory element of AWIS is also relatively "light touch", comprising a single workshop.

*Ex ante strengths of AWIS*

The AWIS intervention has several features that Kosack and Fung (2014) note are associated with success in transparency interventions. Transparency interventions are more successful: when information is contextualized, for example with information about the rights of service users or responsibilities of the providers defined in national standards; when the information includes subjective impressions as well as objective information; and when the intervention establishes clear actions for users or providers that lead to reform. AWIS is designed to have all of these features: The scoring component of AWIS is defined with respect to Bangladeshi government guidelines with respect to school WASH provision; AWIS includes subjective impressions, such as whether or not girls and their parents feel able to influence the quality of support for MHM; and as noted above, AWIS is designed to establish consensus over specific actions that could be taken to improve outcomes. Thus, ex ante, AWIS appears to have features identified by previous literature as predictive of success.

[14]Each school has 575 students with, on average, two more years of secondary education remaining. Had the intervention increased attendance by 2.5 percentage points, a not implausible effect size ex ante, it would have increased total effective schooling by 3.75 years per USD$100 spent, more than any intervention listed by J-PAL (2018) except for deworming in Kenya (Miguel and Kremer, 2004).

**Table 3a.** *Interventions that affected development outcomes*

| Study | Information | Participation | Location | Sector | Time frame |
|---|---|---|---|---|---|
| Pandey et al. (2007) | Rights and responsibilities. | NA | India | Health and Social Services | 1 year |
| Pandey et al. (2009) | Rights and responsibilities. | NA | India | Primary Education | 2-4 months |
| Björkman and Svensson (2009), Nyqvist et al. (2017) | Rights and responsibilities, report cards (survey data). | Two community-service provider meetings workshops, held over 1-2 days, 6 months apart, focus on action plan and monitoring. | Uganda | Health | 2-4 years |
| Barr et al. (2012) | Community scorecard. | Scorecard unification process at school-parent-teacher meeting, three termly follow-up meetings | Uganda | Education | Immediately after interventions. |
| Pradhan et al. (2014) | NA | School management-village council meetings. | Indonesia | Primary Education | About 18 months |
| Mohanan et al. (2020) | Rights and responsibilities, report cards (survey data). | Monthly community-health committee meetings over 1 year, focus on action plans and adherence to plans. | India | Health | 1 year after end of interventions. |
| Raffler et al. (2020) | Rights and responsibilities, report cards (survey data). | Community-service provider workshop, held over 1 day, 6-monthly followups, focus on action plan. | Uganda | Health | 8-20 months |
| Christensen et al. (2021) | Report cards (administrative data). | Community-service provider workshop, three follow-up meetings, focus on action plan. | Sierra Leone | Health | 1.5-4 years. |

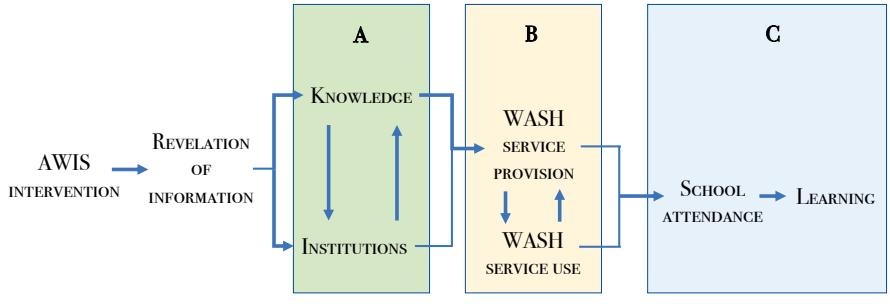
**Table 3b.** *Interventions that did not affect development outcomes*

| Study | Information | Participation | Location | Sector | Time frame |
|---|---|---|---|---|---|
| Olken (2007) | Anonymous comment cards to solicit community views, aggregated by external agents. | Promotion of community-village "accountability" meetings through invitations. | Indonesia | Road construction | Interventions after project, effects through anticipation. |
| Banerjee et al. (2010) | Rights and responsibilities, report cards (based on community members testing childrens' skills). | Community-headteacher-school committee discussion meeting. | India | Education | 1 year |
| Lieberman et al. (2014) | Report cards for individual children (NGO staff assessments), strategies for parents to get involved locally/nationally. | NA | Kenya | Education | 5-6 months |
| Nyqvist et al. (2017) | NA | Two community-service provider meetings workshops, held over 1-2 days, 6 months apart, focus on action plan and monitoring. | Uganda | Health | 2 years |
| Fabbri et al. (2019) | Report cards (survey data) | Separate meetings with service providers and/or communities. | India | Health | 18 months |
| Arkedis et al. (2021) | Report cards (data gathered by field staff) | 6 facilitated community-only meetings, held monthly, focus on action plans. | Indonesia and Tanzania | Health | 18 months |
| BenYishay et al. (2022) | NA | Multi-stakeholder dialogue workshop | Niger | Health and education | 10 months |

# 4 Theory of change

Our evaluation design is guided by a theory of change that connects the AWIS intervention to the outcomes of interest (Figure 3). AWIS is designed to spread information about priority areas for action and potential courses of action, and to strengthen institutions. In turn, these changes should improve WASH service provision and may impact on WASH facility use patterns. Improved WASH service provision may in turn increase school attendance and translate into improved exam performance. Drawing on this theory of change, we define three families of outcomes, designated A, B, and C, which are listed in Table 4 and described later in further detail.

*Figure 3.* Theory of change




The implicit logic underlying transparency interventions is the idea that such interventions can ignite an "action cycle", in which information translates into useful, constructive action that improves service provision (Kosack and Fung, 2014). To initiate such an "action cycle", several criteria must be met. First, the intervention must reveal new information, and the new information must be salient to the users or providers of a service. Second, the revealed information must change the choices or actions of service users or providers. Third, these changes in choices or actions must be sufficient to affect service provision.[15]

To trace out the links between information and action, we provide descriptive evidence about the information that is discussed during the AWIS workshops, actions that are identified during the workshops, and actions that participants report taking after the workshop. Additionally, we evaluate impacts on knowledge about WASH standards and provision, and institutional arrangements (outcome family A). We define institutional arrangements as the presence and quality of institutions for WASH facility management and provision of support to students for MHM. Such institutions include, for instance, whether or not the school has written policies related to WASH issues, whether the school allocates a budget to WASH services, and how frequently school representatives meet with parents. Knowledge and institutions may be self-

[15] We build from a similar discussion in Kosack and Fung (2014).

reinforcing. Improvements to knowledge or institutional arrangements constitute potentially important mechanisms for improvements where we expected that we might detect effects of the intervention even if they are not sufficient to improve service provision over the time frame of our intervention.

We then evaluate impacts on WASH service provision and on use of WASH services (together constituting outcome family B). Understanding effects on service use is essential in order to correctly interpret any effects on WASH service provision because the two are closely related. If WASH service provision improves, use of WASH facilities might increase in response. Increased use might in turn lead to problems with, for example, cleanliness or maintenance that might not have arisen in the absence of changes in use patterns.

The final piece in our theory of change is the link between WASH provision, school attendance, and school attainment (outcome family C). A widespread view is that inadequate WASH provision leads to reduced school attendance, through an increased rate of water-borne disease, and for female students, as a result of students missing school during menstruation (Alam et al., 2017; McMichael, 2019). School attendance is in turn linked to school attainment (e.g., Roby, 2004).

# 5 Data

Data collection was managed by a Bangladeshi non-government organization, NGO Forum for Public Health, who were not involved in implementation.[16] NGO Forum for Public Health employed a team of enumerators who collected baseline data before the intervention, during July and August 2018, and endline data between August and October 2019, between 8 and 12 months after the intervention.

One male enumerator and one female enumerator visited each study school for the primary data collection activities, which took one day in each school. We designed the data collection process to minimize disruption to the school day, with a flexible schedule that allowed enumerators to arrange data collection and interviews around student and teacher activities. The data collection process comprises surveys with three main classes of respondents and four data collection modules at both baseline and endline, which we describe briefly below.

Headteachers gave informed consent to the study as a whole and separately to each of the data collection modules. Headteachers in all study schools consented to the data collection process, although in a few cases they declined consent for one or more of the data collection modules.

Enumerators surveyed headteachers, teachers, and students. Female enumerators interviewed female respondents. All interviewees individually assented or consented to be surveyed, although we did not ask respondents about

[16] See `http://www.ngof.org`.

private or sensitive information. In particular, we phrased all questions related to WASH and MHM practices in general terms, asking respondents to draw on both their own and their peers' experiences in their responses.[17]

Enumerators asked all respondents to describe the quality of the existing WASH facilities in the school; the constraints faced by female students during menstruation and the consequences for school attendance; and their views about how to improve WASH services and practices in the school. Additionally, enumerators asked respondents for information about specific areas of interest, as follows:

*Headteachers*

Enumerators interviewed headteachers regarding school programs and initiatives related to WASH and MHM; student, parent and teacher involvement in WASH facility management; and the school budget and institutional processes, especially with respect to WASH facilities.

*Teachers*

Enumerators interviewed male and female teachers in each school regarding the content of school curricula related to WASH and MHM; the system of responsibilities within the school for cleaning of WASH facilities; and student awareness of WASH and MHM practices. In ten schools, there were no female teachers, so we interviewed only male teachers.

*Students*

Enumerators interviewed male and female students regarding whether and how WASH and MHM practices are discussed and taught in school. To identify students likely to be well-informed and comfortable with an interview, headteachers identified "class captains", a position assigned by teachers to the most responsible and best-performing students. The vast majority of interviewed students were in 10th grade and between 14 and 16 years old.[18]

Enumerators also interviewed school cleaning staff regarding the availability of cleaning products and the cleaning and maintenance schedules. In practice, however, few schools had cleaning staff present on the day of the interview. In particular, at baseline, only 7 schools had any cleaning staff of which which 5 had cleaning staff present on the day of interview.[19]

[17] Appendix B describes the consent process.

[18] Two interviewed students were in 9th grade; three were aged 17 years old; and one was aged 13.

[19] We successfully interviewed a member of the cleaning staff whenever one was present.

Enumerators also collected data directly on WASH facilities, WASH use, and student attendance via school records and direct observations.

*WASH facility census*

Enumerators observed and photographed all WASH facilities in study schools, including drinking water points, toilet cubicles, urinals, and handwashing stations. Enumerators recorded whether each facility complied with basic WASH and MHM standards. We also tested all sources of drinking water available in the school for contamination with fecal bacteria and arsenic. Table 5 enumerates the number of school facilities of each type at baseline and endline.

*Latrine usage*

We used event loggers and sensors to monitor latrine usage over three school days in an anonymous, discreet, and non-intrusive way.[20] We tracked latrine usage in a sample of toilet cubicles that were functioning and accessible to students.[21] We primarily used magnetic sensors to record opening and closing of latrine doors. Processing these data allows us to recover a history of latrine use events, corresponding to periods when the door was closed bookended by periods when the door was open to allow entry to and exit from the latrine. Sensors and event loggers are small and were discreetly packaged. Importantly for privacy, the door sensors can be installed outside of the latrine cubicle. Enumerators installed and removed sensors at times when they were not in use by students, in order to interfere as little as possible with normal latrine use.[22]

[20] Specifically, we collect data over three days, excluding holidays and Fridays, when schools are closed.

[21] This implies that we did not collect data if there was no functioning toilet accessible to students.

[22] At baseline, we used motion sensors in some cases when latrines did not have functional doors. Motion sensors use infra-red sensors to detect the presence of a warm body. See Appendix C for further details.

**Table 5.** *Number of observations: sanitation facilities*

| | Number of observations | | | | | |
|---|---|---|---|---|---|---|
| | Baseline | | | Endline | | |
| Facility type | Male | Female | Mixed | Male | Female | Mixed |
| Toilet cubicle (in use) | 54 | 95 | 7 | 47 | 103 | 6 |
| Toilet cubicle (out of use) | - | - | 70 | - | - | 61 |
| Urinal | 62 | - | - | 54 | - | - |
| Hand-washing station | 63 | 86 | 57 | 50 | 74 | 68 |
| Drinking water point | - | - | 60 | - | - | 60 |
| Total | 179 | 181 | 194 | 151 | 177 | 195 |

*Notes* Table shows number of facilities available for student use enumerated in each category. Toilets that are out of use are listed as for mixed use since gender of users is not defined.

Although we designed this data collection process to minimize intrusion, it is still possible that students might have avoided use of the school latrines during monitoring, a type of "Hawthorne" effect or response to monitoring. However, monitoring was identical in both treated and control schools, and it seems unlikely that students in treated schools would have responded differently to monitoring. As a result, any "Hawthorne" effect should not affect the comparison between treated and control schools.

*Attendance records*

Enumerators scanned and digitized school attendance records for the month in which the survey takes place and additionally for the preceding February and March. We selected these months for comparison across years because no major religious festivals or national exams interrupted school attendance patterns during these months in 2018 or 2019.

*Observed attendance*

To validate the attendance records, enumerators collected direct observations of student attendance on the day of the survey. To observe attendance discreetly, enumerators visited classrooms and asked students to list three improvements they would like to make in their school. Students then voted for their preferred suggestions. Enumerators recorded the number of male and female students voting for each suggestion, allowing them to record the total numbers of male and female students present in class. Enumerators made clear that the voting exercise was purely hypothetical.

Tables 6 and 7 summarize sampling strategies and response rates.[23] Response rates were high, with one exception. The original baseline protocol sampled more teachers than enumerators could feasibly interview on the day of the visit, and the baseline survey form did not enforce compliance with this protocol. Although response rates were low relative to the targeted numbers, enumerators did successfully interview at least one male and at least one female teacher in all study schools with teachers of the given gender. Baseline teacher response rates are similar at baseline in treatment and control schools (Appendix Table K2).[24] We addressed these issues in the endline survey.

We also collected data from three additional sources: administrative data on exam results, workshop records, and semi-structured interviews in a subsample of schools.

*Exam results*
Enumerators downloaded individual exam results for students at study schools from official records publicly available online. Enumerators recorded individual student results (grade point averages, or GPAs) on the most recent standardized national exams for grade 8 (Junior School Certificate, JSC) and grade 10 (Secondary School Certificate, SSC), inferring gender from student names, which are mostly gender-specific in Bangladesh.[25]

*Workshop records*
Staff from DORP, the local implementing NGO, completed a detailed report after each workshop, including participant information, workshop processes, discussion outcomes, and the main lessons learned. DORP staff translated the reports into English. We digitized and classified data from these reports, which we use in descriptive analyses.

*Semi-structured interviews*
To obtain qualitative insights into the strengths and weaknesses of the intervention, enumerators conducted semi-structured phone interviews with up to 4 respondents from a total of 12 schools, evenly sampled at random from treatment and control schools in the two study upazilas. We conducted the interviews in fall 2020, a year after the endline data collection and two years after

[23] Further details are in Appendix D. Our study design also oversampled AWIS student participants, but in practice this led to only two additional student interviews.
[24] Teachers listed later in the roster were less likely to be successfully interviewed (Appendix Table K3).
[25] Both sets of results are publicly available online on the education board's web-based result publication system. School-level records list individual student results and student ID numbers. Detailed individual student data are also publicly available, including student names, searchable using the student ID number.

implementation. Schools were closed for in-person instruction at the time of the semi-structured interviews due to the COVID pandemic. In most schools, we successfully interviewed the headteacher, a parent representative from the school management committee, one male teacher, and one female teacher.[26]

[26]We originally planned to conduct in-person focus group discussions in study schools in spring 2020. Because of the COVID pandemic, we initially postponed these activities and then replaced them with phone interviews to minimize unnecessary travel and in-person contact. School closures prevented us from interviewing students. See Appendix E for further details.

**Table 4.** *Outcomes*

| | Outcome | Unit of observation | Time fixed effects | Unit fixed effects |
|---|---|---|---|---|
| A1 | Head-teacher knowledge | Headteacher | Upazila-year | School |
| A2a | Male teacher knowledge | Teacher | Upazila-year | School |
| A2b | Female teacher knowledge | Teacher | Upazila-year | School |
| A3a | Male student knowledge | Student | Upazila-year | School |
| A3b | Female student knowledge | Student | Upazila-year | School |
| A4 | Quality of institutional processes | School | Upazila-year | School |
| B1a | No. functioning toilets (male student use) | School | Upazila-year | School |
| B1b | No.functioning toilets (female student use) | School | Upazila-year | School |
| B1c | No. functioning toilets (mixed use) | School | Upazila-year | School |
| B2a | Male toilet quality index | Toilet | Upazila-year | School |
| B2b | Female toilet quality index | Toilet | Upazila-year | School |
| B3a | Toilet events/day (male student toilets) | Toilet | Upazila-year | School |
| B3b | Toilet events/day (female student toilets) | Toilet | Upazila-year | School |
| C1a | School attendance (male students) | Student-day | Grade-upazila-date-year | School-grade-day of week |
| C1b | School attendance (female students) | Student-day | Grade-upazila-date-year | School-grade-day of week |
| C2a | Exam results (male students) | Student | Level-upazila-year | School-level |
| C2b | Exam results (female students) | Student | Level-upazila-year | School-level |

*Notes* Date fixed effects refer to day and month fixed effects. Day of week fixed effects comprise fixed effects for each weekday; schools are closed on Fridays. Year fixed effects refer to calendar year fixed effects, corresponding to periods pre and post intervention.

**Table 6.** *Sampling strategies, sample sizes, and response rates: survey modules*

| Module | Unit of observation | Sampling | Number of observations and response rates | | | | Primary reasons for non-response |
|---|---|---|---|---|---|---|---|
| | | | Baseline | | Endline | | |
| | | | Male | Female | Male | Female | |
| Headteacher survey | Headteacher | Census | 58 (100.0%) | 2 (100.0%) | 58 (100.0%) | 2 (100.0%) | N/A |
| Teacher survey | Teacher | Random sample of one M and one F teacher, plus oversampling of WASH/MHM- responsible teachers and AWIS participants. | 124 (66.0%) | 52 (55.3%) | 136 (91.3%) | 76 (100.0%) | Baseline coding/ protocol error, teacher non-consent. |
| Student survey | Student | Random sample of one M and one F class captain from the highest grade with a class captain present during the survey. | 48 (98.0%) | 60 (100.0%) | 49 (100.0%) | 60 (100.0%) | Student non-assent (1 student). |

*Notes* M: male, F: female. All numbers of observations correspond to numbers of successfully completed surveys. Response rates defined relative to number of respondents targeted for interview and shown in brackets. Primary reasons for non-response list the most commonly-occurring reasons.

**Table 7.** *Sampling strategies, sample sizes, and response rates: direct observation modules*

| Module | Unit of observation | Sampling | Number of observations and response rates | | | | Primary reasons for non-response |
|---|---|---|---|---|---|---|---|
| | | | Baseline | | Endline | | |
| | | | Male | Female | Male | Female | |
| WASH facilities | WASH facility | Census of toilets, urinals, handwashing facilities, and water points. | 554 facilities in 60 schools (100.0%) | | 523 facilities in 60 schools (100.0%) | | N/A |
| Latrine usage | Toilet | Up to 4 toilets that are functioning and accessible to students; randomly sampled if more than 4. | 46 toilets (85.2%) | 82 toilets (90.1%) | 32 toilets (74.4%) | 87 toilets (90.6%) | Enumerator or sensor error. |
| Attendance records | Student | One class per grade, randomly sampling between classes if more than one class. | 8532 students in 234 classes (95.5%) | 14085 students in 289 classes (96.3%) | 9791 students in 244 classes (99.6%) | 15183 students in 299 classes (99.7%) | Headteacher non-consent (2 schools). |
| Attendance observations | Student | One class per grade, randomly selecting between classes if more than one class. | 3955 students in 234 classes (95.5%) | 7720 students in 289 classes (96.3%) | 4291 students in 235 classes (95.9%) | 7968 students in 290 classes (96.7%) | Headteacher non-consent (2 schools). |
| Exam results | Student | Administrative data | 5181 students in 49 schools (100.0%) | 5666 students in 60 schools (100.0%) | 5039 students in 49 schools (100.0%) | 6002 students in 60 schools (100.0%) | N/A |

*Notes* Number of observations for attendance records corresponds to number of students for whom we have attendance data in February and/or March of the corresponding year. Response rates shown in brackets. Response rates for WASH facilities and exam results calculated at the school level. Response rates for latrine usage relative to the toilets selected for data collection according to the protocol; mixed gender toilets (7) pooled with male-only toilets for the purposes of reporting sample size and response rates in this table. Response rates for attendance observations are at the class level.

# 6 The experiment

We randomly assigned 30 schools to receive the intervention and 30 schools to a control group that received no intervention via this project, although importantly we did not prevent them from receiving any other interventions. All schools had the same ex ante probability of receiving the program.

We assigned schools to treatment and control at public lotteries, ensuring transparency over the treatment assignment process. We held one treatment assignment lottery in each of the two study upazilas, Bhola Sadar and Ramgati, with 15 schools in each upazila assigned to treatment and 15 to control. The lotteries took place after baseline data collection.

Figures 4 and 5 shows the location of treated and control schools in Bhola Sadar and Ramgati, respectively. Balance tests using baseline data confirm that differences between schools assigned to treatment and control are consistent with differences that could arise due to chance. Table 8 shows that the treatment and control groups are balanced with respect to the main outcomes at baseline.[27]

A potential threat to experimental validity arose because of a second intervention that was carried out in schools during this time period, consisting of distribution of free sanitary products to schools for female students and education about menstrual hygiene. This intervention was carried out in 5 study schools before baseline and 13 study schools between baseline and endline. We successfully coordinated with the implementing organization to ensure that assignment to this intervention was uncorrelated with treatment with the AWIS intervention (Appendix Table K11).

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[27]To report tests of joint significance of differences across outcomes, we collapse all data to weighted school-level means, using sampling weights as we do throughout the analysis. To avoid missing observations, we pool data from male and female respondents, and from toilet facilities used by male, female, and mixed students. When we pool data across genders, we use indices constructed on pooled data. We residualize all outcomes with respect to upazila fixed effects, interacted with gender, grade, exam level, and date fixed effects where relevant, for comparability with the randomization checks on individual outcomes. For latrine use data, we include sensor-type fixed effects as we do for the individual checks. We replace missing data with the upazila-level mean value: for one school for outcome B2, five schools for outcome B3, and one school for outcome C1. Appendix Tables K4 to K10 provide extended baseline comparisons between schools, including additional joint tests of comparability on 20 school-level characteristics.

*Figure 4.* Study schools in Bhola Sadar upazila

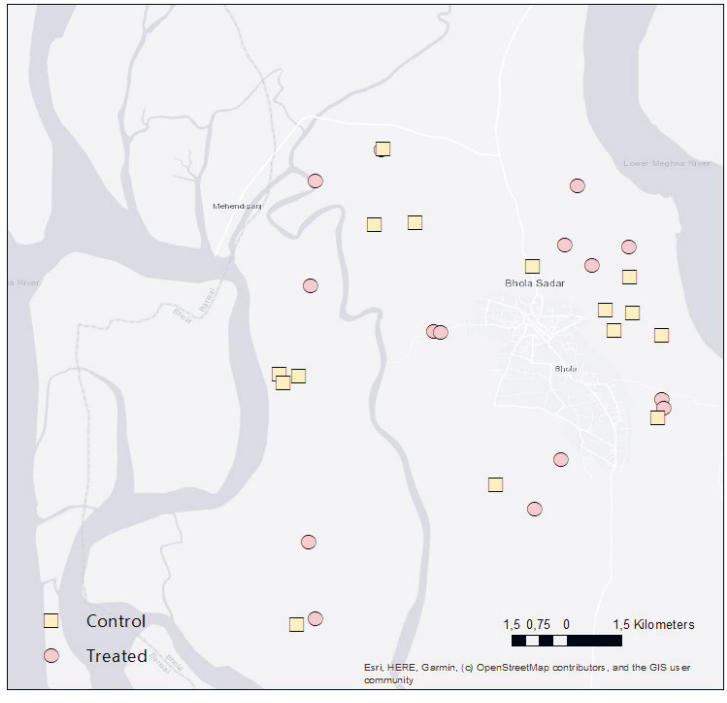



*Figure 5.* Study schools in Ramgati upazila

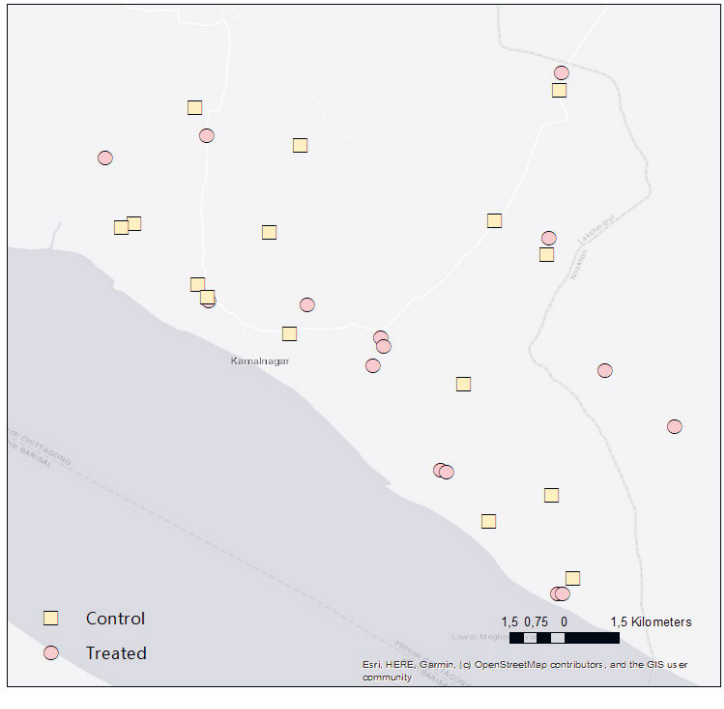


**Table 8.** *Baseline summary statistics*

| | Control | Treated | Obs |
|---|---|---|---|
| A1 Head-teacher knowledge | 0.000 | -0.034 | 60 |
| | (0.113) | (0.100) | |
| A2a Male teacher knowledge | -0.000 | -0.094 | 124 |
| | (0.088) | (0.099) | |
| A2b Female teacher knowledge | -0.001 | 0.378* | 52 |
| | (0.127) | (0.133) | |
| A3a Male student knowledge | -0.013 | -0.073 | 48 |
| | (0.087) | (0.142) | |
| A3b Female student knowledge | 0.000 | -0.098 | 60 |
| | (0.106) | (0.121) | |
| A4 Institutional quality | -0.003 | 0.006 | 60 |
| | (0.066) | (0.061) | |
| B1a No. functioning toilets (male student use) | 1.200 | 1.008 | 49 |
| | (0.256) | (0.199) | |
| B1b No. functioning toilets (female student use) | 1.267 | 1.900** | 60 |
| | (0.165) | (0.165) | |
| B1c No. functioning toilets (mixed use) | 0.200 | 0.033 | 60 |
| | (0.082) | (0.037) | |
| B2a Male toilet quality index | -0.011 | -0.044 | 92 |
| | (0.059) | (0.075) | |
| B2b Female toilet quality index | -0.004 | 0.063 | 127 |
| | (0.054) | (0.056) | |
| B3a Toilet events/day (male student toilets) | 10.113 | 9.000 | 80 |
| | (2.861) | (1.760) | |
| B3b Toilet events/day (female student toilets) | 11.603 | 14.312 | 114 |
| | (2.735) | (3.195) | |
| C1a Daily attendance (male students) | 0.602 | 0.554* | 320988 |
| | (0.020) | (0.021) | |
| C1b Daily attendance (female students) | 0.673 | 0.644 | 552949 |
| | (0.017) | (0.015) | |
| C2a Exam result (male students) | 2.171 | 2.385* | 5131 |
| | (0.083) | (0.079) | |
| C2b Exam result (female students) | 2.296 | 2.251 | 5586 |
| | (0.079) | (0.039) | |
| $p$ value of F-test for joint significance | | 0.456 | 60 |
| $p$ value of Hotelling's T-Squared test | | 0.455 | 60 |

*Notes* Standard errors clustered by school and shown in parentheses. Results from regressions with stratification controls. Unit of observation as listed in Table 4. All observations weighted so that each school (or school and grade/level, for outcomes in family C) counts equally in summary statistics. Attendance comparisons include day-grade-upazila fixed effects; exam result comparisons include exam level-upazila fixed effects; all other regressions include upazila fixed effects. Latrine use comparisons also include sensor type fixed effects. $p$ values from joint F test and Hotelling'ÂÂs T-squared test joint significance of differences on all outcome variables between treatment and control groups. *** $p$ <0.01, ** $p$ <0.05, * $p$ <0.1.

# 7 Empirical approach

*Estimating equation*

We analyze data as a panel or a repeated cross-section. When the unit of observation is the school, the data form a balanced panel. When the unit of observation is more granular, such as at the student level, the data form a repeated cross-section, because both the population and the sample may change between baseline and endline, for example because some students graduate and new students enroll. We estimate the following equation:

$$Y_{ijt} = \alpha_j + \gamma_t + \beta \left(1_{treated,j} \times POST_t\right) + \beta_u \left(U_j \times 1_{treated,j} \times POST_t\right) + \varepsilon_{ijt} \quad (1)$$

where $Y_{ijt}$ is an outcome variable in unit of observation $i$ in school $j$ at time $t$. The $\alpha_j$ term is a unit fixed effect that absorbs baseline differences across schools, and $\gamma_t$ is a time fixed effect that absorbs overall time trends. In the simplest specifications, unit fixed effects are school fixed effects, and time fixed effects are upazila-year fixed effects. When we have more granular data available we additionally account for variation across subgroups or more comprehensively across time (see Table 4). The indicator variable $1_{treated,j}$ takes the value one if school $j$ is assigned to treatment and zero otherwise, the indicator variable $POST_t$ takes the value one at endline and zero at baseline, and the variable $U_j$ is a demeaned upazila dummy that accounts for stratification in random assignment to treatment. The coefficient of interest is $\beta$, the difference-in-differences between treated and control schools. Including the centered and interacted stratification control ensures that $\beta$ is estimated consistently (Lin, 2013; Imbens and Rubin, 2015; Gibbons et al., 2019). Random assignment of treatment to communities ensures that $1_{treated,j}$ is uncorrelated with the error term $\varepsilon_{ijt}$, implying that estimates of $\beta$ have a causal interpretation.

*Hypothesis testing*

The number of schools in our study is small, so we use randomization-based inference to evaluate the statistical significance of the estimated coefficient, $\hat{\beta}$. We simulate the lottery that assigns schools to treatment and control 1000 times. We then estimate the distribution of $\hat{\beta}$ under the sharp null hypothesis of no difference between treated and control schools that is imposed by the simulations. We then compare the realized differences between treated and control schools to the distribution of differences under the null. Because the number of schools in our study is relatively small and the direction of the main hypotheses was clear—the intervention is designed to improve outcomes—we pre-specified one-sided hypothesis tests to maximize power to detect improvements. The one-sided hypothesis tests evaluate whether or not we can reject the null hypothesis that treatment did not improve outcomes. We also report analytical standard errors, accounting for clustering at the school level.

*Weighting*

The number of observations per school differs for different outcome variables. To ensure consistency between analyses, we always weight observations by the inverse of the number of observations per school.[28] This approach ensures that each school counts equally in summary statistics in all analyses.

*Pre-specification*

We pre-specified our analyses, and we follow a pre-analysis plan.[29] We deviate from this pre-analysis plan in one significant respect. We originally pre-specified that our primary hypothesis tests would pool data from male and female respondents, or toilet facilities used by male and female students, and that we would report effects disaggregated by gender as secondary hypotheses. However, male and female respondents are very different, as are toilets used by male and female students. Female respondents are much more knowledgeable about WASH/MHM, and toilets used by female students have higher quality at baseline (Appendix Figures K4a to K4c). Not all schools have male students and not all schools have female teachers, implying that pooling across genders also involves pooling across different populations. Ex post, we prefer the cleaner and more transparent approach of reporting results separately for each gender.[30] The smaller sample sizes for female teachers and male students are reflected in larger confidence intervals for effects in these groups.

[28]For outcomes A2a and A2b, the weights we apply account for differential probabilities of inclusion in the sample for different types of teacher and specifically for oversampling of teachers with special responsibility for WASH or MHM. For outcome B3a and B3b, weights account for differential probabilities of inclusion in the sample for functioning and non-functioning toilets. For outcomes in family C, we weight observations so that each grade (C1a and C1b) or exam type (C2a and C2b) contributes equally, ensuring that the estimated effects are insensitive to changes in the distribution of students across grades or exam types.

[29]We pre-registered hypotheses on July 17th, 2018, during baseline data collection and before implementation took place. We finalized a pre-analysis plan using the baseline data on September 25, 2019, after endline data collection had begun but before analyzing any endline data. RCT ID: AEARCTR-0003111.

[30]Our original approach also created inconsistencies between results in the pooled samples and the gender-separated samples. Weighting schools equally in the gender-separated samples implies weighting each school and gender equally, whereas weighting schools equally in the pooled regression implies that changes in the gender characteristics of the sample are reflected in the estimated changes in outcome variables. In principle, we could have anticipated this issue when we wrote the pre-analysis plan. In practice, the problem did not occur to us until we implemented the analyses after endline. We also deviate from the pre-specified approach in a second, more minor respect. We replace school-grade-date fixed effects with school-grade-day of week fixed effects for outcomes C1a and C1b. We make this change because our original approach effectively dropped data from dates in 2019 that were Fridays in 2018 and vice versa, about one in six observations. Appendix Table K12 shows the originally pre-specified primary hypotheses and regression specifications.

*Multiple hypothesis testing*

We report results for a total of seventeen outcomes in three families. There is thus a relatively high likelihood that we reject the null hypothesis in at least one of these tests due to chance, even if the null hypothesis of no improvement in outcomes is in fact correct. Along with the "naïve" $p$ values, which would be correct for tests of any one of the outcomes in isolation, we also report $p$ values that control the family wise error rate (FWER) within each of our three outcome "families". The FWER-$p$ values we report are obtained from simulating the joint distribution of $p$ values for all outcomes within one family and comparing the "naïve" $p$ value for each outcome to the minimum $p$ value across all outcomes in the family (Westfall and Young, 1993; Anderson, 2008).[31] A drawback of disaggregating all results by gender is that it expands the number of hypotheses we analyze, increasing the risk of erroneously rejecting a null hypothesis that is in fact correct. On the other hand, expanding the number of hypotheses we test makes the already-conservative FWER-adjusted results still more conservative.

*Joint hypothesis tests*

Often, the strongest evidence for impacts comes not from large effects for one outcome but from modest but consistent effects across many outcomes (Young, 2019). We thus also report a joint test for each family of hypotheses which ranks the $p$ values from smallest to largest and compares the distribution of each of these to the distribution of ranked $p$ values obtained from randomization-based inference. This allows us to infer the likelihood of seeing differences at least as large, in relative terms, as the observed differences across all outcomes within each family. We did not pre-specify these joint hypothesis tests.

*Exploratory analyses*

Given that we unexpectedly observed cases in which the intervention worsened outcomes relative to the control group, we also report $p$ values from two-sided hypothesis tests. These $p$ values evaluate the likelihood that differences with the realized absolute magnitude would have occurred under the null hypothesis of no effect. These tests are not pre-specified, and our experiment was not powered to detect negative effects, so these results should be treated as exploratory.

[31] Appendix F provides more details on how we implement multiple and joint hypothesis testing. We note that we pre-specified that we would also report $p$ values that control the FWER across the full set of outcomes, but since none of the results are independently statistically significant when we control the FWER within each of the three families, the more conservative $p$ values do not provide much additional information.

*Index construction*

Throughout, we follow Anderson (2008) and use inverse covariance weighting to combine several sub-indices in summary indices. This approach weights information from several standardized outcomes by the inverse of their covariance matrix in order to maximize the amount of information captured in the index. Following the original formulation in Anderson (2008), we calculate the index across all non-missing data for each observation separately, ensuring that we retain the maximum possible amount of information in construction of the index.[32] In all cases, we normalize the index relative to control schools at baseline, so that changes with respect to baseline and treatment effects are measured with respect to pre-intervention conditions. Note that index variables are thus close to zero—or exactly zero when there are no missing data—for control schools in Table 8 by construction.

*Power*

Although the number of treated and control units is relatively small, constrained by the available budget for both the intervention and data collection, we maximize our ability to detect effects by collecting data wherever possible at a more disaggregated unit of observation (e.g., for individual WASH facilities or latrine cubicles or individual students) or over a greater period of time (e.g., over multiple days). Ex post power calculations, using either analytical standard errors or the distribution of the point estimates obtained from randomization-based inference, suggest that the study was powered to detect effects of reasonable magnitude compared to ex ante expectations, typically an increase of 0.2 to 0.4 standard deviations for the outcome variables that are indexes, characterized as small to medium-sized effects (Cohen, 1988).[33]

*Robustness tests*

We implement a series of pre-specified robustness tests. We re-estimate all the main effects using control variables selected by the Lasso algorithm from a pre-specified group of exogenous control variables comprising baseline school-level characteristics interacted with an endline dummy. The results remain very stable in these alternative specifications.[34] In the interests of brevity, we

[32] We adapt code generously shared by Samii (2018) so that it deals with missing observations.

[33] See Appendix Table K13. We calculate minimum detectable effects for one-sided hypothesis tests at 90% significance and 80% power as 2.12 times the estimated standard error of the point estimate (Djimeu and Houndolo, 2016), using both the analytical standard errors and the distribution of point estimates from the randomization-based inference; both approaches give similar conclusions. Note that while ex post power calculations that use estimated effect sizes are known to be misleading (see, e.g., Gelman, 2019), this problem does not affect ex post power calculations that use only information about standard errors.

[34] See Appendix G for details of how we implemented this approach and results tables estimated using this approach.

summarize the results of all other robustness tests when we discuss the relevant results.[35]

# 8 Results

Motivated by the theory of change described in section 4, we evaluate the effect of the AWIS intervention on the following three outcome families: knowledge, processes and institutions (A); WASH provision and use (B); and educational outcomes (C). In the interests of readability, we integrate discussion of how we construct each specific outcome within each family with the discussion of results.

## A: Knowledge and institutions

We estimate effects on six outcomes in this family. We measure knowledge of WASH and MHM standards for headteachers (A1), male and female teachers (A2a and A2b), and male and female students (A3a and A3b) using indices. Each index is created from several sub-indices that measure the share of correct responses given to a number of questions about best practices for WASH and MHM. For teachers and students, we construct indices separately for male and female respondents. We measure institutional quality using an index capturing the presence and quality of institutions to manage WASH facilities and provide support to students for MHM (A4).[36]

Results are shown in Table 9. Most (5 out of 6) of the indices show small to moderate positive treatment effects. The improvements in knowledge are largely driven by changes in knowledge about MHM while the improvements in institutions are driven by increased frequency of parent meetings and an increase in the likelihood that the school has a budget specifically assigned to cleaning WASH facilities.[37] Two of the individual treatment effects are statistically significant at the 10% level, although neither of these effects are statistically significant when we control the family-wise error rate (FWER). The likelihood of seeing effects of at least this magnitude across all outcomes is relatively low, however. Under the sharp null of no positive effects on any outcome, the probability of seeing this distribution of $p$ values is .031 (see also Appendix Figure K5). Thus the results seem most consistent with modest positive effects on knowledge and institutions across a range of outcomes.

These results are also consistent with our descriptive data. Semi-structured interview respondents and field staff from DORP (the implementing NGO) re-

[35] We also implement robustness tests that we did not prespecify. When we do so, we note that this is the case.

[36] Detailed information on the construction of all outcomes in family A is provided in Appendix H.

[37] Appendix Tables K14 to K17 report effects broken down by subindex.

ported that some schools painted the government guidelines for school WASH on school walls after the AWIS workshop. Participants also reported that the workshop helped them understand their rights and responsibilities with respect to WASH (Appendix Figure K6).

The main sample for male and female teachers comprises teachers selected at random and teachers selected because they have special responsibility for WASH or MHM education. These groups are sampled consistently across treatment and control groups. Just over two-fifths of the teachers in the main analysis in treated schools themselves participated in the AWIS workshop. Improvements in teacher knowledge are driven by the randomly-sampled teachers rather than those with special responsibility for WASH or MHM (Appendix Table K18). The results are not sensitive to excluding new staff from the sample (Appendix Table K19).[38]

**Table 9.** *A: Knowledge and institutions*

| | Head-teacher | Teacher knowledge | | Student knowledge | | Institutional |
|---|---|---|---|---|---|---|
| | knowledge | Male | Female | Male | Female | quality |
| Treatment × post | 0.264 | 0.121 | -0.032 | 0.123 | 0.249 | 0.109 |
| | (0.169) | (0.164) | (0.223) | (0.213) | (0.177) | (0.132) |
| One-sided $p$ | 0.075 | 0.235 | 0.560 | 0.275 | 0.085 | 0.213 |
| FWER one-sided $p$ | 0.371 | 0.628 | 0.628 | 0.628 | 0.371 | 0.628 |
| Two-sided $p$ | 0.133 | 0.466 | 0.889 | 0.553 | 0.166 | 0.430 |
| N | 120 | 226 | 111 | 96 | 120 | 120 |
| Baseline mean | -0.008 | -0.041 | 0.179 | -0.062 | -0.047 | 0.000 |
| Change control | -0.191 | -0.104 | 0.175 | -0.040 | 0.080 | 0.178 |
| Change treated | 0.072 | 0.018 | 0.143 | 0.093 | 0.330 | 0.287 |
| Unit of observation | Headteacher | Teacher | Teacher | Student | Student | School |

*Notes* Standard errors clustered by school and shown in parentheses. Results from regressions that include school and upazila-year fixed effects and stratification controls. Weights applied so that all schools have the same weight in summary statistics. One-sided $p$ values (pre-specified) test the null hypothesis of no positive effects; FWER-controlled $p$ values (pre-specified) hold the possibility of making type I errors within the family to its nominal level. Two-sided $p$ values (exploratory) test the null hypothesis of no effects of any sign. Changes in treated and control schools are raw weighted mean changes between baseline and endline for each group, and the difference between the two may not correspond to the estimated treatment effect.

[38] We did not pre-specify the robustness tests desribed in this paragraph.

## B: WASH service provision and use

We measure effects on seven outcomes within this family (Table 10). First, we enumerate the number of functioning toilets that are available to students in each school, disaggregating these results by the number of toilet cubicles available to male students only (B1a), female students only (B1b), and for mixed use (B1c). A toilet cubicle is an individual stall, seat, squat-plate or drop-hole where a person can defecate in private. Toilets are defined as functioning if they can be used and are not broken, damaged, or full. Relative to the control group, we cannot reject the null hypothesis of no positive impact on the number of functioning toilets available to any group. The number of functioning toilets for female student use actually declines in relative terms, primarily driven by improvements in the control schools rather than absolute declines in the treated schools.

We then examine the quality of toilets available to students. We create an index which measures the average quality of WASH facilities, including functionality, cleanliness, and, for toilets used by female students, the presence of items required for hygienic practices for MHM.[39] Since few schools have mixed gender toilets, we report this index only for toilets available to male students (B2a) and female students (B2b). We cannot reject the null of no quality improvement for either group, and quality declines for toilets available for female students. Again, these effects are primarily driven by relatively large improvements in quality in control schools, although toilet cleanliness declines in absolute terms in treated schools.

We finally turn to use of latrine facilities. Increased use of latrine facilities might be a proxy for improved quality, with the complicating factor that increased use might itself decrease cleanliness. As prespecified, we measure latrine use as the count of toilet use events in each toilet, coding non-functioning toilets as zero use, in toilets available to male students (B3a) and female students (B3b). Again, we cannot reject the null hypothesis of no increase for either group, and usage declines relative to the control group in toilets available to female students. The point estimate for girls remains negative across a range of alternative approaches to processing the sensor data.[40]

Our study was designed and powered to detect effects of the expected (positive) sign. We thus pre-specified one-sided hypothesis tests. However, all three of the treatment effects for toilets available to female students are negative, and two would have been rejected at the 10% level had we pre-specified two-sided hypothesis tests. The likelihood of seeing three negative effects of

[39] The full set of sub-indexes is in Appendix I. Sub-index results are in Appendix Tables K20 and K21.

[40] Appendix Table K22 describes sensitivity to excluding non-functional toilets (not prespecified) and controlling for the type of sensor used to detect use at baseline. Appendix Table K23 shows results using an alternative measure of use, total duration of use events. Appendix Table K24 shows results aggregating latrine use frequency or duration across latrines to the school level.

this magnitude within this outcome family and gender is small ($p$ = .008, Appendix Figure K7). In all three cases, the negative point estimate is driven not by absolute declines in the treated group but by relatively large improvements in the control group.

**Table 10.** *B: WASH provision and use*

| | Number of Functional toilets | | | Index WASH quality | | Toilet event frequency | |
|---|---|---|---|---|---|---|---|
| | Male | Female | Mixed | Male | Female | Male | Female |
| Treatment × post | 0.228 | -0.533 | 0.067 | 0.107 | -0.163 | -0.112 | -8.322 |
| | (0.301) | (0.284) | (0.178) | (0.147) | (0.087) | (4.516) | (6.202) |
| One-sided $p$ | 0.252 | 0.950 | 0.454 | 0.244 | 0.965 | 0.496 | 0.927 |
| FWER one-sided $p$ | 0.889 | 1.000 | 0.967 | 0.889 | 1.000 | 0.967 | 1.000 |
| Two-sided $p$ | 0.461 | 0.083 | 0.876 | 0.492 | 0.082 | 0.987 | 0.175 |
| N | 98 | 120 | 120 | 162 | 238 | 128 | 205 |
| Baseline mean | 1.079 | 1.583 | 0.117 | -0.007 | 0.045 | 5.821 | 11.805 |
| Change control | -0.292 | 0.400 | -0.033 | -0.049 | 0.157 | 1.736 | 14.835 |
| Change treated | -0.040 | -0.133 | 0.033 | 0.020 | -0.043 | -0.523 | 8.750 |
| Unit of observation | School | School | School | Toilet | Toilet | Toilet | Toilet |

*Notes* Standard errors clustered by school and shown in parentheses. Results from regressions that include school and upazila-year fixed effects and stratification controls. Weights applied so that all schools have the same weight in summary statistics. One-sided $p$ values (pre-specified) test the null hypothesis of no positive effects; FWER-controlled $p$ values (pre-specified) hold the possibility of making type I errors within the family to its nominal level. Two-sided $p$ values (exploratory) test the null hypothesis of no effects of any sign. Changes in treated and control schools are raw weighted mean changes between baseline and endline for each group, and the difference between the two may not correspond to the estimated treatment effect.

## C: School attendance and educational attainment

We finally turn to effects on school attendance and educational achievement (Table 11). Our primary measure of attendance is drawn from school records from the months of February and March, two months during which school attendance was relatively uninterrupted by exams or national holidays in 2018 and 2019. We estimate effects on school attendance for male (C1a) and female (C1b) students. In both cases, attendance rates in treated schools increase by a small amount (0.9 percentage points, or p.p.) relative to control schools, but in neither case can we statistically reject the null hypothesis of no improve-

ment.[41] The 90% confidence intervals exclude increases of more than 4.1 p.p for girls. For comparison, data on self-reported absence during menstruation suggest that absence associated with menstruation could reduce female student attendance by 4.3 p.p.,[42] while the extent of absence associated with water-borne disease may be larger but is more uncertain (McMichael, 2019).

One might be concerned that registered attendance is inaccurate or prone to overreporting, especially as female student attendance is incentivized through a stipend scheme.[43] However, observed and registered attendance are similarly distributed and highly correlated (Appendix Figures K8a and K8b). At baseline, overreporting is absent for female students and modest for male students. In schools and grades consisting only of one class—in which we can be sure that both measures correspond to the same group of students—around 1.7 additional male students are recorded present per class than enumerators observe (Appendix Table K26). Overreporting increases between baseline and endline but does not increase differentially in treated and control schools (Appendix Table K27).

Our primary measure of educational achievement is GPA on national standardized tests. We compare results from standardized tests—national Junior and Senior School Certificate exams—taken before and after the intervention. In both cases, we fail to detect increases in test results for either male (C2a) or female (C2b) students. The 90% confidence intervals rule out improvements of 0.16 points on a 5-point scale for female students and 0.07 points for male students. We also do not detect effects on the share of students passing the exam, an alternative measure of achievement (Appendix Table K28).[44]

[41] The results are similar, but make less efficient use of the data, when we use the pre-specified school-grade-date fixed effects instead of school-grade-day of week fixed effects (Appendix Table K25.

[42] Alam et al. (2017) report that 41% of Bangladeshi school girls report missing school due to menstruation and that those who report missing school miss on average 2.8 days per menstrual cycle. Assuming a median cycle length of around 31 days for adolescent girls (Flug et al., 1984) and a 6 day school week, this is equivalent to a 4.3 percentage point reduction in attendance.

[43] The Female Secondary School Stipend Project pays tuition fees and provides monthly stipends for unmarried rural girls up to grade 10 who attend recognized institutions, remain unmarried, and meet attendance and exam result targets.

[44] Students fail the exam if they score a mean GPA of below 1. For these students, the final score is not recorded. In the main analysis, we code this students as having a GPA equal to 0.5, but the results are insensitive to alternative assumptions (Appendix Table K29).

**Table 11.** *C: School attendance and exam results*

| | School attendance | | Exam results | |
|---|---|---|---|---|
| | Male | Female | Male | Female |
| Treatment × post | 0.010 | 0.006 | -0.064 | 0.021 |
| | (0.026) | (0.021) | (0.079) | (0.083) |
| One-sided *p* | 0.367 | 0.431 | 0.774 | 0.411 |
| FWER one-sided *p* | 0.776 | 0.776 | 0.776 | 0.776 |
| Two-sided *p* | 0.714 | 0.795 | 0.442 | 0.827 |
| N | 695623 | 1152921 | 10119 | 11471 |
| Baseline mean | 0.574 | 0.658 | 2.322 | 2.278 |
| Change control | 0.017 | 0.005 | 0.168 | 0.146 |
| Change treated | 0.037 | 0.012 | 0.071 | 0.216 |
| Unit of observation | Student-day | Student-day | Student | Student |
| Time fixed effects | Grade-upazila-date-year | Grade-upazila-date-year | Level-upazila-year | Level-upazila-year |
| Unit fixed effects | School-grade-day of week | School-grade-day of week | School-level | School-level |

*Notes* Standard errors clustered by school and shown in parentheses. Results from regressions that include school and time and unit fixed effects as specified and stratification controls. Weights applied so that schools and grades (for school attendance) and schools and exam levels (for exam results) have the same weight in summary statistics. One-sided *p* values (pre-specified) test the null hypothesis of no positive effects; FWER-controlled *p* values (pre-specified) hold the possibility of making type I errors within the family to its nominal level. Two-sided *p* values (exploratory) test the null hypothesis of no effects of any sign. Changes in treated and control schools are raw weighted mean changes between baseline and endline for each group, and the difference between the two may not correspond to the estimated treatment effect.

A caveat to these results is that students not enrolled in the school or grade can sit the School Certificate exams. This might in principle attenuate any treatment effects of the program, because some students sitting the exams will not have been exposed to treatment. Also, not all enrolled students sit the exam. However, the number of students sitting the exams prior to the baseline survey is highly correlated with current class size in the relevant grade, suggesting that any attenuation may be small (Appendix Figures K9a and K9b).

The results in outcome family C could be affected if the intervention led treated schools to increase or decrease enrolment relative to the control group. Male student enrolment increases modestly and by similar amounts in both treatment and control schools. Female student enrolment actually declines in treated schools and increases in control schools (Appendix Table K30). The total number of students sitting exams is relatively stable over time and does not change differentially in treated and control schools (Appendix Table K31).

# 9 Discussion

The results show that the AWIS intervention led to modest improvements in knowledge and institutions for governing and managing WASH services but did not improve WASH service provision. Exploratory analysis suggests that treated schools may have improved WASH service provision for female students less successfully than control schools. In this section, we suggest four potential reasons why the intervention did not improve WASH service provision despite the positive effects on knowledge and institutions. One reason relates to the research design, and three relate to the intervention design. We draw upon three sources of descriptive data: 1) participant evaluations collected during the AWIS workshop; 2) retrospective survey questions about the workshop and participant behaviour afterward; and 3) qualitative data from the semi-structured interviews. We also compare our results with previous literature to shed light on the generalizability of our findings.

*Time frame of the evaluation*

We evaluate the effects of the intervention in the year following the workshop. This time frame is not atypical compared to previous studies (Tables 3a and 3b). However, some potential effects may continue to evolve over longer time frames, such as the construction of new WASH facilities or the incorporation of improved WASH facilities in new construction. Our results leave open the possibility that the improvements we see in knowledge and institutions might still lead to meaningful impacts on WASH provision over longer time frames.

We find some evidence to support this view in our descriptive data. The intervention increases the similarity of answers given by different respondents to questions about the problems faced by female students related to menstrual hygiene management (MHM), suggesting an improved common understanding of problems with WASH provision at study schools.[45] Almost all workshop participants we surveyed at endline also reported taking action after the workshop, and almost all felt that their actions were successful (Appendix Figure K10). In one study school, semi-structured interview respondents consistently describe transformational changes in WASH provision, including physical improvements to infrastructure and wide-ranging improvements in collective facility management. These changes appear to have taken place after the endline survey. Interview respondents attribute the impetus for these changes to the AWIS intervention. This pattern of results is also consistent with findings after a similar intervention conducted in Indonesia and Tanzania: focus groups in a substantial minority of communities described sustained, concrete efforts

[45] We calculate a similarity index across dyads of survey respondents including headteachers, male teachers, and female students. These are the classes of respondents that we interview in all study schools. For each dyad, we calculate a simple matching coefficient that measures similarity of responses across groups of questions. We estimate an increase in dyad-level similarity in responses to questions about how MHM impacts female students. More details are in Appendix J.

two years later, despite the intervention doing little on average to improve service provision in measurable terms (Kosack et al., 2019; Arkedis et al., 2021).

On the other hand, these self-reported data could reflect experimenter demand effects or other types of reporting bias. While the implementing NGO was not involved in data collection, the NGO that collected data might have been associated with the intervention through their role in organizing the treatment assignment lotteries. The dichotomy between the self-reported and measured efficacy of actions taken by participants could also arise through attribution bias if participants mistakenly attribute unrelated progress to the effects of their actions (Kosack et al., 2019). Other evidence also cautions against too much optimism. Some potential improvements, such as changes to cleaning protocols, are quite low-cost in terms of budget, albeit not necessarily in terms of time or effort, but few schools appear to have successfully implemented similar improvements. Interviewees speak of limited financial resources and the difficulty of raising additional funds when students' families are poor. Facilities also fall rapidly into disrepair. One semi-structured interview respondent speaks of new facilities falling into disrepair in less than eighteen months. The number of schools with no functioning toilets available to students is actually higher at endline (6) than at baseline (2), in both cases evenly divided between treated and control schools. Another notes the challenges of maintaining knowledge and informal institutions in the context of a rapidly changing school population.

*Time-limited nature of the intervention*

One potential reason why the intervention we study might not improve WASH service provision is that it is limited to a single, one-off workshop. Such an intervention design would be attractive, if successful, as cost-effectiveness would then be high. However, the "action cycle" of Kosack and Fung (2014) implicitly requires new information about the consequences of actions to be fed back to stakeholders. Otherwise, stakeholders will not know whether or not their actions were effective, and efforts may not be sustained. A natural question is whether a one-off workshop is sufficient to achieve this. Workshop participants themselves raised concerns that the intervention was too short, too isolated, too limited in scale, and lacked follow-up engagement (Appendix Figure K11). Potentially as a consequence of its time-limited nature, the intervention might have been more successful in increasing understanding of problems than agreement over solutions: at endline, there is *less* consensus about priorities for action in treated schools than in control schools.[46] This suggests that the intervention might have successfully generated a range of ideas about possible solutions but not built consensus over a specific and unified action plan. Transparency interventions may be more effective if coupled with longer-term engagement, for example, through a series of workshops in

[46]Details in Appendix J.

which participants evaluate not only baseline conditions but progress relative to previously identified action plans. Consensus-building over action plans may be as important as consensus-building over problem diagnosis. Additionally, transparency interventions might be more effective if coupled with accountability measures that incentivize adherence to action plans. Consistent with this hypothesis, common features of many successful interventions are a specific focus on creating concrete action plans and a longer time frame that includes follow-up meetings to monitor progress (Table 3a). In contrast, unsuccessful interventions often either fail to bring service providers and users together at all or are characterized by more time-limited interventions (Table 3b).

*Challenges of scale*

For transparency interventions to successfully ignite "action cycles" (Kosack and Fung, 2014), changes made by actors must affect provision. This also implies that interventions must influence actors who have the capacity to affect provision. It is not clear whether this is the case for the AWIS intervention. Semi-structured interview respondents report dependence on external organizations for financial support for major ("hard") infrastructure improvements. But local administrators, who have limited and fragmented responsibility for school WASH services, did not attend the AWIS workshops (see Table 2). In contrast, the actions that students, teachers, and headteachers report taking were primarily "soft" (Appendix Figure K12). Raffler et al. (2020) show suggestive evidence that a similar intervention was more successful when local officials participated in community-service provider workshops, although local officials only participated in about 30% of workshops. Additionally, the only explicit mentions of corruption or trust problems in the semi-structured interviews are with respect to contractors doing shoddy or incomplete work. This class of governance problems is not within the locus of control of school management or leadership.

In contexts like these, transparency interventions may need to think explicitly about how to influence other actors, perhaps using two-stage workshops at different administrative scales. Local administrators might be more likely to attend a district-wide workshop than multiple school-level workshops. Alternatively, transparency interventions might be coupled with financial support that frees schools from dependence on external decision-makers. In at least one successful precedent, facilitating collaboration between schools and community leadership increased the impact of a grant (Pradhan et al., 2014).

*Misleading expectations*

In this study, control schools appeared to more successfully improve WASH service provision for female students than treated schools, with the caveat that our experiment was not designed to measure negative effects. However, intervention designers and researchers may want to consider the possibility that

stand-alone transparency or accountability interventions might create misleading expectations about future financial support, potentially discouraging participants from trying to solve their own problems in anticipation of this support. In a semi-structured interview, a teacher who participated in the AWIS workshop mentions informing DORP (the local implementing NGO) that the school needed a new well during the AWIS workshop, seemingly with the expectation that DORP would help provide one. While the negative effects we observe appear unique in the published literature, many studies report null effects, and Arkedis et al. (2021) note that expectations about "how these projects work" may shape the choices people make about whether or not to dedicate time and effort to project activities.

The evidence is inconsistent with another potential explanation, that the intervention failed because it antagonized participants or was simply altogether ineffective. The intervention did improve knowledge and institutions. Ex post, survey respondents expressed satisfaction with the intervention, albeit not universally: a substantial share of students and teachers felt that some participants were more influential than others, and about a third of teachers did not feel free to express their opinions freely (Appendix Figure K13). Participants also praised the educational nature of the intervention and the systematic, constructive, and participatory discussion (Appendix Figure K14).

# 10 Conclusion

This study evaluates the impact of a transparency intervention on WASH service provision in schools in rural Bangladesh. While interventions designed to increase transparency are currently "in vogue", evidence from similar interventions in other sectors finds only mixed effects (Joshi, 2010; Kosack and Fung, 2014; Fox, 2015). The transparency intervention we study is a deliberative multi-stakeholder workshop initiated with a community scoring exercise, closely resembling one of a set of "exemplary transparency interventions" identified by Kosack and Fung (2014). Eight to twelve months after implementation, the intervention improves knowledge of WASH and MHM standards and best practices, and WASH management institutions, but does not improve WASH service provision. Although our study was not designed to detect negative effects, we find suggestive evidence that control schools improve WASH service provision for female students more successfully than treated schools. We find no detectable effects on school attendance or exam performance.

Drawing on extensive descriptive data, we identify four potential explanations for the absence of a measurable positive effect on WASH service provision despite the positive effects on knowledge and institutions: the relatively

short time frame of the study; the time-limited nature of the intervention; the challenges of implementing transparency interventions at a decentralized level when actors are dependent on external sources of support; and the potential for such interventions to create misleading expectations of future support, discouraging stakeholders from independently seeking solutions.

Designers of transparency interventions may wish to consider how to address the potential limitations we outline above. In particular, our results suggest that interventions may need longer time horizons to be effective, for example incorporating repeated workshops to build consensus about concrete action plans and monitor progress. Future studies should also examine how the effects of transparency interventions vary over time.

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